

Senegal Tax Audits: Comments and Revisions

February, 2026

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1 New exhibits for paper

1.1 Regression Analysis - Description of Approach

Objective. Using predicted evasion estimates generated by the Random Forest model, we test whether execution of algorithm-assigned cases depends on whether the "first-in-line" for execution algorithm case(s) on a given list are high- vs. low-potential. The "list" unit differs by audit type: for desk audits it is defined at the inspector $\times$ selection-year level, while for full audits it is defined at the bureau (detailed) $\times$ selection-year level. We compare execution rates of algorithm cases across these lists.

Steps.

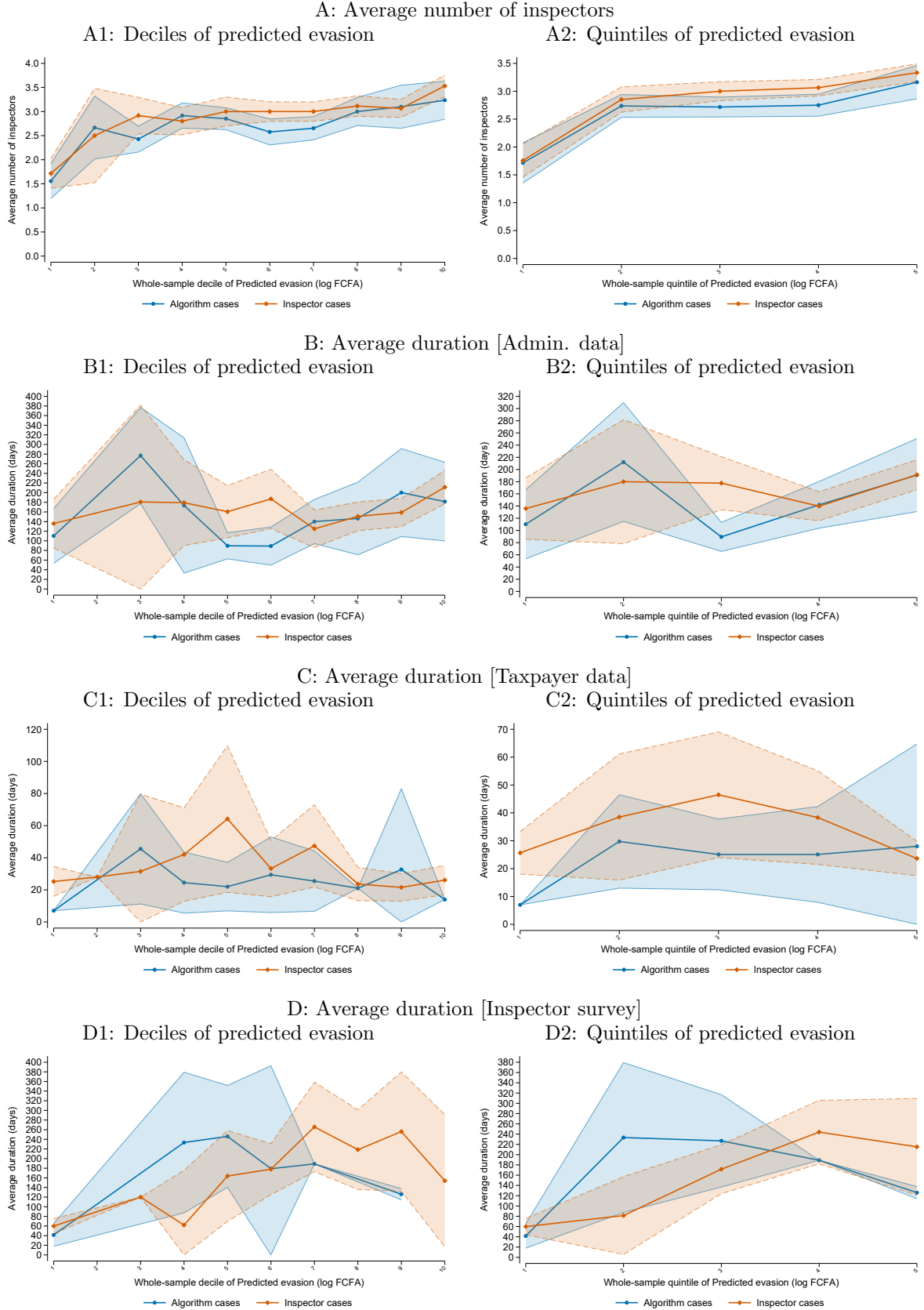
1. **Load predicted evasion estimates data which includes estimates for selected audits, and restrict by audit type (full or desk audit).** We use the dataset containing predicted evasion (`yhatrf`) and execution outcomes, and split the analysis by audit type (`x2==0` desk; `x2==1` full).
2. **Construct predicted-evasion quantile groups (audit-type specific).** We compute quintiles of predicted evasion `yhatrf` in the following way:
 - **Desk audits:** quintiles within `verificateur_selection` $\times$ `selectionyear`.
 - **Full audits:** quintiles within `bureau_detailed` $\times$ `selectionyear`.
3. **Restrict to algorithm-assigned cases.** We keep only algorithm-selected audits (`algorithm==1`) so that all outcomes and tags refer to the execution of algorithm-assigned cases.
4. **Define "top of the list" using execution sequence among algorithm cases.** Within each list, we order algorithm cases by the sequencing variable (`sequencing`) to define early algorithm cases (e.g., Top 1/Top 2, first 10% or first 20% of the algorithm sequence).
5. **Tag lists based on the potential of first-in-line algorithm cases.** For each list, we create indicators for whether an first-in-line algorithm case is: (i) **high-potential** (top quintile of `yhatrf`, `yhatrf_q5==5`), or (ii) **low-potential** (bottom quintile, `yhatrf_q5==1`). This generates list-level groups such as "high-potential-first" vs "low-potential-first".
6. **Collapse to the list level and compute execution rates.** We collapse the data to the list level and compute the mean execution rate of assigned algorithm cases within each list (mean of `y2` among `algorithm==1`).
7. **Compare execution outcomes across list groups.** We present (i) boxplots of list-level algorithm execution rates (unweighted and weighted by the number of assigned algorithm cases), and (ii) regression estimates comparing execution rates across "high-potential-first" and "low-potential-first" lists (with fixed effects varying by audit type).

Table 1: Execution of algorithm-assigned audits and high predicted evasion cases at the top of the list [Full audits]

Outcome: Execution Rate of Algorithm-Selected Cases	No FE			Selection year FE			Bureau FE		
	Top 2	First 10% seq	First 20% seq	Top 2	First 10% seq	First 20% seq	Top 2	First 10% seq	First 20% seq
Panel A: Discrete measures									
High potential evasion case in list =1	0.218* (0.124)	0.289** (0.123)	0.207* (0.108)	0.176 (0.105)	0.224** (0.102)	0.199** (0.087)	0.144 (0.185)	0.245 (0.173)	0.165 (0.153)
Low potential evasion case in list =1	-0.079 (0.108)	0.044 (0.112)	-0.108 (0.116)	-0.079 (0.105)	-0.007 (0.110)	-0.072 (0.112)	-0.067 (0.122)	0.085 (0.127)	-0.049 (0.171)
Selection Year FE	No	No	No	Yes	Yes	Yes	No	No	No
Bureau FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	29	29	29	29	29	29	29	29	29
R-squared	0.129	0.118	0.208	0.366	0.350	0.420	0.424	0.431	0.430
Panel B: Continuous measure									
Share of high predicted evasion case at the top	0.532** (0.225)	0.530** (0.240)	0.556*** (0.187)	0.457** (0.190)	0.403 (0.238)	0.481** (0.216)	0.398 (0.325)	0.492 (0.333)	0.444 (0.310)
Selection Year FE	No	No	No	Yes	Yes	Yes	No	No	No
Bureau FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	29	29	29	29	29	29	29	29	29
R-squared	0.112	0.076	0.097	0.350	0.313	0.343	0.416	0.422	0.417

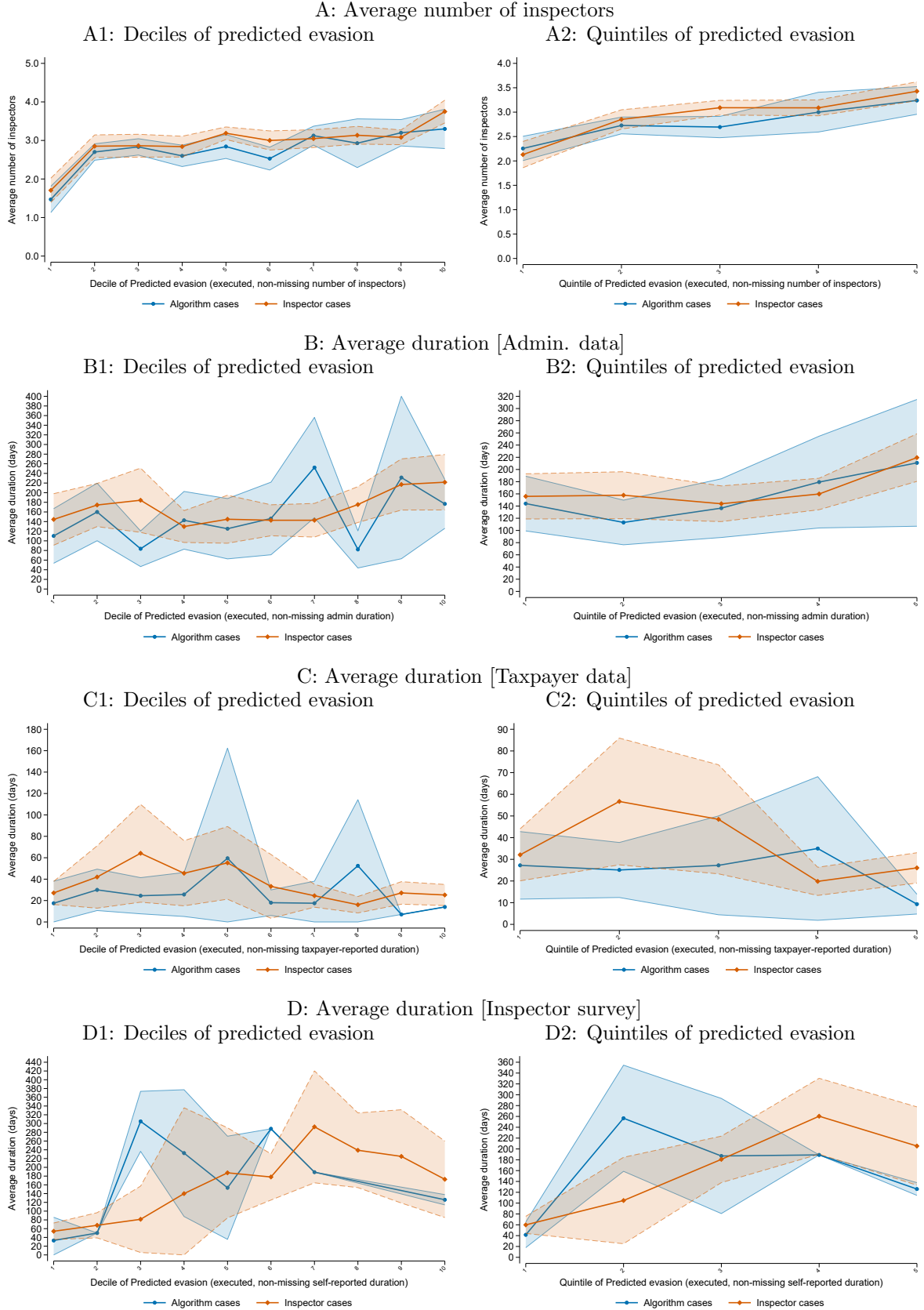
Notes: This table reports regression estimates of how the presence of high- or low-predicted-evasion (high-/low-potential) cases at the top of an algorithm list is associated with the execution rate of algorithm-assigned audits. The unit of observation is a bureau-by-selection-year list (b, t) . The dependent variable is y_{bt} , the execution rate of algorithm-selected cases in bureau b and selection year t (executed algorithm-selected cases divided by assigned algorithm-selected cases). To classify the audit case potential we use the estimated predicted evasion from the Random Forest model and partitioned it into quintiles within tax office $\times$ selection year; a "high" predicted evasion corresponds to the top quintile and "low" predicted evasion to the bottom quintile. Panel A presents the results of the discrete specification of an OLS regression through the following equation: $y_{bt} = \alpha + \beta_H \text{HighTop}_{bt} + \beta_L \text{LowTop}_{bt} + \delta_t + \mu_b + \varepsilon_{bt}$, where $\text{HighTop}_{bt} = 1$ if the top of the ranking contains a high-predicted-evasion case under the relevant top-of-list definition (Top 2, first 10% of the sequence, first 20% of the sequence), and 0 otherwise; LowTop_{bt} is defined analogously for low-predicted-evasion cases. Panel B presents the results of an alternative specification using a continuous measure of high potential cases, by estimating: $y_{bt} = \alpha + \theta \text{ShareHighTop}_{bt} + \delta_t + \mu_b + \varepsilon_{bt}$, where ShareHighTop_{bt} is the share of high-predicted-evasion cases at the top of the list (same top-of-list definitions as in Panel A). Columns labeled No FE omit fixed effects, Selection Year FE include δ_t only, and Bureau FE include μ_b only. Standard errors are robust. Stars denote significance at $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Figure 1: Predicted evasion vs. number of inspectors and duration, Full audits [Executed cases sample – preferred]



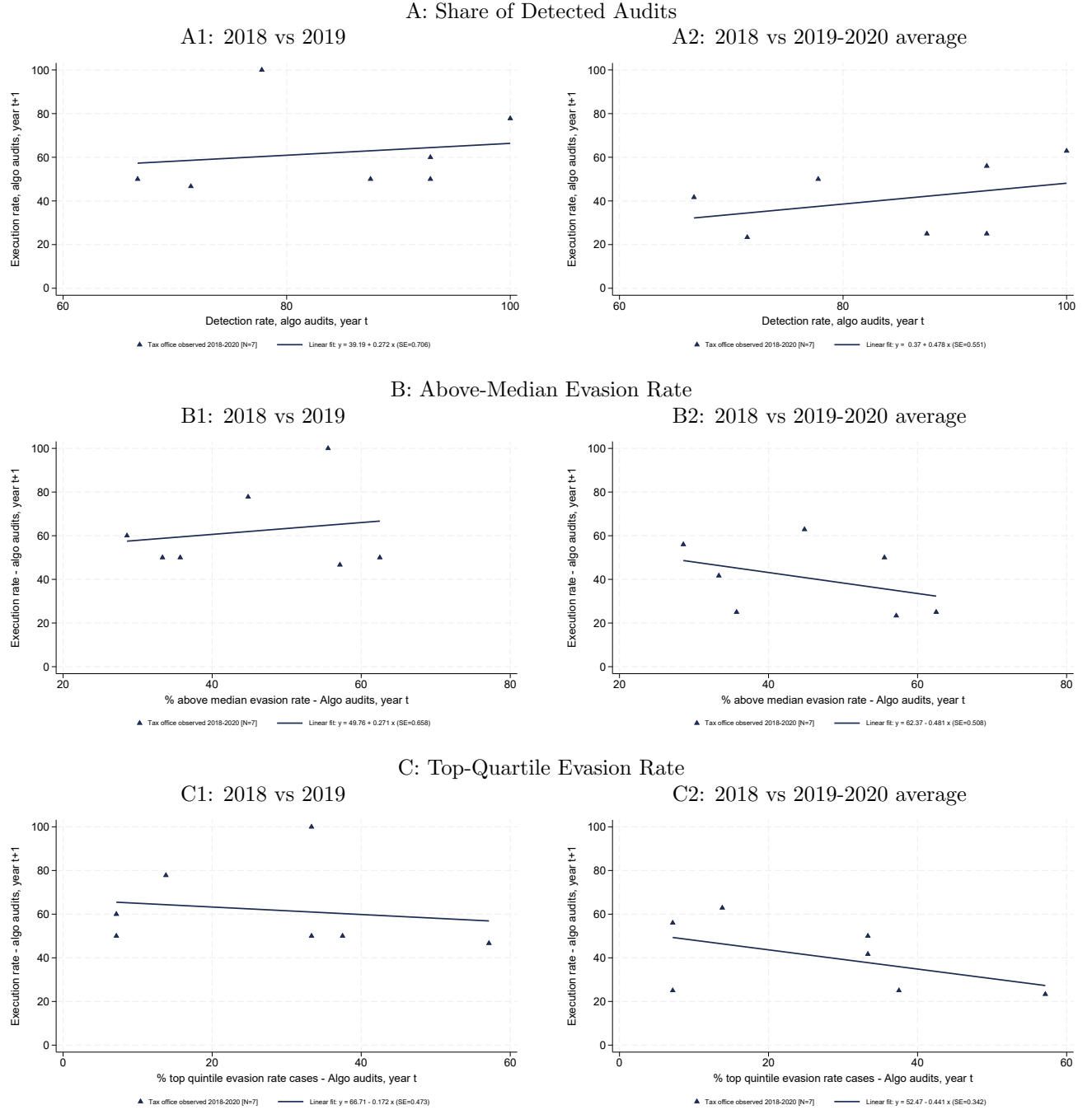
Notes: Predicted-evasion bins are defined using the pooled sample of selected full audits and are then kept fixed when computing outcomes for executed cases. Each panel reports mean outcomes by selection method and predicted-evasion bin, comparing algorithm-selected and inspector-selected cases. Panel A shows the number of inspectors; Panel B administrative duration; Panel C taxpayer-reported duration; and Panel D inspector/self-reported duration. Shaded bands indicate 95% confidence intervals for each method-by-bin cell. Column 1 uses predicted-evasion deciles, while Column 2 uses predicted-evasion quintiles.

Figure 2: Predicted evasion vs. number of inspectors and duration, Full audits [Outcome-specific executed sample]



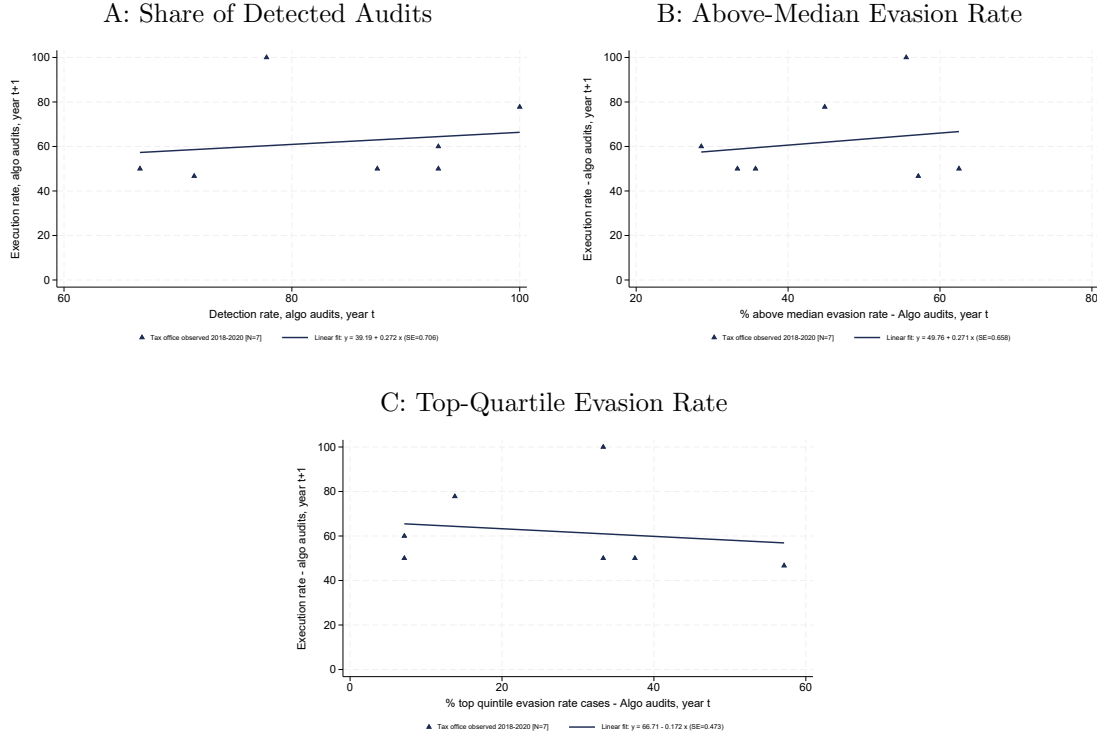
Notes: Predicted-evasion bins are defined separately for each outcome using the pooled eligible sample of executed full audits with non-missing predicted evasion and non-missing outcome data, and these outcome-specific cutoffs are then applied to both algorithm-selected and inspector-selected cases. Panel A shows the number of inspectors; Panel B administrative duration; Panel C taxpayer-reported duration; and Panel D inspector/self-reported duration. Shaded bands indicate 95% confidence intervals for each method-by-bin cell. Column 1 uses predicted-evasion deciles, while Column 2 uses predicted-evasion quintiles.

Figure 3: Execution Rates and Audit Success, Algorithm-Selected Full Audits



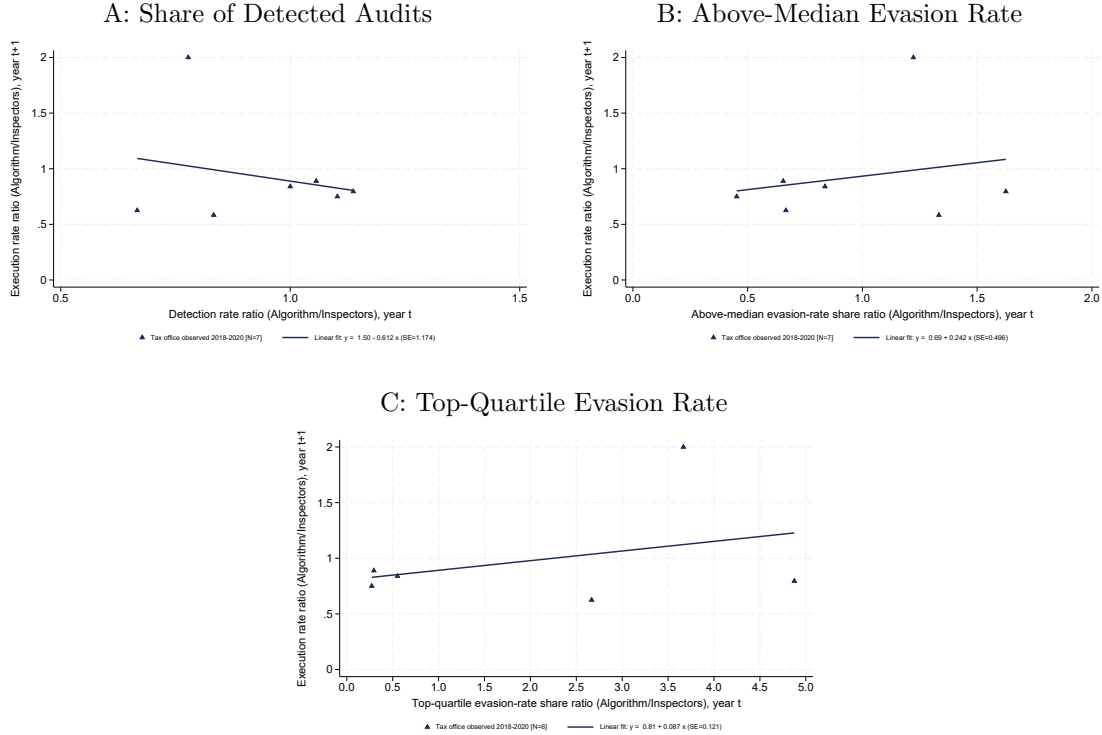
Notes: This figure relates tax office-level exposure in year t to successful algorithm-selected audits to the execution rate of algorithm-selected audits in year $t + 1$. The sample includes selected full audits only (excluding safety cases) and is aggregated at the tax office $\times$ year level. The estimation sample keeps offices observed in at least two periods in the relevant comparison window (2018 vs 2019, or 2018 vs the 2019–2020 average specification). An audit is classified as detected when it is executed and uncovered evasion value is strictly positive (complement of a void audit). Evasion rate is defined for executed audits as uncovered evasion value divided by uncovered evasion value plus average turnover; average turnover is constructed as the average reported sales (2014–2020). Panel A uses the share of detected audits; Panel B the share of audits above the median evasion-rate cutoff; Panel C the share in the top quartile of the evasion-rate distribution. Median and quartile evasion-rate cutoffs are computed within selection-year $\times$ selection method. For Panels A–C, shares on the horizontal axis use executed algorithm-selected audits as denominators. The vertical-axis execution measure is the algorithm execution rate, defined as executed algorithm-selected audits divided by assigned algorithm-selected audits. Column 1 compares 2018 and 2019. Column 2 compares 2018 with the average of 2019 and 2020, computed as the row mean. Points represent tax offices observed during the whole analysis period (2018–2020), and lines show unweighted linear fits.

Figure 4: Execution Rates and Audit Success, Algorithm-Selected Full Audits: 2018 vs 2019



Notes: This figure relates tax office-level exposure in year t to successful algorithm-selected audits to the execution rate of algorithm-selected audits in year $t+1$. The sample includes selected full audits only (excluding safety cases) and is aggregated at the tax office $\times$ year level. The estimation sample keeps offices observed in both 2018 and 2019. An audit is classified as detecting evasion when it is executed and the uncovered evasion value is strictly positive. Evasion rate is defined for executed audits as uncovered evasion value divided by uncovered evasion value plus average turnover; average turnover is constructed as the average reported sales (2014–2020). Panel A uses the share of audits that detect evasion, conditional on audit implementation; Panel B the share of audits above the median evasion-rate cutoff; and Panel C the share in the top quartile of the evasion-rate distribution. Median and quartile evasion-rate cutoffs are computed within tax office $\times$ selection year. In all panels, shares on the horizontal axis use executed algorithm-selected audits as denominators. The vertical-axis execution measure is the algorithm execution rate, defined as executed algorithm-selected audits divided by assigned algorithm-selected audits. Each plot compares 2018 and 2019. Points represent tax offices observed in both years, and lines show unweighted linear fits.

Figure 5: Execution Rates and Audit Success relative to Inspector-selected Cases, Full Audits: 2018 vs 2019



Notes: This figure relates tax office-level exposure in year t to successful algorithm-selected audits to the execution rate of algorithm-selected audits in year $t+1$. The sample includes selected full audits only (excluding safety cases) and is aggregated at the tax office $\times$ year level. The estimation sample keeps offices observed in both 2018 and 2019. An audit is classified as detecting evasion when it is executed and the uncovered evasion value is strictly positive. Evasion rate is defined for executed audits as uncovered evasion value divided by uncovered evasion value plus average turnover; average turnover is constructed as the average reported sales (2014–2020). Panel A uses the share of audits that detect evasion, conditional on audit implementation, relative to the corresponding share among inspector-selected audits; Panel B uses the share of audits above the median evasion-rate cutoff, relative to the corresponding share among inspector-selected audits; and Panel C uses the share in the top quartile of the evasion-rate distribution, relative to the corresponding share among inspector-selected audits. Median and quartile evasion-rate cutoffs are computed within tax office $\times$ selection year. In all panels, the horizontal axis measures the algorithm-selected share relative to the corresponding inspector-selected share within the same tax office and year. The vertical-axis execution measure is the algorithm execution rate, defined as executed algorithm-selected audits divided by assigned algorithm-selected audits. Each plot compares 2018 and 2019. Points represent tax offices observed in both years, and lines show unweighted linear fits.

2 Audit productivity

2.1 Table 5 with predicted evasion controls

This section presents the full-audit Table 5 top-panel variants for the audit-productivity discussion. The tables below keep the original sample and layout and add predicted-evasion controls in the robustness specifications.

Table 2: Resources Allocated to Audits and Audit Productivity, Full Audits (Panels A1 and B1 only)

	A1: Resource Outcomes				B1: Productivity Outcomes			
	Number of Agents (1)	Duration in Days (Taxpayer Survey) (2)	Days from Start to Conf. (Admin Data) (3)	Days Working on Case (Self-Reported) (4)	Evasion/ Number of Agents (5)	Evasion/Duration (Taxpayer Survey) (6)	Evasion/Duration (Admin. Data) (7)	Evasion/Days Working (Self-Reported) (8)
Algorithm	-0.26*** (0.08)	-8.54** (3.72)	-30.18** (13.75)	-27.26 (30.60)	-0.55*** (0.18)	-0.12 (0.39)	-0.50* (0.26)	-0.74 (0.49)
Inspectors x Overlap	0.04 (0.17)	-25.57 (21.89)	-9.83 (27.66)		0.30 (0.44)	0.96 (2.20)	-0.01 (0.57)	
N	508	214	282	51	457	98	253	40
R2	0.21	0.25	0.30	0.25	0.31	0.24	0.25	0.40
Mean outcome	2.87	28.06	160.07	177.07	18.19	15.90	15.00	13.62

Notes: This table replicates the top panels (A1 and B1) of the full-audit version of Table 5 using the original specification. Panel A1 reports resource outcomes and Panel B1 reports productivity outcomes. The sample includes selected full audits and excludes safety cases. Column 1 reports the number of agents recorded on the case. Column 2 uses the taxpayer-survey duration measure. Column 3 uses the administrative measure of days from the start of the audit to confirmation and includes the original date availability controls. Column 4 uses inspectors' self-reported days worked on the case. Columns 5 to 8 use the log of evasion divided by the corresponding resource measure. All regressions include Algorithm, Inspectors x Overlap, Random, and Replacement indicators and absorb inspectorclusteryear fixed effects. Robust standard errors are shown in parentheses. Stars denote significance at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Resources Allocated to Audits and Audit Productivity, Full Audits, with Predicted Evasion Control (Panels A1 and B1 only)

	A1: Resource Outcomes				B1: Productivity Outcomes			
	Number of Agents (1)	Duration in Days (Taxpayer Survey) (2)	Days from Start to Conf. (Admin Data) (3)	Days Working on Case (Self-Reported) (4)	Evasion/ Number of Agents (5)	Evasion/Duration (Taxpayer Survey) (6)	Evasion/Duration (Admin. Data) (7)	Evasion/Days Working (Self-Reported) (8)
Algorithm	-0.19*** (0.06)	-8.18** (3.78)	-28.65** (14.01)	-11.43 (25.35)	0.12 (0.10)	0.86*** (0.24)	0.16 (0.17)	-0.22 (0.27)
Inspectors x Overlap	-0.09 (0.13)	-25.98 (22.09)	-10.64 (27.78)		0.23 (0.24)	1.36 (1.46)	0.21 (0.34)	
Predicted evasion	0.15*** (0.01)	0.40 (0.59)	1.72 (1.96)	13.35*** (1.84)	1.01*** (0.04)	1.31*** (0.10)	1.03*** (0.06)	1.04*** (0.11)
N	508	214	282	51	457	98	253	40
R2	0.52	0.25	0.30	0.51	0.78	0.74	0.71	0.70
Mean outcome	2.87	28.06	160.07	177.07	18.19	15.90	15.00	13.62

Notes: * 0.10 ** 0.05 *** 0.01 significance levels. OLS results of regressions of outcomes on the selection method of the case, with fixed effects at the inspector-cluster $\times$ year level. The specifications additionally control for predicted evasion. The sample includes all full audit cases selected from the 2018, 2019, and 2020 audit programs, excluding safety cases, for which we had data on the outcome variables. Panel A1 shows the results for the resource outcomes, and Panel B1 shows the results for the productivity outcomes. Column 1 shows the results for the number of agents involved in the audit, which is obtained from the audit records. Column 2 shows the results for audit duration as reported by taxpayers in the taxpayer survey. The survey asked them about their last full audit experience. Column 3 uses the difference between the date of initial notice and the date of the start of the audit (either verification announcement or information request or start date of audit) as a proxy for the duration of the audit. This measure is available only for cases that had an initial notice. We drop negative values, trim the data at the 99th percentile, and control for dummies indicating availability of date of initial notice and final notice. These controls are helpful because the definition of the start date may vary depending on the availability of information. Column 4 measures the duration of the audit using the self-reported number of days inspectors worked on a case, reported on a spreadsheet that inspectors needed to fill out during the 2019 audit program. Columns 5 to 8 use as outcomes the log of the ratio of evasion over the different cost measures used in Columns 1 to 4. We multiplied the answer by the number of inspectors that worked in the case. Robust standard errors (Huber-White formula) are shown in parentheses.

Table 4: Resources Allocated to Audits and Audit Productivity, Full Audits, with Predicted Evasion Quadratic Control (Panels A1 and B1 only)

	A1: Resource Outcomes				B1: Productivity Outcomes			
	Number of Agents (1)	Duration in Days (Taxpayer Survey) (2)	Days from Start to Conf. (Admin Data) (3)	Days Working on Case (Self-Reported) (4)	Evasion/ Number of Agents (5)	Evasion/Duration (Taxpayer Survey) (6)	Evasion/Duration (Admin. Data) (7)	Evasion/Days Working (Self-Reported) (8)
Algorithm	-0.18*** (0.06)	-10.33** (4.18)	-26.46* (14.24)	-16.02 (24.60)	0.12 (0.10)	0.85*** (0.24)	0.18 (0.17)	-0.25 (0.26)
Inspectors x Overlap	-0.09 (0.13)	-26.27 (21.84)	-9.54 (27.64)		0.25 (0.24)	1.72 (1.45)	0.23 (0.35)	
Predicted evasion	0.08 (0.06)	10.55** (5.07)	-8.71 (12.33)	47.60 (31.52)	0.36 (0.43)	-1.75 (1.26)	-0.28 (0.52)	-2.68 (2.11)
Predicted evasion squared	0.00 (0.00)	-0.34* (0.17)	0.37 (0.45)	-1.34 (1.25)	0.02 (0.01)	0.08** (0.04)	0.04** (0.01)	0.10* (0.06)
N	508	214	282	51	457	98	253	40
R2	0.52	0.27	0.30	0.52	0.78	0.75	0.72	0.71
Mean outcome	2.87	28.06	160.07	177.07	18.19	15.90	15.00	13.62

Notes: * 0.10 ** 0.05 *** 0.01 significance levels. OLS results of regressions of outcomes on the selection method of the case, with fixed effects at the inspector-cluster $\times$ year level. The specifications additionally control for predicted evasion and its square. The sample includes all full audit cases selected from the 2018, 2019, and 2020 audit programs, excluding safety cases, for which we had data on the outcome variables. Panel A1 shows the results for the resource outcomes, and Panel B1 shows the results for the productivity outcomes. Column 1 shows the results for the number of agents involved in the audit, which is obtained from the audit records. Column 2 shows the results for audit duration as reported by taxpayers in the taxpayer survey. The survey asked them about their last full audit experience. Column 3 uses the difference between the date of initial notice and the date of the start of the audit (either verification announcement or information request or start date of audit) as a proxy for the duration of the audit. This measure is available only for cases that had an initial notice. We drop negative values, trim the data at the 99th percentile, and control for dummies indicating availability of date of initial notice and final notice. These controls are helpful because the definition of the start date may vary depending on the availability of information. Column 4 measures the duration of the audit using the self-reported number of days inspectors worked on a case, reported on a spreadsheet that inspectors needed to fill out during the 2019 audit program. Columns 5 to 8 use as outcomes the log of the ratio of evasion over the different cost measures used in Columns 1 to 4. We multiplied the answer by the number of inspectors that worked in the case. Robust standard errors (Huber-White formula) are shown in parentheses.

Table 5: Resources Allocated to Audits and Audit Productivity, Full Audits, with Predicted Evasion Decile Controls (Panels A1 and B1 only)

	A1: Resource Outcomes				B1: Productivity Outcomes			
	Number of Agents (1)	Duration in Days (Taxpayer Survey) (2)	Days from Start to Conf. (Admin Data) (3)	Days Working on Case (Self-Reported) (4)	Evasion/ Number of Agents (5)	Evasion/Duration (Taxpayer Survey) (6)	Evasion/Duration (Admin. Data) (7)	Evasion/Days Working (Self-Reported) (8)
Algorithm	-0.20*** (0.06)	-10.35** (4.68)	-23.59 (14.49)	-21.35 (21.61)	0.05 (0.11)	0.84*** (0.26)	0.11 (0.19)	-0.46 (0.33)
Inspectors x Overlap	-0.10 (0.13)	-28.01 (20.74)	-5.10 (27.54)		0.21 (0.27)	0.61 (1.90)	0.23 (0.37)	
N	508	214	282	51	457	98	253	40
R2	0.51	0.29	0.32	0.57	0.76	0.71	0.66	0.76
Mean outcome	2.87	28.06	160.07	177.07	18.19	15.90	15.00	13.62

Notes: * 0.10 ** 0.05 *** 0.01 significance levels. OLS results of regressions of outcomes on the selection method of the case, with fixed effects at the inspector-cluster $\times$ year level. The specifications additionally control for predicted-evasion deciles formed on the pooled regression-eligible full-audit sample. The sample includes all full audit cases selected from the 2018, 2019, and 2020 audit programs, excluding safety cases, for which we had data on the outcome variables. Panel A1 shows the results for the resource outcomes, and Panel B1 shows the results for the productivity outcomes. Column 1 shows the results for the number of agents involved in the audit, which is obtained from the audit records. Column 2 shows the results for audit duration as reported by taxpayers in the taxpayer survey. The survey asked them about their last full audit experience. Column 3 uses the difference between the date of initial notice and the date of the start of the audit (either verification announcement or information request or start date of audit) as a proxy for the duration of the audit. This measure is available only for cases that had an initial notice. We drop negative values, trim the data at the 99th percentile, and control for dummies indicating availability of date of initial notice and final notice. These controls are helpful because the definition of the start date may vary depending on the availability of information. Column 4 measures the duration of the audit using the self-reported number of days inspectors worked on a case, reported on a spreadsheet that inspectors needed to fill out during the 2019 audit program. Columns 5 to 8 use as outcomes the log of the ratio of evasion over the different cost measures used in Columns 1 to 4. We multiplied the answer by the number of inspectors that worked in the case. Robust standard errors (Huber-White formula) are shown in parentheses.

Table 6: Resources Allocated to Audits and Audit Productivity, Full Audits, with Predicted Evasion Quintile Controls (Panels A1 and B1 only)

	A1: Resource Outcomes				B1: Productivity Outcomes			
	Number of Agents (1)	Duration in Days (Taxpayer Survey) (2)	Days from Start to Conf. (Admin Data) (3)	Days Working on Case (Self-Reported) (4)	Evasion/ Number of Agents (5)	Evasion/Duration (Taxpayer Survey) (6)	Evasion/Duration (Admin. Data) (7)	Evasion/Days Working (Self-Reported) (8)
Algorithm	-0.20*** (0.06)	-8.56** (4.08)	-27.06* (14.34)	-7.38 (21.59)	-0.02 (0.12)	0.76*** (0.28)	-0.02 (0.19)	-0.37 (0.30)
Inspectors x Overlap	-0.10 (0.13)	-27.41 (21.58)	-6.37 (27.54)		0.21 (0.28)	0.44 (1.72)	0.14 (0.39)	
N	508	214	282	51	457	98	253	40
R2	0.50	0.27	0.32	0.54	0.72	0.66	0.62	0.75
Mean outcome	2.87	28.06	160.07	177.07	18.19	15.90	15.00	13.62

Notes: * 0.10 ** 0.05 *** 0.01 significance levels. OLS results of regressions of outcomes on the selection method of the case, with fixed effects at the inspector-cluster $\times$ year level. The specifications additionally control for predicted-evasion quintiles formed on the pooled regression-eligible full-audit sample. The sample includes all full audit cases selected from the 2018, 2019, and 2020 audit programs, excluding safety cases, for which we had data on the outcome variables. Panel A1 shows the results for the resource outcomes, and Panel B1 shows the results for the productivity outcomes. Column 1 shows the results for the number of agents involved in the audit, which is obtained from the audit records. Column 2 shows the results for audit duration as reported by taxpayers in the taxpayer survey. The survey asked them about their last full audit experience. Column 3 uses the difference between the date of initial notice and the date of the start of the audit (either verification announcement or information request or start date of audit) as a proxy for the duration of the audit. This measure is available only for cases that had an initial notice. We drop negative values, trim the data at the 99th percentile, and control for dummies indicating availability of date of initial notice and final notice. These controls are helpful because the definition of the start date may vary depending on the availability of information. Column 4 measures the duration of the audit using the self-reported number of days inspectors worked on a case, reported on a spreadsheet that inspectors needed to fill out during the 2019 audit program. Columns 5 to 8 use as outcomes the log of the ratio of evasion over the different cost measures used in Columns 1 to 4. We multiplied the answer by the number of inspectors that worked in the case. Robust standard errors (Huber-White formula) are shown in parentheses.

2.2 Table 6 with predicted evasion controls

This subsection presents the baseline Table 6 replication together with variants that keep the original sample and layout and add predicted-evasion controls.

Table 7: Algorithm Selection, Perceived Audit Efficiency and Corruption

A: Surveyed Firms (Self-Reporting a Recent Audit)						
Outcome:	Efficiency Index			Corruption Index		
	Full Audits (1)	Desk Audits (2)	All Audits (3)	Full Audits (4)	Desk Audits (5)	All Audits (6)
Algorithm	-0.41*** (0.15)	0.01 (0.14)	-0.17 (0.10)	0.05 (0.15)	-0.05 (0.14)	-0.02 (0.10)
Inspectors x Overlap	0.45** (0.18)	0.04 (0.26)	0.16 (0.21)	-0.56* (0.29)	-0.12 (0.27)	-0.18 (0.20)
Algorithm x Random		-0.09 (0.19)	-0.05 (0.18)		-0.08 (0.18)	-0.04 (0.16)
N	199	273	472	199	272	471
R2	0.10	0.02	0.03	0.11	0.10	0.05
Mean outcome	0.00	0.00	0.00	-0.07	0.05	0.00
B: Only Audited Firms (as per Administrative Audit Data)						
	(1)	(2)	(3)	(4)	(5)	(6)
Algorithm	-0.36 (0.26)	0.37 (0.23)	-0.02 (0.17)	0.29 (0.24)	-0.24 (0.22)	0.02 (0.17)
Inspectors x Overlap	0.19 (0.24)	0.40 (0.35)	-0.07 (0.29)	-0.33 (0.48)	-0.13 (0.31)	0.09 (0.29)
Algorithm x Random		-0.01 (0.27)	0.08 (0.24)		-0.03 (0.26)	-0.04 (0.23)
N	89	120	211	88	122	212
R2	0.10	0.08	0.03	0.16	0.13	0.05
Mean outcome	0.00	-0.06	-0.02	-0.14	-0.03	-0.08

Notes: * 0.10 ** 0.05 *** 0.01 significance levels. OLS results with fixed effects at the year and tax office level for outcomes from the taxpayer survey. Given the lower number of observations in the survey, we do not run the desk audit regression with year $\times$ inspector fixed effects, as is done with outcomes from the administrative data. Robust standard errors are in parentheses. The table shows the association between audit selection method and taxpayers' perception of the efficiency of audits (Columns 1 to 3), and of the incidence of corruption during audits (Columns 4 to 6). The sample of surveyed taxpayers is a subsample of the audit cases selected by the yearly audit programs between 2018 and 2020. Panel A shows the results for all firms interviewed in the survey, conditional on them self-reporting an audit in recent years. Panel B is restricted to firms for which we can confirm that an audit as part of the yearly audit program took place, according to the administrative audit reports. The two outcomes are (1) an index of perceived efficiency of audits, and (2) an index of perceived corruption of audits. Each measure is constructed by combining three questions into a normalized index following Anderson (2008). To measure efficiency, we combine the questions on taxpayers' assessment of the auditor's (i) technical knowledge, (ii) efficiency during the audit, and (iii) capacity to uncover all evaded taxes. For corruption, we combine the questions on the perception of (i) the auditor's dishonesty during the audit, (ii) the frequency of bribes paid to inspectors among firms similar to the respondent's, and (iii) the existence of a preferential tax audit treatment for firms connected with the tax administration. The coefficients are measured in terms of standard deviation of the respective indexes.

Table 8: Algorithm Selection, Perceived Audit Efficiency and Corruption, with Predicted Evasion Control

A: Surveyed Firms (Self-Reporting a Recent Audit)						
Outcome:	Efficiency Index			Corruption Index		
	Full Audits (1)	Desk Audits (2)	All Audits (3)	Full Audits (4)	Desk Audits (5)	All Audits (6)
Algorithm	-0.40** (0.16)	-0.01 (0.14)	-0.18* (0.10)	-0.00 (0.15)	-0.06 (0.13)	-0.04 (0.10)
Inspectors x Overlap	0.43** (0.18)	0.05 (0.26)	0.17 (0.21)	-0.46 (0.30)	-0.12 (0.27)	-0.17 (0.20)
Algorithm x Random		-0.09 (0.19)	-0.06 (0.18)		-0.08 (0.18)	-0.05 (0.16)
Predicted evasion	0.02 (0.04)	-0.02 (0.02)	-0.01 (0.01)	-0.06 (0.03)	-0.01 (0.02)	-0.02 (0.01)
N	199	273	472	199	272	471
R2	0.10	0.03	0.03	0.12	0.10	0.06
Mean outcome	0.00	0.00	0.00	-0.07	0.05	0.00
B: Only Audited Firms (as per Administrative Audit Data)						
	(1)	(2)	(3)	(4)	(5)	(6)
Algorithm	-0.36 (0.27)	0.31 (0.23)	-0.05 (0.17)	0.27 (0.25)	-0.24 (0.22)	0.00 (0.16)
Inspectors x Overlap	0.18 (0.24)	0.40 (0.36)	-0.06 (0.30)	-0.27 (0.49)	-0.13 (0.31)	0.10 (0.28)
Algorithm x Random		-0.00 (0.27)	0.07 (0.24)		-0.03 (0.25)	-0.04 (0.23)
Predicted evasion	0.01 (0.06)	-0.03 (0.02)	-0.02 (0.02)	-0.04 (0.06)	0.00 (0.02)	-0.01 (0.02)
N	89	120	211	88	122	212
R2	0.10	0.10	0.04	0.16	0.13	0.05
Mean outcome	0.00	-0.06	-0.02	-0.14	-0.03	-0.08

Notes: * 0.10 ** 0.05 *** 0.01 significance levels. OLS results with fixed effects at the year and tax office level for outcomes from the taxpayer survey. The specifications additionally control for predicted evasion. Given the lower number of observations in the survey, we do not run the desk audit regression with year $\times$ inspector fixed effects, as is done with outcomes from the administrative data. Robust standard errors are in parentheses. The table shows the association between audit selection method and taxpayers' perception of the efficiency of audits (Columns 1 to 3), and of the incidence of corruption during audits (Columns 4 to 6). The sample of surveyed taxpayers is a subsample of the audit cases selected by the yearly audit programs between 2018 and 2020. Panel A shows the results for all firms interviewed in the survey, conditional on them self-reporting an audit in recent years. Panel B is restricted to firms for which we can confirm that an audit as part of the yearly audit program took place, according to the administrative audit reports. The two outcomes are (1) an index of perceived efficiency of audits, and (2) an index of perceived corruption of audits. Each measure is constructed by combining three questions into a normalized index following Anderson (2008). To measure efficiency, we combine the questions on taxpayers' assessment of the auditor's (i) technical knowledge, (ii) efficiency during the audit, and (iii) capacity to uncover all evaded taxes. For corruption, we combine the questions on the perception of (i) the auditor's dishonesty during the audit, (ii) the frequency of bribes paid to inspectors among firms similar to the respondent's, and (iii) the existence of a preferential tax audit treatment for firms connected with the tax administration. The coefficients are measured in terms of standard deviation of the respective indexes.

Table 9: Algorithm Selection, Perceived Audit Efficiency and Corruption, with Predicted Evasion Quadratic Control

A: Surveyed Firms (Self-Reporting a Recent Audit)						
Outcome:	Efficiency Index			Corruption Index		
	Full Audits (1)	Desk Audits (2)	All Audits (3)	Full Audits (4)	Desk Audits (5)	All Audits (6)
Algorithm	-0.39** (0.16)	-0.01 (0.14)	-0.17* (0.11)	-0.02 (0.15)	-0.06 (0.13)	-0.06 (0.10)
Inspectors x Overlap	0.42** (0.18)	0.05 (0.27)	0.18 (0.21)	-0.45 (0.31)	-0.13 (0.27)	-0.19 (0.20)
Algorithm x Random		-0.09 (0.19)	-0.06 (0.18)		-0.09 (0.18)	-0.06 (0.16)
Predicted evasion	-0.05 (0.30)	-0.00 (0.07)	-0.04 (0.05)	0.05 (0.23)	0.04 (0.07)	0.07 (0.06)
Predicted evasion squared	0.00 (0.01)	-0.00 (0.00)	0.00 (0.00)	-0.00 (0.01)	-0.00 (0.00)	-0.00* (0.00)
N	199	273	472	199	272	471
R2	0.10	0.03	0.03	0.12	0.10	0.06
Mean outcome	0.00	0.00	0.00	-0.07	0.05	0.00
B: Only Audited Firms (as per Administrative Audit Data)						
	(1)	(2)	(3)	(4)	(5)	(6)
Algorithm	-0.39 (0.27)	0.31 (0.23)	-0.03 (0.17)	0.33 (0.25)	-0.23 (0.22)	-0.00 (0.17)
Inspectors x Overlap	0.17 (0.24)	0.40 (0.35)	-0.03 (0.29)	-0.25 (0.49)	-0.12 (0.31)	0.09 (0.28)
Algorithm x Random		0.02 (0.26)	0.11 (0.24)		0.00 (0.26)	-0.06 (0.24)
Predicted evasion	0.14 (0.37)	-0.08 (0.13)	-0.11 (0.08)	-0.34 (0.32)	-0.08 (0.12)	0.02 (0.08)
Predicted evasion squared	-0.00 (0.01)	0.00 (0.01)	0.00 (0.00)	0.01 (0.01)	0.00 (0.01)	-0.00 (0.00)
N	89	120	211	88	122	212
R2	0.11	0.10	0.04	0.18	0.13	0.06
Mean outcome	0.00	-0.06	-0.02	-0.14	-0.03	-0.08

Notes: * 0.10 ** 0.05 *** 0.01 significance levels. OLS results with fixed effects at the year and tax office level for outcomes from the taxpayer survey. The specifications additionally control for predicted evasion and its square. Given the lower number of observations in the survey, we do not run the desk audit regression with year $\times$ inspector fixed effects, as is done with outcomes from the administrative data. Robust standard errors are in parentheses. The table shows the association between audit selection method and taxpayers' perception of the efficiency of audits (Columns 1 to 3), and of the incidence of corruption during audits (Columns 4 to 6). The sample of surveyed taxpayers is a subsample of the audit cases selected by the yearly audit programs between 2018 and 2020. Panel A shows the results for all firms interviewed in the survey, conditional on them self-reporting an audit in recent years. Panel B is restricted to firms for which we can confirm that an audit as part of the yearly audit program took place, according to the administrative audit reports. The two outcomes are (1) an index of perceived efficiency of audits, and (2) an index of perceived corruption of audits. Each measure is constructed by combining three questions into a normalized index following Anderson (2008). To measure efficiency, we combine the questions on taxpayers' assessment of the auditor's (i) technical knowledge, (ii) efficiency during the audit, and (iii) capacity to uncover all evaded taxes. For corruption, we combine the questions on the perception of (i) the auditor's dishonesty during the audit, (ii) the frequency of bribes paid to inspectors among firms similar to the respondent's, and (iii) the existence of a preferential tax audit treatment for firms connected with the tax administration. The coefficients are measured in terms of standard deviation of the respective indexes.

Table 10: Algorithm Selection, Perceived Audit Efficiency and Corruption, with Predicted Evasion Decile Controls

A: Surveyed Firms (Self-Reporting a Recent Audit)						
Outcome:	Efficiency Index			Corruption Index		
	Full Audits (1)	Desk Audits (2)	All Audits (3)	Full Audits (4)	Desk Audits (5)	All Audits (6)
Algorithm	-0.36** (0.16)	-0.03 (0.15)	-0.16 (0.11)	-0.01 (0.15)	-0.06 (0.14)	-0.06 (0.10)
Inspectors x Overlap	0.42** (0.20)	0.12 (0.26)	0.20 (0.21)	-0.36 (0.34)	-0.19 (0.27)	-0.21 (0.19)
Algorithm x Random		-0.05 (0.19)	-0.02 (0.17)		-0.10 (0.18)	-0.08 (0.16)
N	199	273	472	199	272	471
R2	0.13	0.07	0.05	0.13	0.13	0.08
Mean outcome	0.00	0.00	0.00	-0.07	0.05	0.00
B: Only Audited Firms (as per Administrative Audit Data)						
	(1)	(2)	(3)	(4)	(5)	(6)
Algorithm	-0.38 (0.27)	0.30 (0.23)	-0.05 (0.17)	0.21 (0.25)	-0.22 (0.23)	0.03 (0.17)
Inspectors x Overlap	0.21 (0.26)	0.66* (0.37)	0.03 (0.29)	-0.33 (0.51)	-0.13 (0.47)	0.01 (0.31)
Algorithm x Random		-0.04 (0.27)	0.03 (0.24)		-0.02 (0.27)	-0.04 (0.24)
N	89	120	211	88	122	212
R2	0.16	0.20	0.08	0.23	0.17	0.10
Mean outcome	0.00	-0.06	-0.02	-0.14	-0.03	-0.08

Notes: * 0.10 ** 0.05 *** 0.01 significance levels. OLS results with fixed effects at the year and tax office level for outcomes from the taxpayer survey. The specifications additionally control for predicted-evasion deciles formed on the pooled regression-eligible survey-analysis sample. Given the lower number of observations in the survey, we do not run the desk audit regression with year $\times$ inspector fixed effects, as is done with outcomes from the administrative data. Robust standard errors are in parentheses. The table shows the association between audit selection method and taxpayers' perception of the efficiency of audits (Columns 1 to 3), and of the incidence of corruption during audits (Columns 4 to 6). The sample of surveyed taxpayers is a subsample of the audit cases selected by the yearly audit programs between 2018 and 2020. Panel A shows the results for all firms interviewed in the survey, conditional on them self-reporting an audit in recent years. Panel B is restricted to firms for which we can confirm that an audit as part of the yearly audit program took place, according to the administrative audit reports. The two outcomes are (1) an index of perceived efficiency of audits, and (2) an index of perceived corruption of audits. Each measure is constructed by combining three questions into a normalized index following Anderson (2008). To measure efficiency, we combine the questions on taxpayers' assessment of the auditor's (i) technical knowledge, (ii) efficiency during the audit, and (iii) capacity to uncover all evaded taxes. For corruption, we combine the questions on the perception of (i) the auditor's dishonesty during the audit, (ii) the frequency of bribes paid to inspectors among firms similar to the respondent's, and (iii) the existence of a preferential tax audit treatment for firms connected with the tax administration. The coefficients are measured in terms of standard deviation of the respective indexes.

Table 11: Algorithm Selection, Perceived Audit Efficiency and Corruption, with Predicted Evasion Quintile Controls

A: Surveyed Firms (Self-Reporting a Recent Audit)						
Outcome:	Efficiency Index			Corruption Index		
	Full Audits (1)	Desk Audits (2)	All Audits (3)	Full Audits (4)	Desk Audits (5)	All Audits (6)
Algorithm	-0.41*** (0.16)	0.01 (0.14)	-0.16 (0.10)	0.02 (0.15)	-0.10 (0.14)	-0.05 (0.10)
Inspectors x Overlap	0.43** (0.20)	0.04 (0.26)	0.17 (0.20)	-0.42 (0.31)	-0.11 (0.26)	-0.19 (0.18)
Algorithm x Random		-0.08 (0.20)	-0.05 (0.18)		-0.07 (0.18)	-0.07 (0.17)
N	199	273	472	199	272	471
R2	0.10	0.03	0.03	0.12	0.11	0.07
Mean outcome	0.00	0.00	0.00	-0.07	0.05	0.00
B: Only Audited Firms (as per Administrative Audit Data)						
	(1)	(2)	(3)	(4)	(5)	(6)
Algorithm	-0.41 (0.26)	0.32 (0.24)	-0.04 (0.17)	0.28 (0.25)	-0.32 (0.23)	-0.00 (0.16)
Inspectors x Overlap	0.21 (0.25)	0.45 (0.33)	0.03 (0.29)	-0.31 (0.50)	-0.07 (0.40)	0.05 (0.29)
Algorithm x Random		0.04 (0.28)	0.08 (0.25)		-0.02 (0.27)	-0.06 (0.24)
N	89	120	211	88	122	212
R2	0.12	0.11	0.04	0.21	0.14	0.07
Mean outcome	0.00	-0.06	-0.02	-0.14	-0.03	-0.08

Notes: * 0.10 ** 0.05 *** 0.01 significance levels. OLS results with fixed effects at the year and tax office level for outcomes from the taxpayer survey. The specifications additionally control for predicted-evasion quintiles formed on the pooled regression-eligible survey-analysis sample. Given the lower number of observations in the survey, we do not run the desk audit regression with year $\times$ inspector fixed effects, as is done with outcomes from the administrative data. Robust standard errors are in parentheses. The table shows the association between audit selection method and taxpayers' perception of the efficiency of audits (Columns 1 to 3), and of the incidence of corruption during audits (Columns 4 to 6). The sample of surveyed taxpayers is a subsample of the audit cases selected by the yearly audit programs between 2018 and 2020. Panel A shows the results for all firms interviewed in the survey, conditional on them self-reporting an audit in recent years. Panel B is restricted to firms for which we can confirm that an audit as part of the yearly audit program took place, according to the administrative audit reports. The two outcomes are (1) an index of perceived efficiency of audits, and (2) an index of perceived corruption of audits. Each measure is constructed by combining three questions into a normalized index following Anderson (2008). To measure efficiency, we combine the questions on taxpayers' assessment of the auditor's (i) technical knowledge, (ii) efficiency during the audit, and (iii) capacity to uncover all evaded taxes. For corruption, we combine the questions on the perception of (i) the auditor's dishonesty during the audit, (ii) the frequency of bribes paid to inspectors among firms similar to the respondent's, and (iii) the existence of a preferential tax audit treatment for firms connected with the tax administration. The coefficients are measured in terms of standard deviation of the respective indexes.

3 Inspector effort analysis

3.1 Definition of void audit cases

Table 12: Number of void desk audit cases

Void Audit	Algorithm	Inspectors	Random	Total
Evasion Detected	432	656	116	1204
No evasion detected	117	143	55	315
Total	549	799	171	1519

Table 13: Number of null cases and their notification/confirmation status

	Desk audits			Full audits		
	Evasion Detected	No evasion detected	Total	Evasion Detected	No evasion detected	Total
Only Notified	286	4	290	151	18	169
Only Confirmed	53	4	57	39	0	39
Notified and Confirmed	408	2	410	267	2	269
Neither	0	254	254	0	31	31
Total	747	264	1011	457	51	508
N	1011			508		

Table 14: Number of notified desk audits according to penalty/droit simple registered.

	Desk audits				Full audits		
	Algorithm	Inspectors	Random	Total	Algorithm	Inspectors	Total
has penalités only	4	0	0	4	1	0	1
has both	239	339	112	690	149	268	417
has neither	3	2	1	6	6	14	20
Total	246	341	113	700	156	282	438

From what I understand of the paper, the workflow of inspectors is as follows:

- Irregularities detected $\rightarrow$ evasion value $\rightarrow$ notification to the taxpayer
- Taxpayer may dispute or not $\rightarrow$ confirmation

However, after doing some tabs I noticed that:

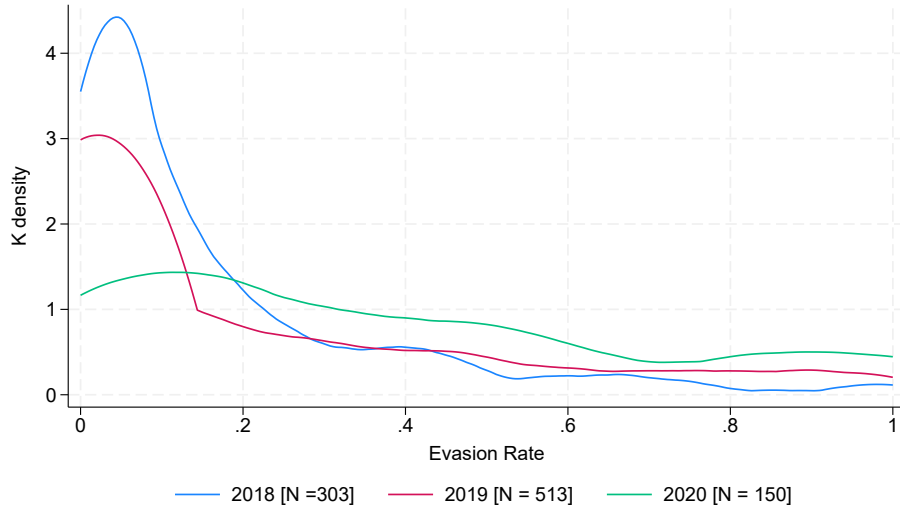
- Some executed cases with positive evasion value skipped the notification step and went directly to confirmation, which shouldn't be an issue if using $y3==0$ for void audit.
- Some cases with zero evasion were either notified or confirmed, which is weird right? If there was no evasion detected ($\text{evasionvalue}==0$ or $y4==0$) no notification or confirmation should've taken place.
- Note: not all inspectors that were assigned desk audits executed an algorithm case.

3.2 Definitions of low evasion rates cases

steps

1. Estimate evasion rate of each case as $\frac{\text{Evasion Value}}{\text{Evasion value} + \text{Mean Turnover (2018-2020)}}$
2. Estimate within inspector cluster year (list) - method [Alg + Rand / Inspectors] quartiles according to evasion rate.
3. Tag cases in bottom 25% (bad) or below median evasion rate

Figure 6: Distribution of evasion rates across years



4. Estimate the shares of cases on the bottom or below median evasion rate per inspector, selection method and year.
5. Run scatterplots per inspector and given method, comparing share of low evasion rate cases in year t and execution rate of that method in year $t + 1$.

4 Approaches to assess inspectors effort

4.1 Approach 1: pre-post void audit case [discarded for now]

Issues encountered:

- Few desk audit inspectors have a valid open - closure audit date for all of their executed cases: 41 out of 123.
- Only 34 out 123 have experienced a void audit case
- Only 2 inspectors have both: i) valid dates for all audits + ii) encountered a void audit

Possible solutions to missing dates?

- Impute starting date values + 1 month
- use average duration? subsampling by void versus non-void?

4.2 Approach 2: Comparing previous year performance with next year's execution rate

In this subsection we compare the inspectors' performance between two periods based on selection year. We proceed to plot per selection method, i) the share of executed cases from year t vrs $t + 1$, and ii) the share of void audits in t versus the share of execution of assigned audits in $t + 1$.

- Sample: Only desk - audit inspectors that are observable for at least two periods. Furthermore, we divide the sample between inspectors that changed tax office in the next period and those that remained in the same tax office.

Definition of shares

Let Alg = algorithm + random cases. Let N denote the number of cases per inspector i at (selection) year t . Then,

Share of void cases:

- Share of all void cases $_{it} = \frac{N_{it}^{\text{Total Void}}}{N_{it}^{\text{Total Executed}}}$
- Share of void alg cases $_{it} = \frac{N_{it}^{\text{Void Alg}}}{N_{it}^{\text{Total Alg Executed}}}$
- Share of void Inspector cases $_{it} = \frac{N_{it}^{\text{Void Inspector}}}{N_{it}^{\text{Total Inspector Executed}}}$

Share of executed cases

- Share of all executed cases $_{it} = \frac{N_{it}^{\text{Total Executed}}}{N_{it}^{\text{Total Assigned}}}$
- Share of executed Alg cases $_{it} = \frac{N_{it}^{\text{Executed Alg}}}{N_{it}^{\text{Total Alg Assigned}}}$
- Share of executed Inspector cases $_{it} = \frac{N_{it}^{\text{Executed Inspector}}}{N_{it}^{\text{Total Inspector Assigned}}}$

Table 15: Number of inspectors by period and subsample (Desk audits)

	2018–2019			2019–2020			2018 – avg(2019–2020)		
	(1) Same bureau	(2) Changed bureau	(3) Total	(4) Same bureau	(5) Changed bureau	(6) Total	(7) Same bureau	(8) Changed bureau	(9) Total
Total Inspectors	81	1	82	86	2	88	96	6	102
Inspectors observable two periods	13	1	14	23	2	25	51	6	57

Table 16: Number of inspectors that registered execution and void audits by period and sub-sample (Desk audits)

	2018-2019			2019-2020			2018 – avg(2019-2020)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Same bureau	Changed bureau	Total	Same bureau	Changed bureau	Total	Same bureau	Changed bureau	Total
Year t									
Executed any audit	13	1	14	23	2	25	30	5	35
Executed algo-selected audit	13	1	14	22	2	24	28	5	33
Executed inspector-selected audit	11	1	12	23	2	25	24	5	29
Reported any void audit	3	0	3	21	2	23	6	2	8
Reported void algo audit	0	0	0	18	2	20	3	1	4
Reported void inspector audit	3	0	3	17	1	18	3	1	4
Reported any below-median evasion rate	13	1	14	21	2	23	27	5	32
Reported below-median evasion rate algo audit	11	1	12	20	2	22	21	5	26
Reported below-median evasion rate inspector audit	10	1	11	18	2	20	18	5	23
Reported any bot-quartile evasion rate	10	1	11	19	2	21	22	5	27
Reported bot-quartile evasion rate algo audit	9	1	10	17	2	19	17	4	21
Reported bot-quartile evasion rate inspector audit	7	1	8	16	1	17	12	5	17
Year t+1									
Executed any audit	13	1	14	21	2	23	45	5	50
Executed algo-selected audit	12	1	13	17	0	17	32	3	35
Executed inspector-selected audit	13	1	14	17	2	19	45	5	50
Reported any void audit	10	0	10	1	0	1	26	2	28
Reported void algo audit	9	0	9	1	0	1	22	2	24
Reported void inspector audit	8	0	8	0	0	0	21	1	22
Reported any below-median evasion rate	12	1	13	17	1	18	41	4	45
Reported below-median evasion rate algo audit	12	0	12	13	0	13	30	2	32
Reported below-median evasion rate inspector audit	11	1	12	9	1	10	36	4	40
Reported any bot-quartile evasion rate	12	0	12	12	0	12	38	3	41
Reported bot-quartile evasion rate algo audit	12	0	12	7	0	7	28	2	30
Reported bot-quartile evasion rate inspector audit	10	0	10	7	0	7	31	2	33

Table 17: Inspector average execution and void case rates by period and sub-sample (Desk audits)

	2018-2019			2019-2020			2018 – avg(2019-2020)		
	(1) Same bureau	(2) Changed bureau	(3) Total	(4) Same bureau	(5) Changed bureau	(6) Total	(7) Same bureau	(8) Changed bureau	(9) Total
Year t									
Exec rate, all audits	43.08 (21.01)	53.33 (.)	43.81 (20.37)	61.59 (26.51)	43.52 (9.166)	60.14 (25.94)	36.50 (21.58)	58 (11.45)	39.33 (21.71)
Exec rate, algo audits	44.12 (25.20)	61.11 (.)	45.33 (24.64)	60.82 (28.25)	50.00 (23.57)	59.96 (27.63)	37.83 (27.28)	54.44 (14.91)	40.02 (26.46)
Exec rate, inspector audits	41.82 (28.58)	41.67 (.)	41.81 (27.45)	62.53 (28.64)	37.50 (5.893)	60.53 (28.31)	34.15 (28.19)	63.33 (17.28)	37.99 (28.63)
Void rate, all audits	2.585 (4.940)	0 (.)	2.401 (4.796)	41.14 (25.24)	28.33 (16.50)	40.11 (24.65)	4.731 (11.66)	3.152 (4.400)	4.506 (10.89)
Void rate, algo audits	0 (0)	0 (.)	0 (0)	41.57 (28.61)	22.50 (3.536)	39.98 (27.87)	6.548 (21.44)	3.333 (7.454)	6.061 (19.90)
Void rate, inspector audits	8.636 (16.45)	0 (.)	7.917 (15.88)	38.55 (29.55)	30 (42.43)	37.87 (29.68)	3.958 (11.70)	4 (8.944)	3.966 (11.13)
Share of below median ev. rate, all audits	48.17 (22.64)	43.75 (.)	47.86 (21.78)	44.06 (23.03)	53.33 (18.86)	44.80 (22.53)	49.85 (28.53)	47.37 (12.09)	49.49 (26.69)
Share of below median ev. rate, algo audits	43.99 (27.95)	45.45 (.)	44.10 (26.86)	48.44 (22.87)	47.50 (38.89)	48.36 (23.31)	44.41 (33.37)	50.76 (24.03)	45.37 (31.89)
Share of below median ev. rate, inspector audits	47.47 (26.75)	40 (.)	46.85 (25.60)	41.34 (27.68)	55 (7.071)	42.43 (26.81)	51.27 (38.64)	44.56 (21.13)	50.11 (36.01)
Share of bottom quartile ev. rate, all audits	23.80 (16.26)	25 (.)	23.88 (15.63)	35.51 (23.59)	45 (7.071)	36.27 (22.78)	29.92 (27.36)	23.98 (10.91)	29.07 (25.63)
Share of bottom quartile ev. rate, algo audits	22.74 (17.58)	27.27 (.)	23.06 (16.94)	40.34 (27.91)	47.50 (38.89)	40.94 (27.95)	23.59 (24.96)	22.12 (18.39)	23.37 (23.84)
Share of bottom quartile ev. rate, inspector audits	24.48 (30.23)	20 (.)	24.11 (28.85)	31.09 (23.45)	30 (42.43)	31.01 (24.07)	32.40 (42.23)	25.89 (11.27)	31.28 (38.59)
Year t+1									
Exec rate, all audits	54.84 (19.38)	41.67 (.)	53.90 (18.95)	19.34 (11.93)	8.333 (0)	18.46 (11.82)	32.07 (23.09)	22.66 (15.59)	31.08 (22.50)
Exec rate, algo audits	53.21 (25.71)	50 (.)	52.98 (24.72)	19.39 (18.45)	0 (0)	17.84 (18.46)	29.04 (27.29)	16.67 (21.08)	27.73 (26.82)
Exec rate, inspector audits	57.68 (21.02)	33.33 (.)	55.94 (21.22)	19.29 (18.81)	11.11 (0)	18.63 (18.15)	33.36 (24.46)	23.15 (15.61)	32.29 (23.79)
Void rate, all audits	37.14 (29.19)	0 (.)	34.48 (29.75)	1.587 (7.274)	0 (0)	1.449 (6.950)	19.11 (25.20)	5.667 (8.788)	17.77 (24.35)
Void rate, algo audits	39.35 (29.74)	0 (.)	36.32 (30.49)	1.961 (8.085)	. (.)	1.961 (8.085)	26.52 (27.27)	15 (13.23)	25.54 (26.44)
Void rate, inspector audits	35.37 (37.39)	0 (.)	32.84 (37.14)	0 (0)	0 (0)	17.89 (0)	17.89 (25.12)	6 (13.42)	16.70 (24.38)
Share of below median ev. rate, all audits	49.55 (21.22)	20 (.)	47.44 (21.86)	42.62 (26.69)	50 (70.71)	43.26 (29.65)	52.12 (28.71)	39.67 (37.09)	50.88 (29.44)
Share of below median ev. rate, algo audits	57.41 (15.53)	0 (.)	52.99 (21.78)	56.08 (39.41)	. (.)	56.08 (39.41)	55.37 (25.68)	31.67 (38.84)	53.34 (27.11)
Share of below median ev. rate, inspector audits	44.27 (25.84)	50 (.)	44.68 (24.87)	33.82 (36.38)	50 (70.71)	35.53 (38.47)	48.87 (34.15)	46 (31.90)	48.58 (33.63)
Share of bottom quartile ev. rate, all audits	38.64 (19.27)	0 (.)	35.88 (21.20)	21.27 (21.17)	0 (0)	19.42 (21.10)	37.75 (28.12)	24 (30.70)	36.37 (28.36)
Share of bottom quartile ev. rate, algo audits	49.77 (18.91)	0 (.)	45.94 (22.77)	27.84 (39.16)	. (.)	27.84 (39.16)	41.73 (28.76)	31.67 (38.84)	40.87 (29.17)
Share of bottom quartile ev. rate, inspector audits	29.35 (23.21)	0 (.)	27.26 (23.64)	16.18 (21.54)	0 (0)	14.47 (20.94)	32.66 (30.14)	21 (32.86)	31.49 (30.27)

Table 18: Number of tax offices by period and subsample (Full - Audits)

	(1) 2018–2019	(2) 2019–2020	(3) 2018 – avg(2019–2020)
Total Bureaux	11	11	11
Inspectors observable two periods	7	11	11

Table 19: Number of Tax offices that registered execution and void audits by period and sub-sample (Full - Audits)

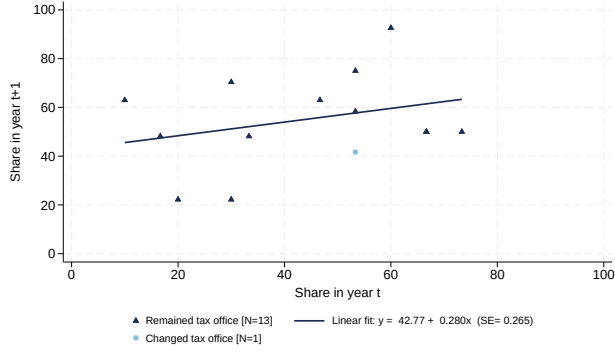
	(1) 2018–2019	(2) 2019–2020	(3) 2018 – avg(2019–2020)
Year t			
Executed any audit	7	11	7
Executed algo-selected audit	7	11	7
Executed inspector-selected audit	7	11	7
Reported any void audit	6	9	6
Reported void algo audit	6	4	6
Reported void inspector audit	4	9	4
Reported any below-median evasion rate	7	11	7
Reported below-median evasion rate algo audit	7	7	7
Reported below-median evasion rate inspector audit	7	11	7
Reported any bot-quartile evasion rate	7	11	7
Reported bot-quartile evasion rate algo audit	7	7	7
Reported bot-quartile evasion rate inspector audit	7	10	7
Year t+1			
Executed any audit	7	10	11
Executed algo-selected audit	7	6	11
Executed inspector-selected audit	7	10	11
Reported any void audit	6	2	9
Reported void algo audit	4	0	4
Reported void inspector audit	6	2	9
Reported any below-median evasion rate	7	8	11
Reported below-median evasion rate algo audit	6	4	8
Reported below-median evasion rate inspector audit	7	7	11
Reported any bot-quartile evasion rate	7	7	11
Reported bot-quartile evasion rate algo audit	6	3	7
Reported bot-quartile evasion rate inspector audit	6	7	10

Table 20: Tax office average execution and void case rates by period and sub-sample (Full - Audits)

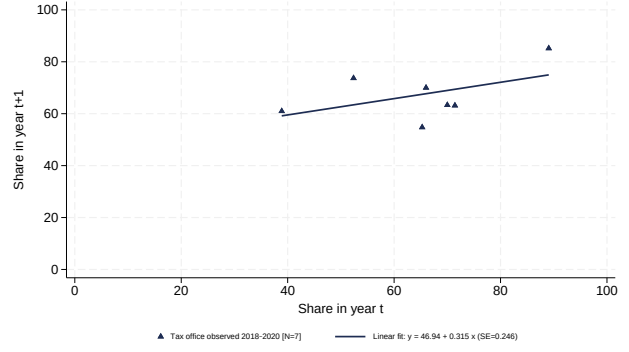
	(1) 2018–2019	(2) 2019–2020	(3) 2018 – avg(2019–2020)
Year t			
Exec rate, all audits	64.72 (15.74)	56.90 (18.92)	64.72 (15.74)
Exec rate, algo audits	52.84 (22.50)	50.10 (26.63)	52.84 (22.50)
Exec rate, inspector audits	75.67 (15.50)	63.46 (18.65)	75.67 (15.50)
Void rate, all audits	11.42 (6.523)	16.86 (14.18)	11.42 (6.523)
Void rate, algo audits	15.84 (12.40)	11.30 (20.82)	15.84 (12.40)
Void rate, inspector audits	9.325 (9.347)	18.93 (16.64)	9.325 (9.347)
Share of below median ev. rate, all audits	48.20 (8.663)	43.65 (21.50)	48.20 (8.663)
Share of below median ev. rate, algo audits	50.58 (10.96)	46.04 (42.06)	50.58 (10.96)
Share of below median ev. rate, inspector audits	47.39 (14.02)	43.21 (17.15)	47.39 (14.02)
Share of bottom quartile ev. rate, all audits	24.64 (10.84)	25.74 (15.07)	24.64 (10.84)
Share of bottom quartile ev. rate, algo audits	26.47 (12.44)	29.11 (31.33)	26.47 (12.44)
Share of bottom quartile ev. rate, inspector audits	24.16 (14.78)	23.89 (14.41)	24.16 (14.78)
Year t+1			
Exec rate, all audits	67.30 (9.983)	28.43 (23.68)	42.66 (15.00)
Exec rate, algo audits	62.06 (19.85)	21.86 (23.85)	35.98 (16.30)
Exec rate, inspector audits	71.67 (15.44)	34.43 (28.66)	48.94 (18.79)
Void rate, all audits	15.54 (13.95)	3.958 (10.51)	11.74 (10.19)
Void rate, algo audits	17.76 (24.26)	0 (0)	10.32 (20.83)
Void rate, inspector audits	13.31 (11.13)	4.386 (10.70)	12.11 (8.835)
Share of below median ev. rate, all audits	52.40 (16.66)	39.24 (22.74)	41.93 (18.23)
Share of below median ev. rate, algo audits	58.06 (35.14)	43.91 (40.09)	48.76 (37.01)
Share of below median ev. rate, inspector audits	48.61 (17.37)	39.49 (32.64)	41.50 (16.81)
Share of bottom quartile ev. rate, all audits	26.05 (9.876)	20.19 (17.15)	23.19 (11.46)
Share of bottom quartile ev. rate, algo audits	31.46 (19.18)	15.17 (23.26)	26.70 (24.44)
Share of bottom quartile ev. rate, inspector audits	21.12 (14.77)	24.71 (29.64)	23.18 (17.33)

Figure 7: Share of execution of all audits_t vrs Share of execution of all audits_{t+1}

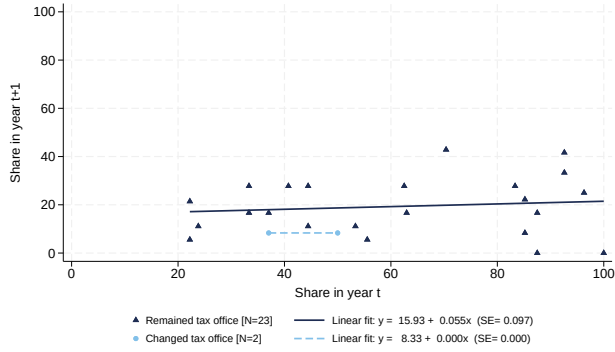
(a) 2018 vrs 2019 (Desk Audits)



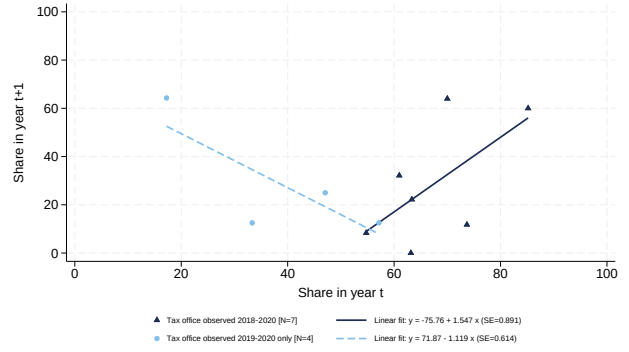
(b) 2018 vrs 2019 (Full Audits)



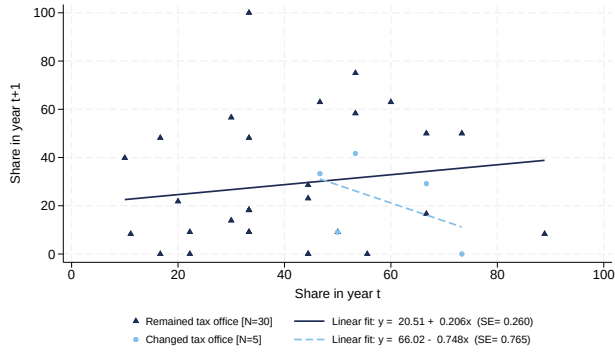
(c) 2019 vrs 2020 (Desk Audits)



(d) 2019 vrs 2020 (Full Audits)



(e) 2018 vrs (2019–2020) (Desk Audits)



(f) 2018 vrs (2019–2020) (Full Audits)

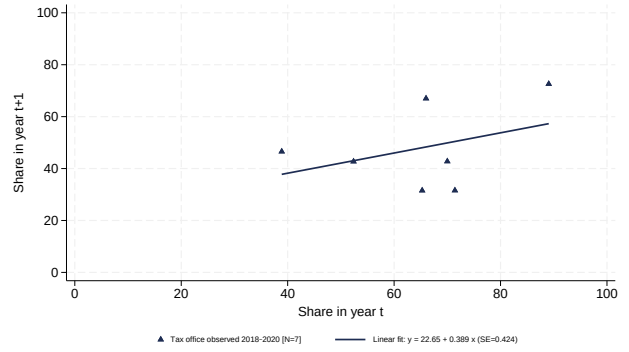
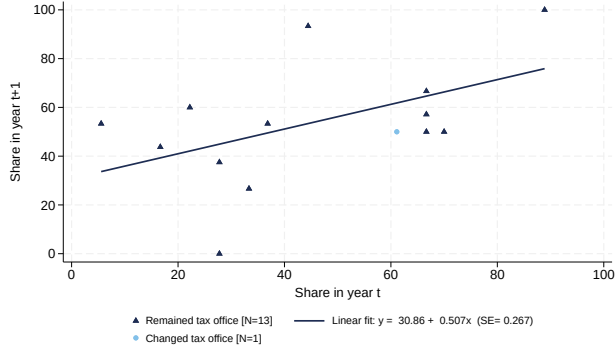
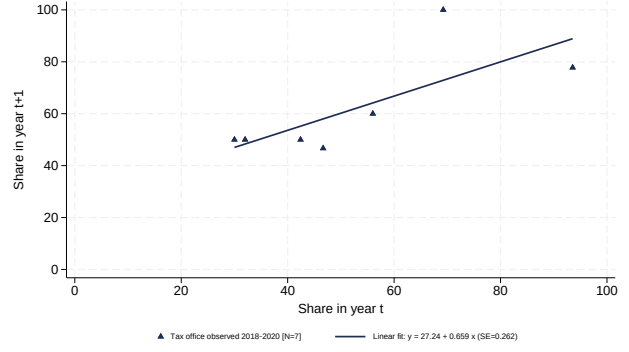


Figure 8: Share of execution of Algorithm_t vs Share of execution of Algorithm_{t+1}

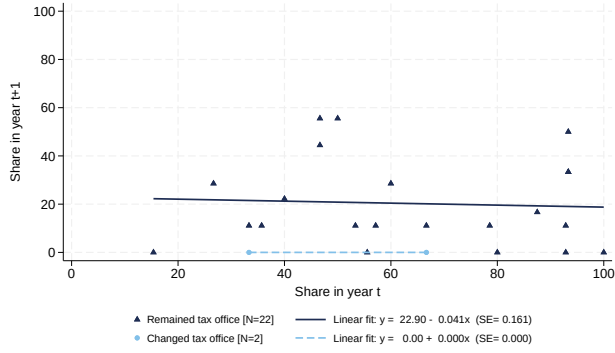
(a) 2018 vrs 2019 (Desk Audits)



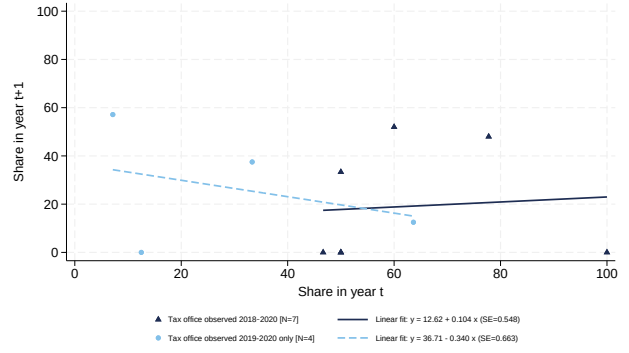
(b) 2018 vrs 2019 (Full Audits)



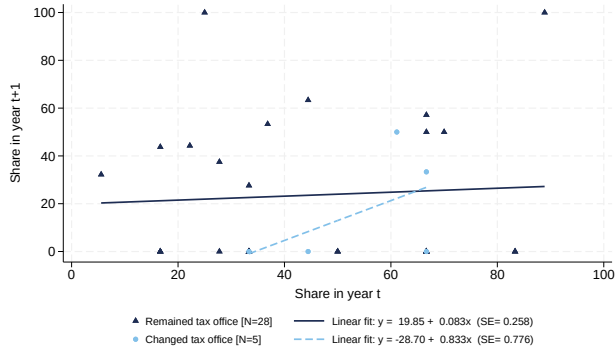
(c) 2019 vrs 2020 (Desk Audits)



(d) 2019 vrs 2020 (Full Audits)



(e) 2018 vrs (2019–2020) (Desk Audits)



(f) 2018 vrs (2019–2020) (Full Audits)

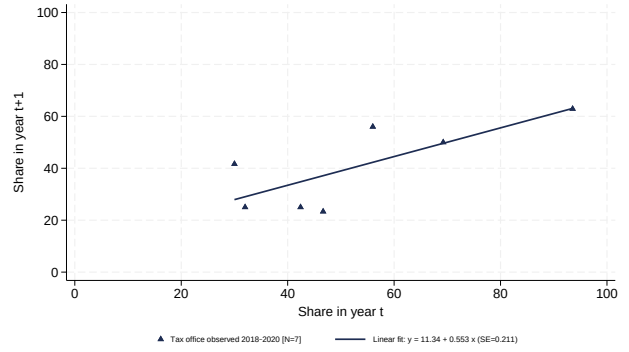
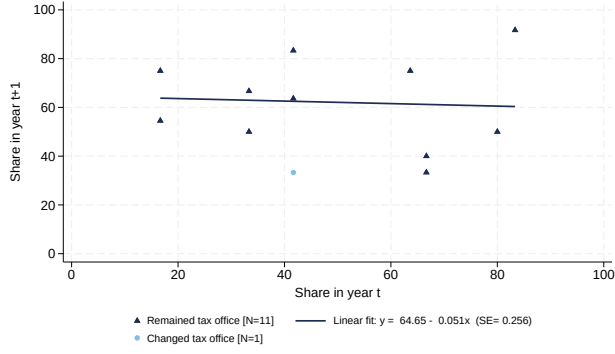
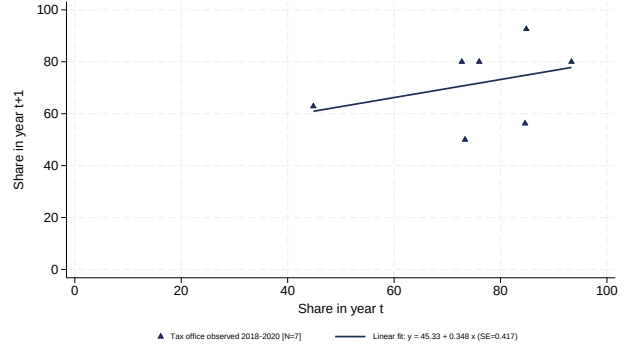


Figure 9: Share of execution of Inspector_t vs Share of execution of Inspector_{t+1}

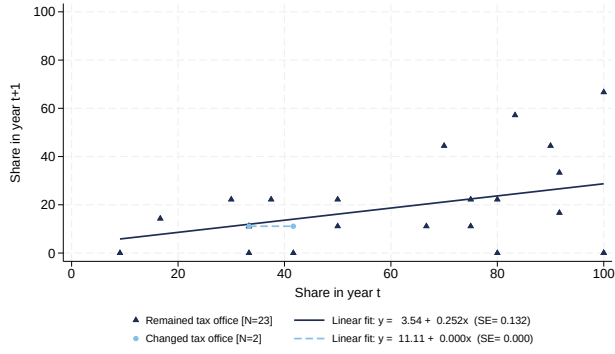
(a) 2018 vrs 2019 (Desk Audits)



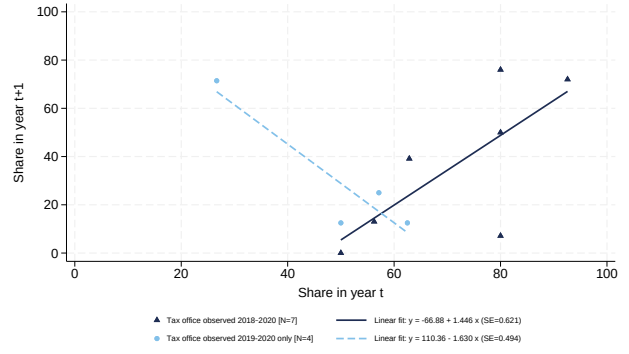
(b) 2018 vrs 2019 (Full Audits)



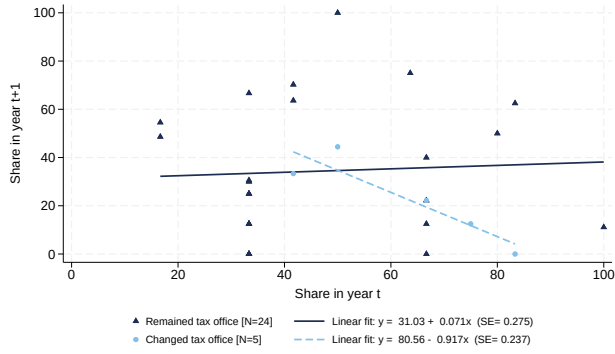
(c) 2019 vrs 2020 (Desk Audits)



(d) 2019 vrs 2020 (Full Audits)



(e) 2018 vrs (2019–2020) (Desk Audits)



(f) 2018 vrs (2019–2020) (Full Audits)

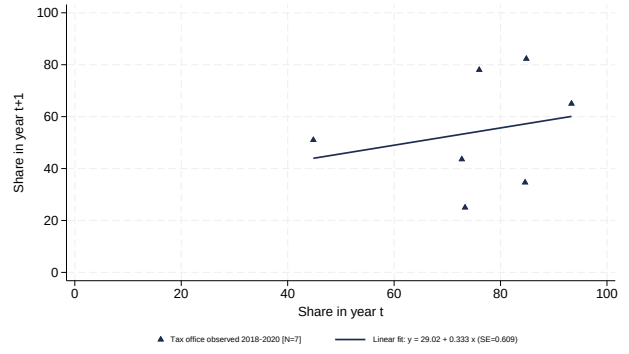
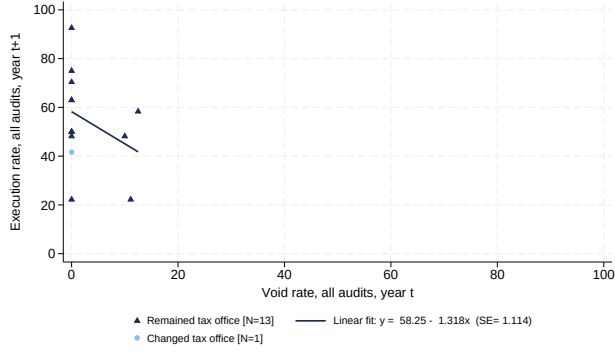
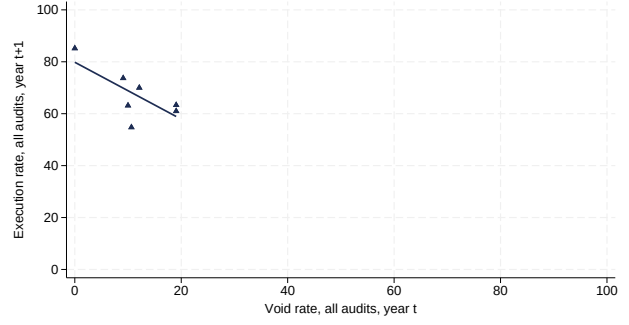


Figure 10: Share of total void_t vrs Share of total execution of assigned t_{+1}

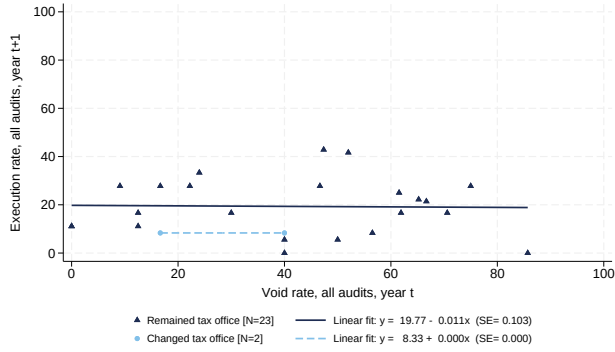
(a) 2018 vrs 2019 (Desk Audits)



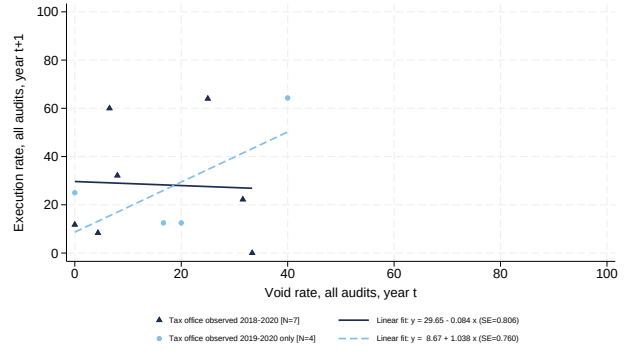
(b) 2018 vrs 2019 (Full Audits)



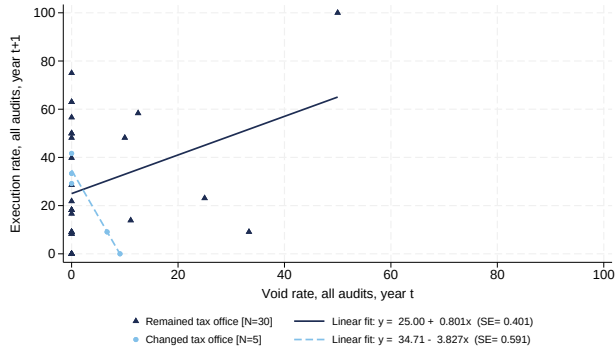
(c) 2019 vrs 2020 (Desk Audits)



(d) 2019 vrs 2020 (Full Audits)



(e) 2018 vrs (2019–2020) (Desk Audits)



(f) 2018 vrs (2019–2020) (Full Audits)

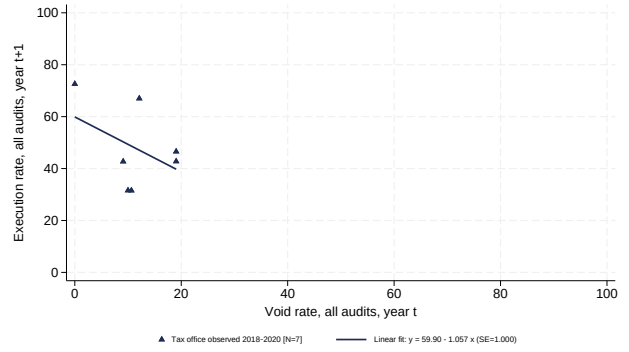
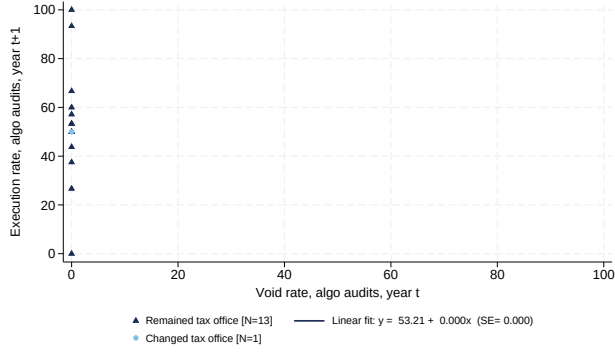
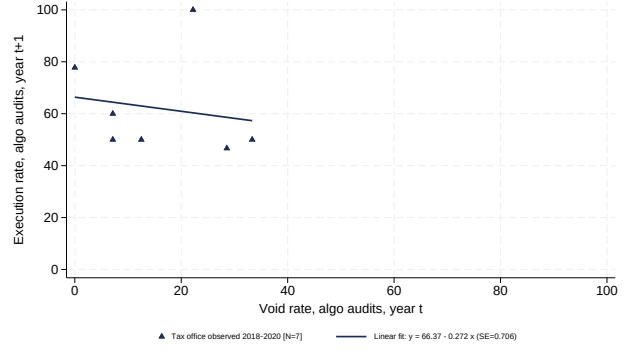


Figure 11: Share of void algo_t vrs share of execution of algo_{t+1}

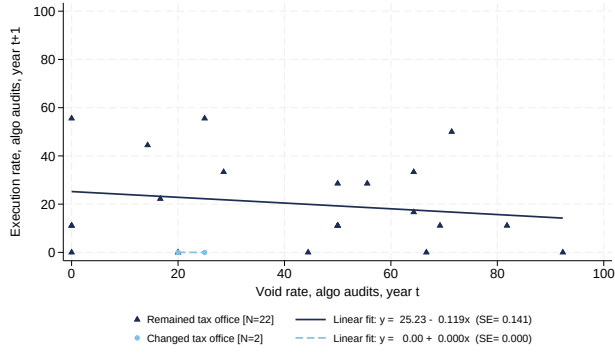
(a) 2018 vrs 2019 (Desk Audits)



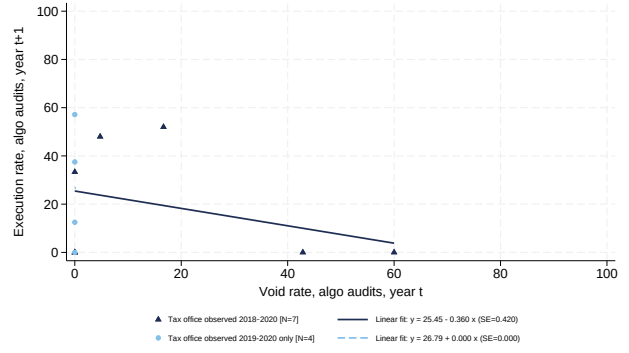
(b) 2018 vrs 2019 (Full Audits)



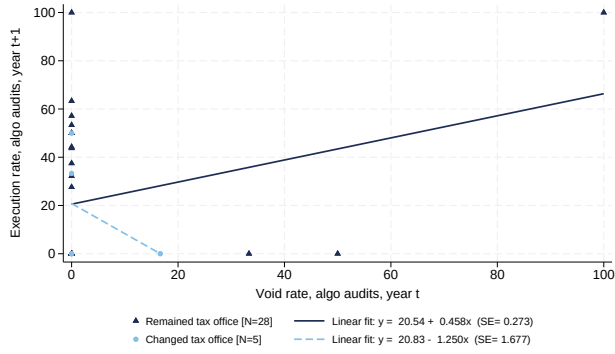
(c) 2019 vrs 2020 (Desk Audits)



(d) 2019 vrs 2020 (Full Audits)



(e) 2018 vrs (2019–2020) (Desk Audits)



(f) 2018 vrs (2019–2020) (Full Audits)

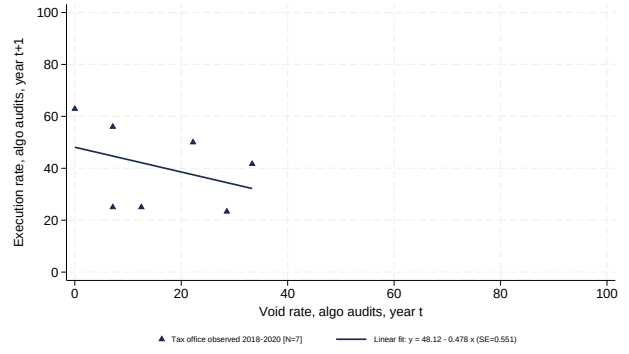
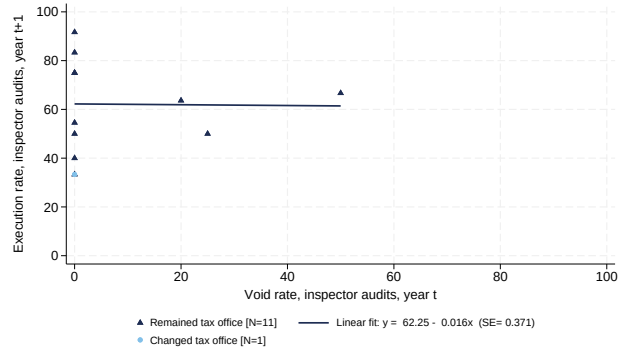
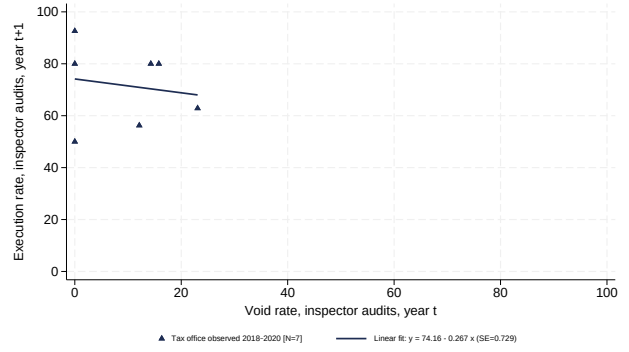


Figure 12: Share of void Inspector-selected_t vrs share of execution of Inspector-selected_{t+1}

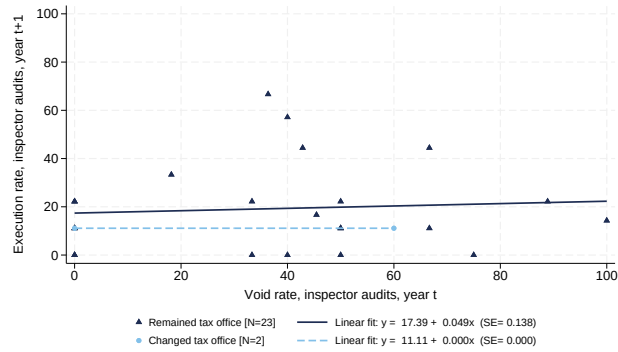
(a) 2018 vrs 2019 (Desk Audits)



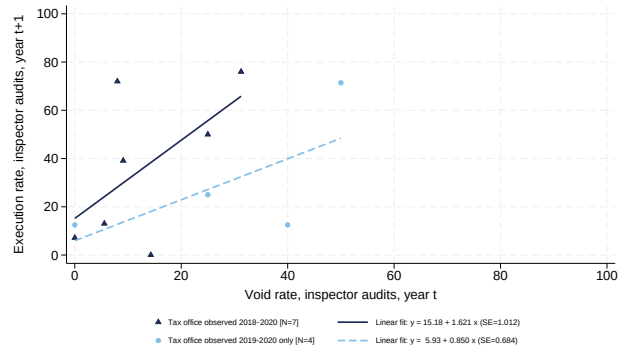
(b) 2018 vrs 2019 (Full Audits)



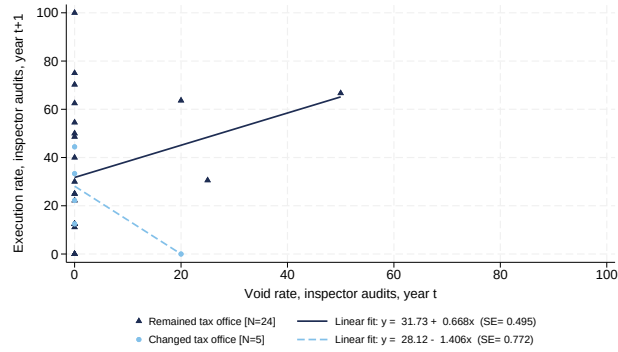
(c) 2019 vrs 2020 (Desk Audits)



(d) 2019 vrs 2020 (Full Audits)



(e) 2018 vrs (2019–2020) (Desk Audits)



(f) 2018 vrs (2019–2020) (Full Audits)

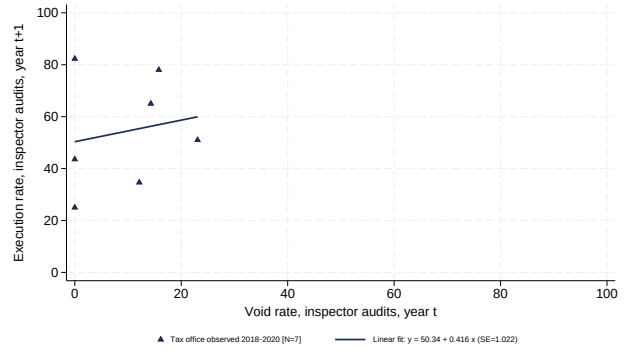
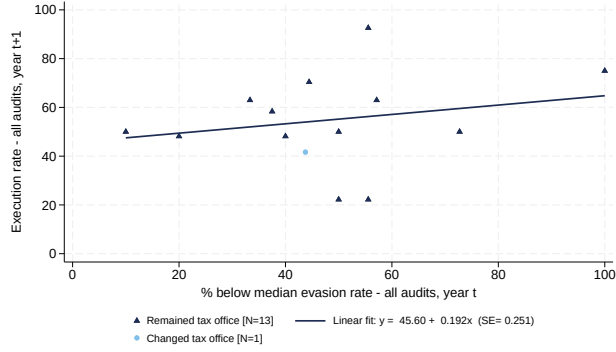
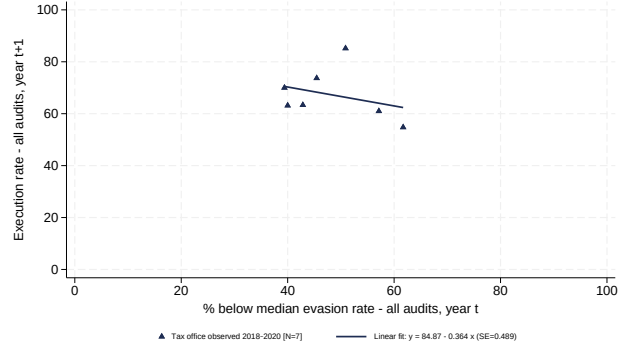


Figure 13: Share of total below median evasion rate cases_t vrs Share of total execution of assigned_{t+1}

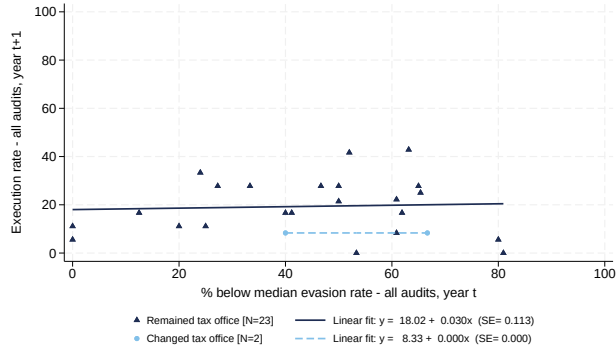
(a) 2018 vrs 2019 (Desk Audits)



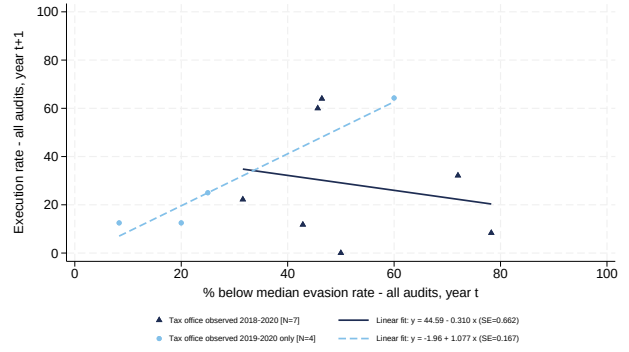
(b) 2018 vrs 2019 (Full Audits)



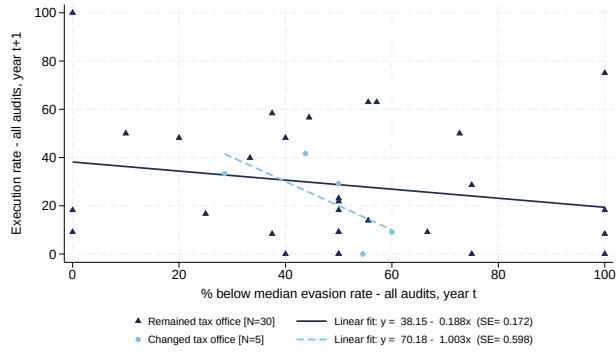
(c) 2019 vrs 2020 (Desk Audits)



(d) 2019 vrs 2020 (Full Audits)



(e) 2018 vrs (2019–2020) (Desk Audits)



(f) 2018 vrs (2019–2020) (Full Audits)

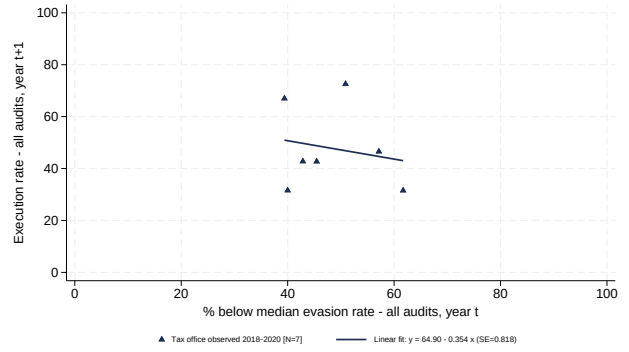
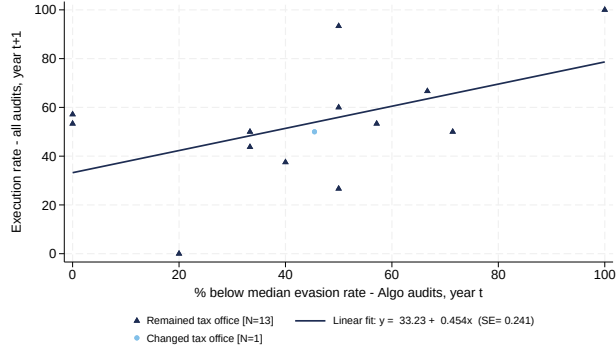
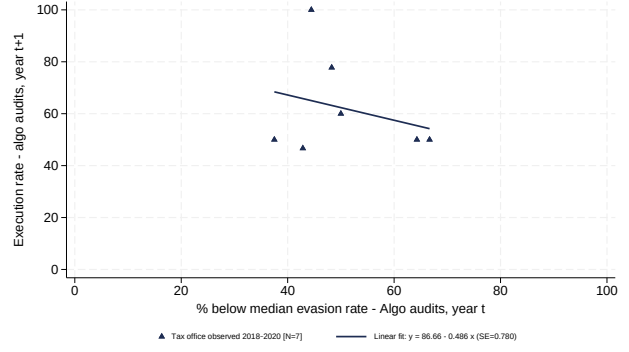


Figure 14: Share of below median evasion rate algo_t vrs share of execution of algo_{t+1}

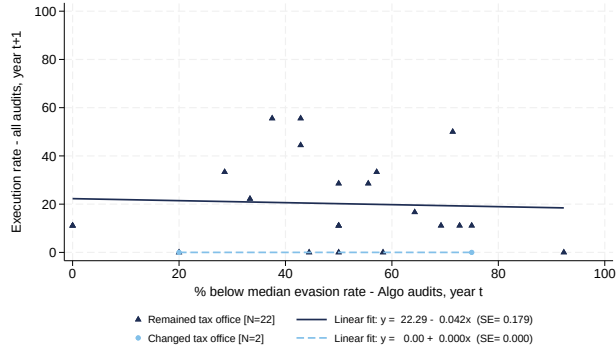
(a) 2018 vrs 2019 (Desk Audits)



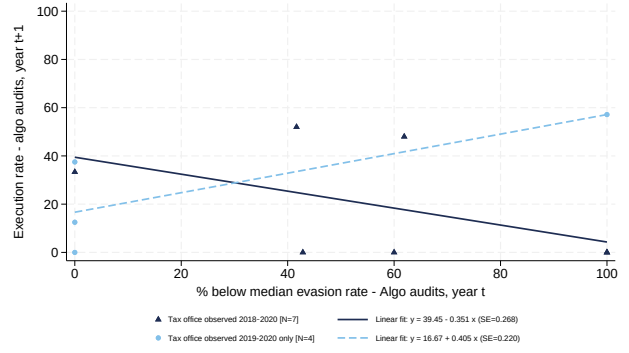
(b) 2018 vrs 2019 (Full Audits)



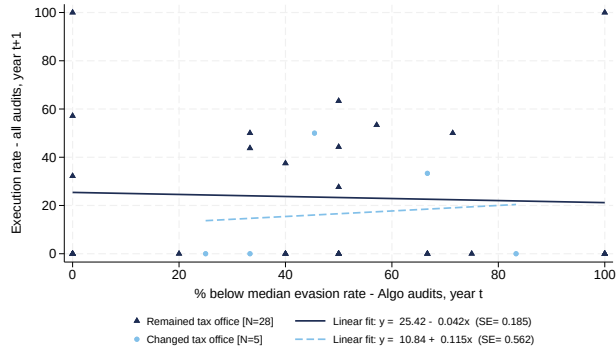
(c) 2019 vrs 2020 (Desk Audits)



(d) 2019 vrs 2020 (Full Audits)



(e) 2018 vrs (2019–2020) (Desk Audits)



(f) 2018 vrs (2019–2020) (Full Audits)

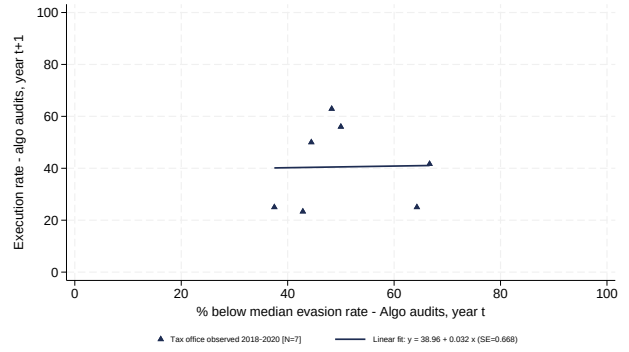


Figure 15: Share of below median evasion rate Inspector-selected_t vrs share of execution of Inspector-selected_{t+1}

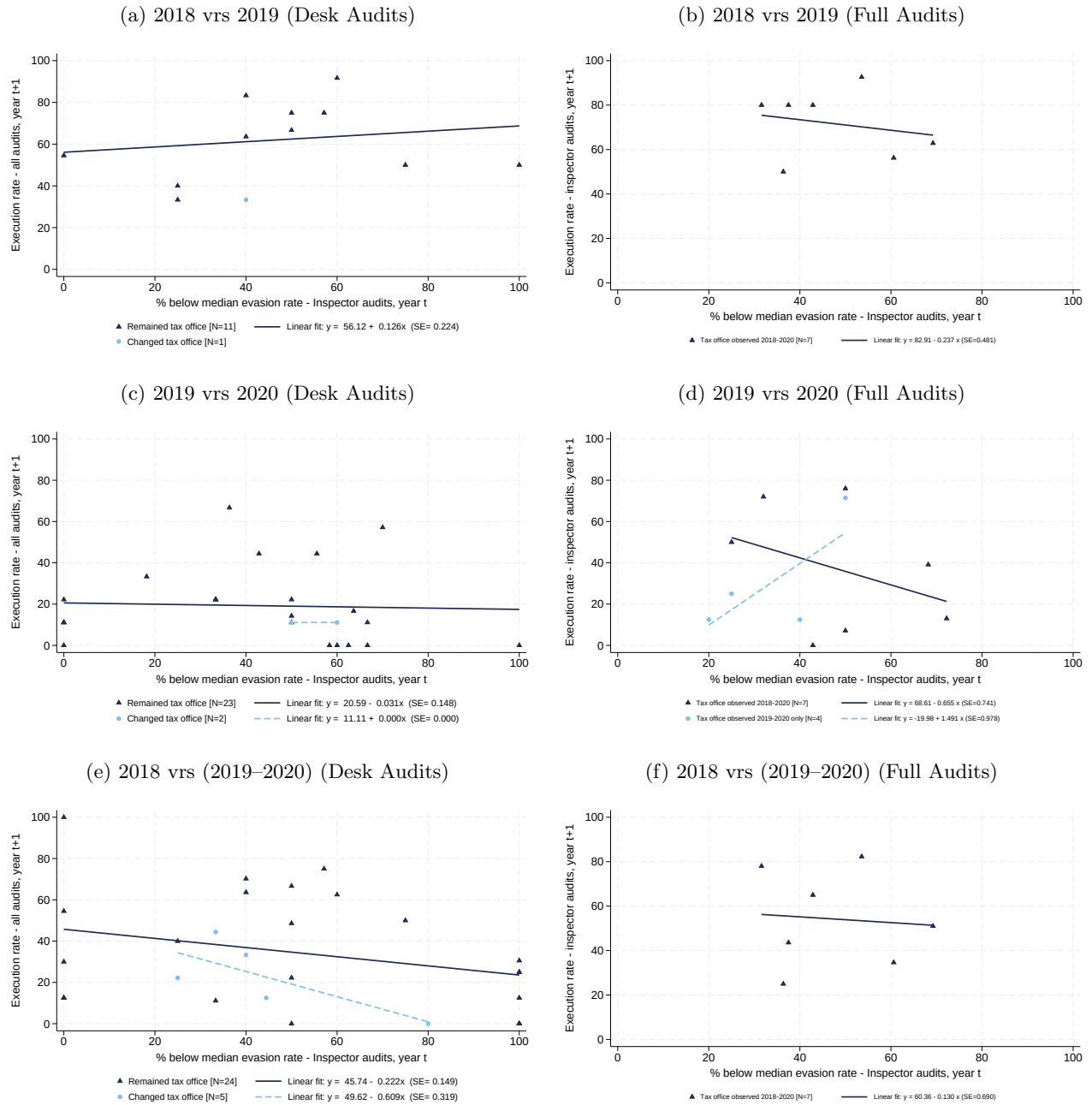
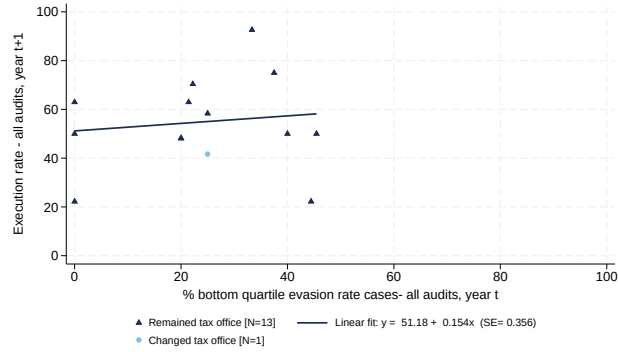
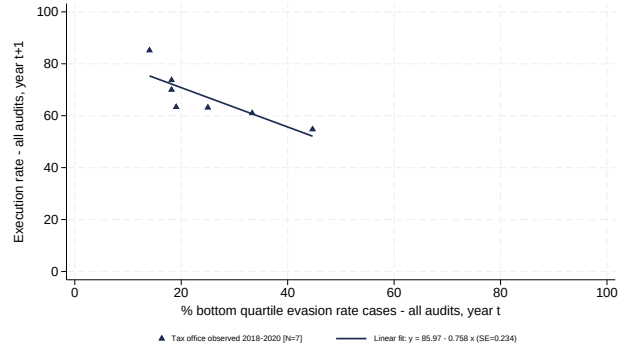


Figure 16: Share of total bottom quartile evasion rate cases_t vrs Share of total execution of assigned_{t+1}

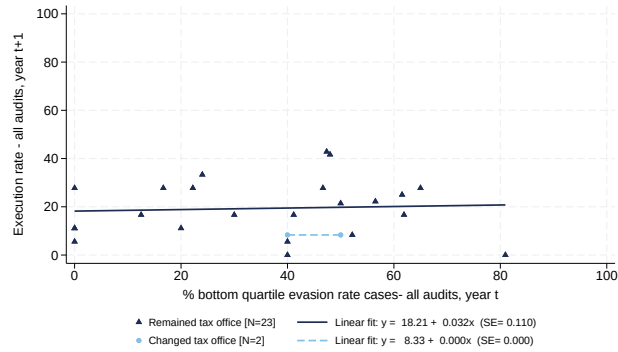
(a) 2018 vrs 2019 (Desk Audits)



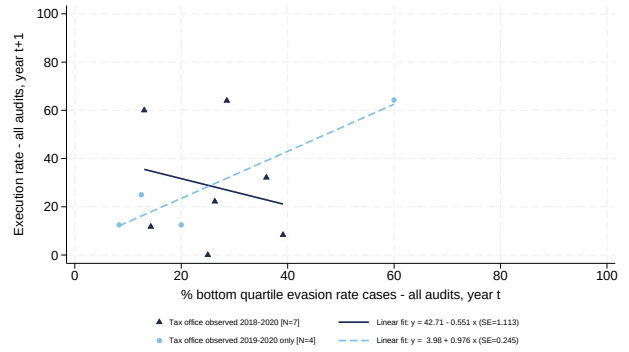
(b) 2018 vrs 2019 (Full Audits)



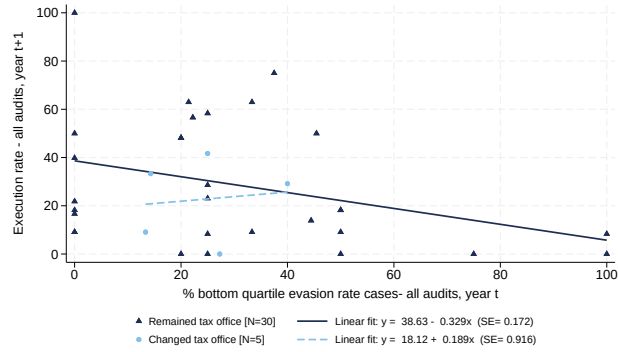
(c) 2019 vrs 2020 (Desk Audits)



(d) 2019 vrs 2020 (Full Audits)



(e) 2018 vrs (2019–2020) (Desk Audits)



(f) 2018 vrs (2019–2020) (Full Audits)

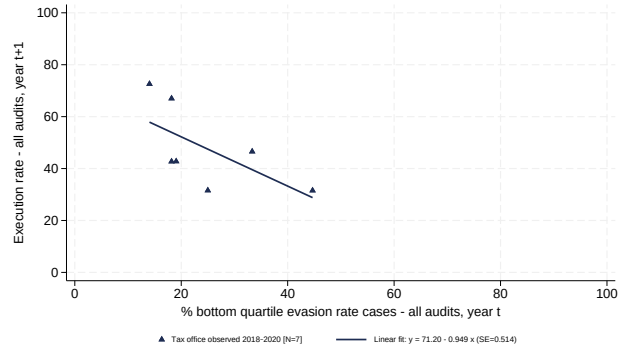
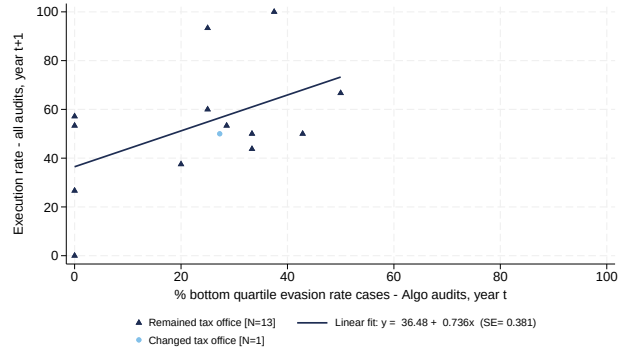
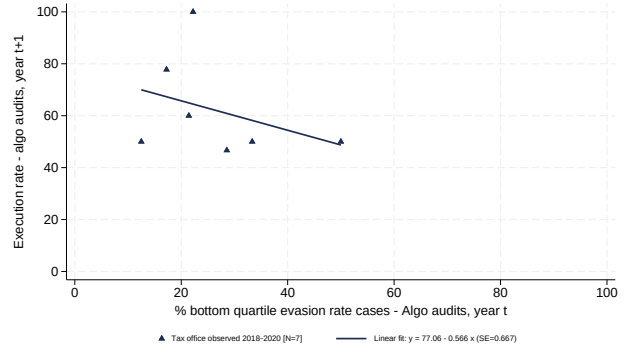


Figure 17: Share of bottom quartile evasion rate algo_t vrs share of execution of algo_{t+1}

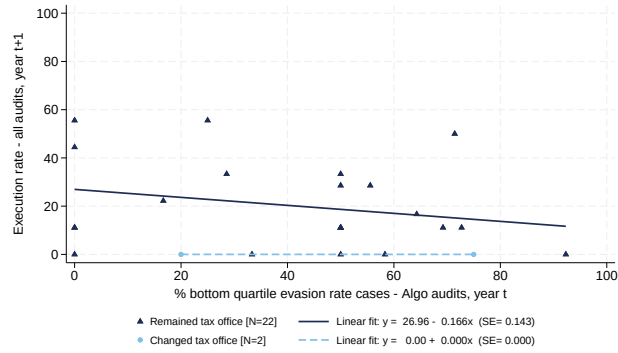
(a) 2018 vrs 2019 (Desk Audits)



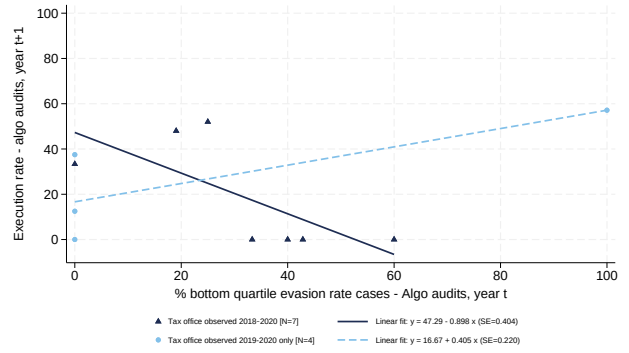
(b) 2018 vrs 2019 (Full Audits)



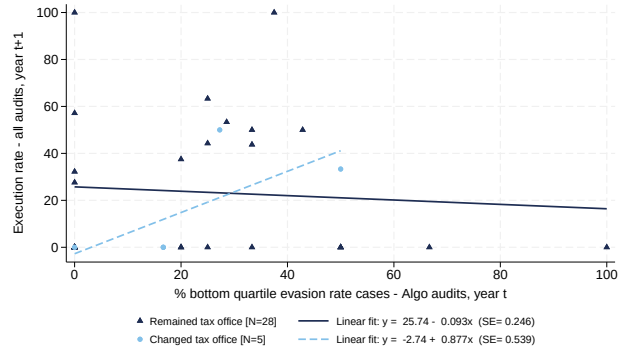
(c) 2019 vrs 2020 (Desk Audits)



(d) 2019 vrs 2020 (Full Audits)



(e) 2018 vrs (2019–2020) (Desk Audits)



(f) 2018 vrs (2019–2020) (Full Audits)

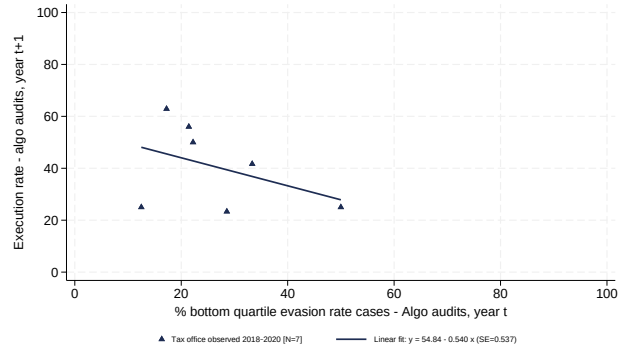
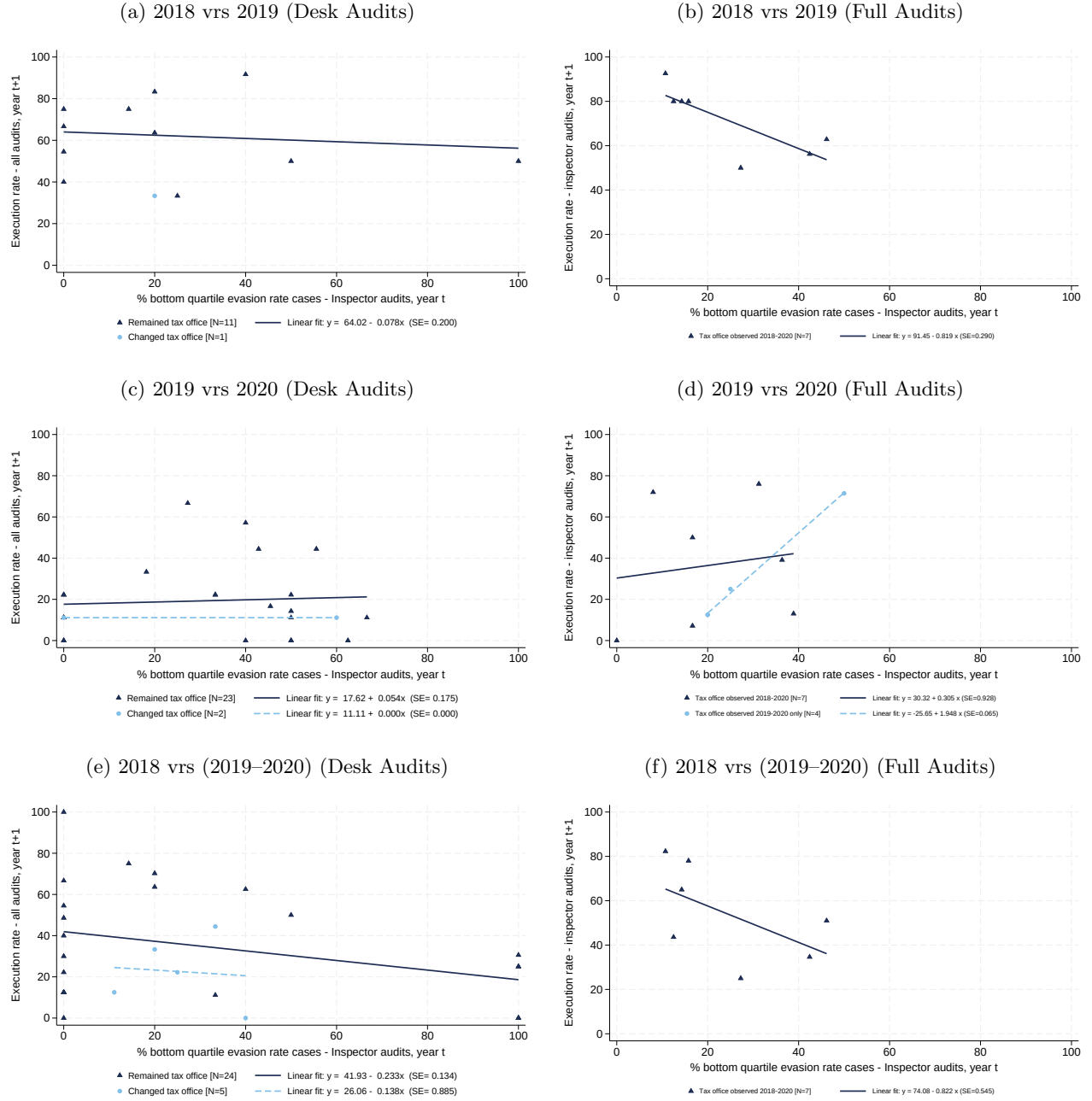


Figure 18: Share of bottom quartile evasion rate $\text{Inspector-selected}_t$ vrs share of execution of $\text{Inspector-selected}_{t+1}$



4.3 Approach 3: Top-listed high-potential algorithm-selected audits

Objective. Using predicted evasion estimates generated by the Random Forest model, we test whether execution of algorithm-assigned cases depends on whether the "first-in-line" for execution algorithm case(s) on a given list are high- vs. low-potential. The "list" unit differs by audit type: for desk audits it is defined at the inspector $\times$ selection-year level, while for full audits it is defined at the bureau (detailed) $\times$ selection-year level. We compare execution rates of algorithm cases across these lists.

Steps.

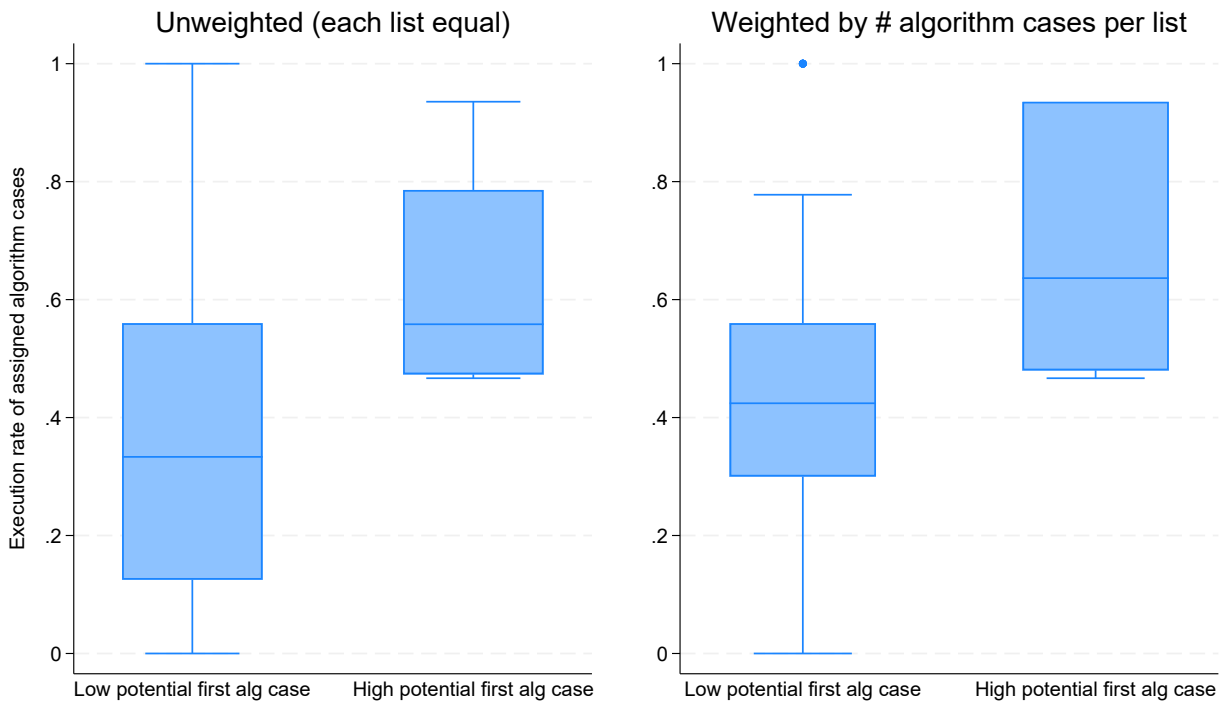
1. Load predicted evasion estimates data which includes estimates for selected audits, and restrict by audit type (full or desk audit). We use the dataset containing predicted evasion (yhatrf)

and execution outcomes, and split the analysis by audit type (`x2==0` desk; `x2==1` full).

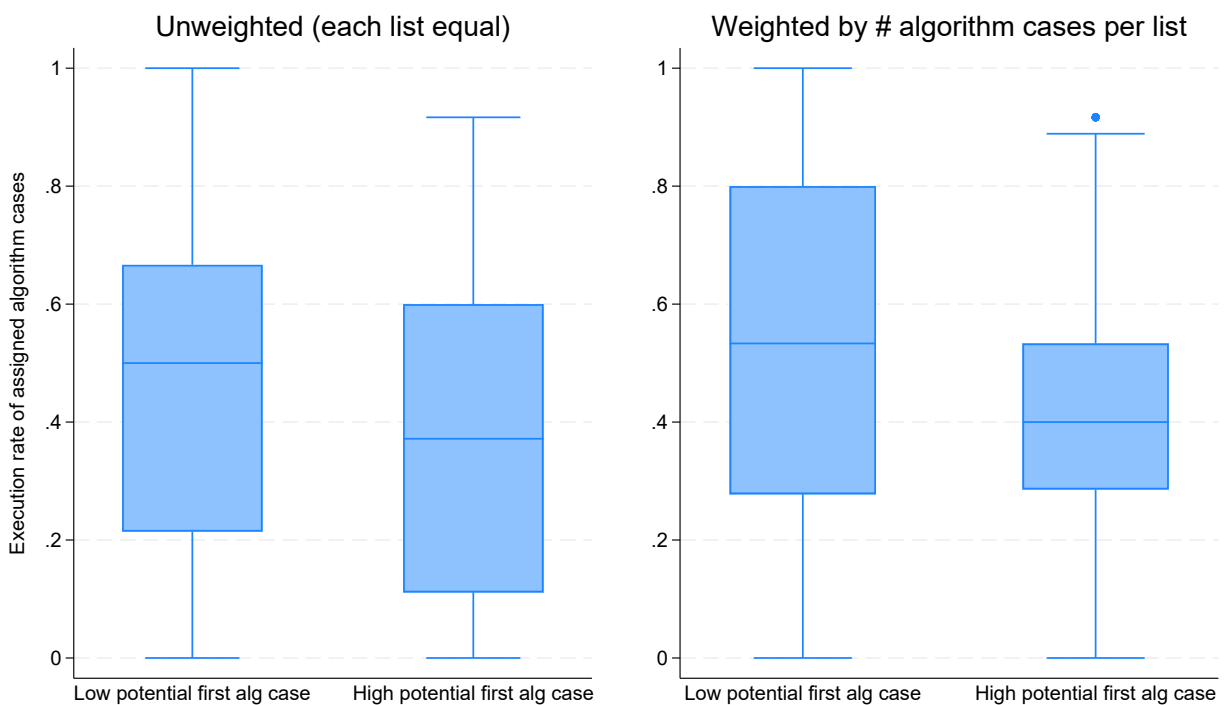
2. **Construct predicted-evasion quantile groups (audit-type specific).** We compute quintiles of predicted evasion `yhatrf` in the following way:
 - **Desk audits:** quintiles within `verificateur.selection × selectionyear`.
 - **Full audits:** quintiles within `bureau.detailed × selectionyear`.
3. **Restrict to algorithm-assigned cases.** We keep only algorithm-selected audits (`algorithm==1`) so that all outcomes and tags refer to the execution of algorithm-assigned cases.
4. **Define "top of the list" using execution sequence among algorithm cases.** Within each list, we order algorithm cases by the sequencing variable (`sequencing`) to define early algorithm cases (e.g., Top 1/Top 2, first 10% or first 20% of the algorithm sequence).
5. **Tag lists based on the potential of first-in-line algorithm cases.** For each list, we create indicators for whether an first-in-line algorithm case is: (i) **high-potential** (top quintile of `yhatrf`, `yhatrf_q5==5`), or (ii) **low-potential** (bottom quintile, `yhatrf_q5==1`). This generates list-level groups such as "high-potential-first" vs "low-potential-first".
6. **Collapse to the list level and compute execution rates.** We collapse the data to the list level and compute the mean execution rate of assigned algorithm cases within each list (mean of `y2` among `algorithm==1`).
7. **Compare execution outcomes across list groups.** We present (i) boxplots of list-level algorithm execution rates (unweighted and weighted by the number of assigned algorithm cases), and (ii) regression estimates comparing execution rates across "high-potential-first" and "low-potential-first" lists (with fixed effects varying by audit type).

Figure 19: Execution rates when the first sequenced algorithm case has high potential evasion versus low potential evasion.

(a) Full audits



(b) Desk audits



4.3.1 Regression estimates

Table 21: Execution of algorithm-assigned audits and high predicted evasion cases at the top of the list [Full audits]

Outcome: Execution Rate of Algorithm-Selected Cases	No FE			Selection year FE			Bureau FE		
	Top 2	First 10% seq	First 20% seq	Top 2	First 10% seq	First 20% seq	Top 2	First 10% seq	First 20% seq
Panel A: Discrete measures									
High potential evasion case in list =1	0.218* (0.124)	0.289** (0.123)	0.207* (0.108)	0.176 (0.105)	0.224** (0.102)	0.199** (0.087)	0.144 (0.185)	0.245 (0.173)	0.165 (0.153)
Low potential evasion case in list =1	-0.079 (0.108)	0.044 (0.112)	-0.108 (0.116)	-0.079 (0.105)	-0.007 (0.110)	-0.072 (0.112)	-0.067 (0.122)	0.085 (0.127)	-0.049 (0.171)
Selection Year FE	No	No	No	Yes	Yes	Yes	No	No	No
Bureau FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	29	29	29	29	29	29	29	29	29
R-squared	0.129	0.118	0.208	0.366	0.350	0.420	0.424	0.431	0.430
Panel B: Continuous measure									
Share of high predicted evasion case at the top	0.532** (0.225)	0.530** (0.240)	0.556*** (0.187)	0.457** (0.190)	0.403 (0.238)	0.481** (0.216)	0.398 (0.325)	0.492 (0.333)	0.444 (0.310)
Selection Year FE	No	No	No	Yes	Yes	Yes	No	No	No
Bureau FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	29	29	29	29	29	29	29	29	29
R-squared	0.112	0.076	0.097	0.350	0.313	0.343	0.416	0.422	0.417

Notes: Panel A reports discrete specifications where the dependent variable is the overall execution rate of algorithm cases (amongst the total assigned algorithm cases) and the regressors are indicators for whether the top of the algorithm ranking contains a high-potential evasion case (and analogously a low-potential case), under alternative definitions of "top of the list" (Top 2, first 10% of sequence, first 20% of the sequence). Panel B reports the continuous specification using the share of high-potential cases at the top of the list. The unit of observation is the audit list (Bureau (detail) × year). Standard errors are robust. Stars denote significance at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 22: Existence of High potential case vrs overall execution rate of algorithm cases [Full audits]

	No FE			Selection year FE			Bureau FE		
	(1) Top 2	(2) First 10% seq	(3) First 20% seq	(4) Top 2	(5) First 10% seq	(6) First 20% seq	(7) Top 2	(8) First 10% seq	(9) First 20% seq
Dependent variable: 'Algorithm Execution rate									
High potential evasion case in list =1	0.218* (0.124)	0.289** (0.123)	0.207* (0.108)	0.176 (0.105)	0.224** (0.102)	0.199** (0.087)	0.144 (0.185)	0.245 (0.173)	0.165 (0.153)
Low potential evasion case in list =1	-0.079 (0.108)	0.044 (0.112)	-0.108 (0.116)	-0.079 (0.105)	-0.007 (0.110)	-0.072 (0.112)	-0.067 (0.122)	0.085 (0.127)	-0.049 (0.171)
Selection Year FE	No	No	No	Yes	Yes	Yes	No	No	No
Bureau FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	29	29	29	29	29	29	29	29	29
R-squared	0.129	0.118	0.208	0.366	0.350	0.420	0.424	0.431	0.430

Notes: The regression was estimated at the list level (Tax office x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is the execution rate of algorithm-selected cases (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 23: Existence of High potential case vrs execution rate of bottom 20% cases [Full audits]

	No FE			Selection year FE			Bureau FE		
	(1) Top 2	(2) First 10% seq	(3) First 20% seq	(4) Top 2	(5) First 10% seq	(6) First 20% seq	(7) Top 2	(8) First 10% seq	(9) First 20% seq
Dependent variable: 'Execution rate of bottom-20% cases									
High potential evasion case in list =1	0.169 (0.207)	0.331* (0.182)	0.313 (0.196)	0.112 (0.188)	0.246 (0.154)	0.320* (0.169)	0.070 (0.363)	0.240 (0.325)	0.390 (0.285)
Low potential evasion case in list =1	-0.081 (0.198)	0.193 (0.179)	-0.079 (0.209)	-0.077 (0.191)	0.128 (0.184)	-0.014 (0.180)	-0.034 (0.249)	0.262 (0.208)	-0.007 (0.250)
Selection Year FE	No	No	No	Yes	Yes	Yes	No	No	No
Bureau FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	27	27	27	27	27	27	27	27	27
R-squared	0.047	0.088	0.168	0.242	0.255	0.339	0.408	0.476	0.494

Notes: The regression was estimated at the list level (Tax office x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is the share of algorithm-selected cases at the bottom 20% of the predicted evasion distribution over the number of assigned bad cases (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 24: Existence of High potential case vrs Executed bottom-20% assigned (all) [Full audits]

	No FE			Selection year FE			Bureau FE		
	(1) Top 2	(2) First 10% seq	(3) First 20% seq	(4) Top 2	(5) First 10% seq	(6) First 20% seq	(7) Top 2	(8) First 10% seq	(9) First 20% seq
Dependent variable: 'Executed bottom-20% assigned (all)									
High potential evasion case in list =1	0.086 (0.057)	0.107** (0.050)	0.127** (0.050)	0.061 (0.059)	0.077 (0.053)	0.120** (0.056)	0.047 (0.080)	0.083 (0.072)	0.166** (0.075)
Low potential evasion case in list =1	0.041 (0.058)	0.089 (0.056)	0.047 (0.062)	0.037 (0.063)	0.067 (0.061)	0.060 (0.066)	0.048 (0.068)	0.138** (0.063)	0.113 (0.083)
Selection Year FE	No	No	No	Yes	Yes	Yes	No	No	No
Bureau FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	29	29	29	29	29	29	29	29	29
R-squared	0.043	0.115	0.119	0.202	0.236	0.276	0.443	0.589	0.556

Notes: The regression was estimated at the list level (Tax office x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is the execution rate of algorithm-executed cases in the bottom 20% of predicted evasion distribution over the total number of algo cases assigned (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 25: Share of High potential case vrs overall execution rate [Full audits]

	No FE			Selection year FE			Bureau FE		
	(1) Top 2	(2) First 10% seq	(3) First 20% seq	(4) Top 2	(5) First 10% seq	(6) First 20% seq	(7) Top 2	(8) First 10% seq	(9) First 20% seq
Dependent variable: 'Algorithm Execution rate									
Share of potential evasion case at the top	0.532** (0.225)	0.530** (0.240)	0.556*** (0.187)	0.457** (0.190)	0.403 (0.238)	0.481** (0.216)	0.398 (0.325)	0.492 (0.333)	0.444 (0.310)
Selection Year FE	No	No	No	Yes	Yes	Yes	No	No	No
Bureau FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	29	29	29	29	29	29	29	29	29
R-squared	0.112	0.076	0.097	0.350	0.313	0.343	0.416	0.422	0.417

Notes: The regression was estimated at the list level (Tax office x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is the execution rate of algorithm-selected cases (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 26: Share of High potential case vrs execution rate of bottom 20% cases [Full audits]

	No FE			Selection year FE			Bureau FE		
	(1) Top 2	(2) First 10% seq	(3) First 20% seq	(4) Top 2	(5) First 10% seq	(6) First 20% seq	(7) Top 2	(8) First 10% seq	(9) First 20% seq
Dependent variable: 'Execution rate of bottom-20% cases									
Share of potential evasion case at the top	0.443 (0.298)	0.501 (0.371)	0.796** (0.295)	0.331 (0.247)	0.329 (0.359)	0.708** (0.293)	0.194 (0.575)	0.321 (0.672)	0.717 (0.492)
Selection Year FE	No	No	No	Yes	Yes	Yes	No	No	No
Bureau FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	27	27	27	27	27	27	27	27	27
R-squared	0.039	0.034	0.099	0.235	0.228	0.292	0.407	0.412	0.449

Notes: The regression was estimatated at the list level (Tax office x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is share of algorithm-selected cases at the bottom 20% of the predicted evasion distribution over the number of assigned bad cases (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 27: Share of High potential case vrs Executed bottom-20% assigned (all) [Full audits]

	No FE			Selection year FE			Bureau FE		
	(1) Top 2	(2) First 10% seq	(3) First 20% seq	(4) Top 2	(5) First 10% seq	(6) First 20% seq	(7) Top 2	(8) First 10% seq	(9) First 20% seq
Dependent variable: 'Executed bottom-20% assigned (all)									
Share of potential evasion case at the top	0.122 (0.096)	0.100 (0.099)	0.156 (0.101)	0.072 (0.086)	0.032 (0.106)	0.123 (0.116)	0.015 (0.101)	0.022 (0.113)	0.089 (0.104)
Selection Year FE	No	No	No	Yes	Yes	Yes	No	No	No
Bureau FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	29	29	29	29	29	29	29	29	29
R-squared	0.024	0.011	0.030	0.187	0.180	0.198	0.426	0.427	0.432

Notes: The regression was estimatated at the list level (Tax office x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is the execution rate of algorithm-executed cases in the bottom 20% of predicted evasion distribution over the total number of algo cases assigned (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 28: Existence of High potential case vrs overall execution rate [Desk audits]

	(1) Top 1	(2) Top 2	(3) First 10% seq	(4) First 20% seq
Dependent variable: 'Algorithm Execution rate				
High potential evasion case in list =1	0.042 (0.049)	0.057 (0.048)	0.055 (0.046)	0.088* (0.045)
Low potential evasion case in list =1	0.018 (0.044)	0.071* (0.041)	0.068 (0.045)	0.064 (0.042)
Selection Year FE	Yes	Yes	Yes	Yes
Bureau FE	Yes	Yes	Yes	Yes
Observations	165	165	165	165
R-squared	0.521	0.528	0.527	0.531

Notes: The regression was estimated at the list level (Inspector x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is the execution rate of algorithm-selected cases (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 29: Existence of High potential case vrs execution rate of bottom 20% cases [Desk audits]

	(1) Top 1	(2) Top 2	(3) First 10% seq	(4) First 20% seq
Dependent variable: 'Execution rate of bottom-20% cases				
High potential evasion case in list =1	0.113 (0.095)	0.183** (0.073)	0.136 (0.091)	0.167** (0.080)
Low potential evasion case in list =1	0.088 (0.072)	0.150** (0.068)	0.155** (0.076)	0.108 (0.076)
Selection Year FE	Yes	Yes	Yes	Yes
Bureau FE	Yes	Yes	Yes	Yes
Observations	144	144	144	144
R-squared	0.422	0.441	0.436	0.431

Notes: The regression was estimated at the list level (Inspector x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is the share of algorithm-selected cases at the bottom 20% of the predicted evasion distribution over the number of assigned bad cases (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 30: Existence of High potential case vrs Executed bottom-20% assigned (all) [Desk audits]

	(1) Top 1	(2) Top 2	(3) First 10% seq	(4) First 20% seq
Dependent variable: ‘Executed bottom-20% assigned (all)’				
High potential evasion case in list =1	0.027 (0.022)	0.059*** (0.022)	0.057** (0.025)	0.052** (0.024)
Low potential evasion case in list =1	0.076** (0.032)	0.081*** (0.025)	0.104*** (0.030)	0.068** (0.027)
Selection Year FE	Yes	Yes	Yes	Yes
Bureau FE	Yes	Yes	Yes	Yes
Observations	165	165	165	165
R-squared	0.491	0.495	0.513	0.485

Notes: The regression was estimated at the list level (Inspector x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is the execution rate of algorithm-executed cases in the bottom 20% of predicted evasion distribution over the total number of algo cases assigned (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 31: Share of High potential case vrs overall execution rate [Desk audits]

	(1) Top 1	(2) Top 2	(3) First 10% seq	(4) First 20% seq
Dependent variable: ‘Algorithm Execution rate’				
Share of high predicted evasion case at the top	0.038 (0.048)	0.008 (0.064)	0.046 (0.048)	0.077 (0.062)
Selection Year FE	Yes	Yes	Yes	Yes
Bureau FE	Yes	Yes	Yes	Yes
Observations	165	165	165	165
R-squared	0.521	0.519	0.521	0.524

Notes: The regression was estimated at the list level (Inspector x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is the execution rate of algorithm-selected cases (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 32: Share of High potential case vrs execution rate of bottom 20% cases [Desk audits]

	(1) Top 1	(2) Top 2	(3) First 10% seq	(4) First 20% seq
Dependent variable: 'Execution rate of bottom-20% cases				
Share of high predicted evasion case at the top	0.081 (0.094)	0.140 (0.113)	0.095 (0.099)	0.151 (0.120)
Selection Year FE	Yes	Yes	Yes	Yes
Bureau FE	Yes	Yes	Yes	Yes
Observations	144	144	144	144
R-squared	0.415	0.417	0.416	0.419

Notes: The regression was estimated at the list level (Inspector x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is the share of algorithm-selected cases at the bottom 20% of the predicted evasion distribution over the number of assigned bad cases (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 33: Share of High potential case vrs Executed bottom-20% assigned (all) [Desk audits]

	(1) Top 1	(2) Top 2	(3) First 10% seq	(4) First 20% seq
Dependent variable: 'Executed bottom-20% assigned (all)				
Share of high predicted evasion case at the top	0.006 (0.022)	0.020 (0.036)	0.024 (0.024)	0.033 (0.031)
Selection Year FE	Yes	Yes	Yes	Yes
Bureau FE	Yes	Yes	Yes	Yes
Observations	165	165	165	165
R-squared	0.464	0.465	0.466	0.467

Notes: The regression was estimated at the list level (Inspector x Selection year), considering only the lists that had assigned algorithm cases in any of the years. The dependent variable is the execution rate of algorithm-executed cases in the bottom 20% of predicted evasion distribution over the total number of algo cases assigned (for each list). All specifications include Selection Year and Bureau fixed effects as indicated. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 34: Probability of execution according to ranking sequence

	Sequence			Rank/ Total assigned ratio			Top 5 in sequence=1		
	(1) Full audits	(2) Desk audits	(3) Desk audits	(4) Full audits	(5) Desk audits	(6) Desk audits	(7) Full audits	(8) Desk audits	(9) Desk audits
Rank in list (seq)	-0.002** (0.001)	-0.001 (0.001)	-0.001 (0.001)						
Rank / total assigned				-0.109** (0.049)	-0.029 (0.033)	-0.032 (0.032)			
Seq $\leq 5=1$							0.057 (0.042)	0.011 (0.019)	0.008 (0.018)
Algorithm	-0.168*** (0.030)	-0.045** (0.018)	-0.043** (0.017)	-0.167*** (0.030)	-0.046** (0.018)	-0.044** (0.017)	-0.169*** (0.030)	-0.046** (0.018)	-0.044** (0.017)
Overlap	0.157** (0.067)	0.049 (0.035)	0.047 (0.036)	0.156** (0.068)	0.049 (0.035)	0.048 (0.036)	0.152** (0.068)	0.049 (0.035)	0.047 (0.036)
Random		-0.001 (0.029)	-0.000 (0.028)		-0.001 (0.029)	-0.000 (0.028)		-0.001 (0.029)	-0.000 (0.028)
Mean of dep. var.	0.53	0.37	0.37	0.53	0.37	0.37	0.53	0.37	0.37
N	944	2731	2731	944	2731	2731	944	2731	2731
Tax center $\times$ year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Inspector $\times$ year FE	No	No	Yes	No	No	Yes	No	No	Yes

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. OLS results of a regression of the main audit outcomes on the selection method of the case, for full and desk audits separately. Specifications for full audits control for tax office $\times$ year fixed effects, which corresponds to the selection list for full audits. For desk audits, we show specifications with tax office $\times$ year fixed effects, and inspector $\times$ year fixed effects, the latter corresponding to the list for desk audits

5 Measures of inspector effort by predicted evasion

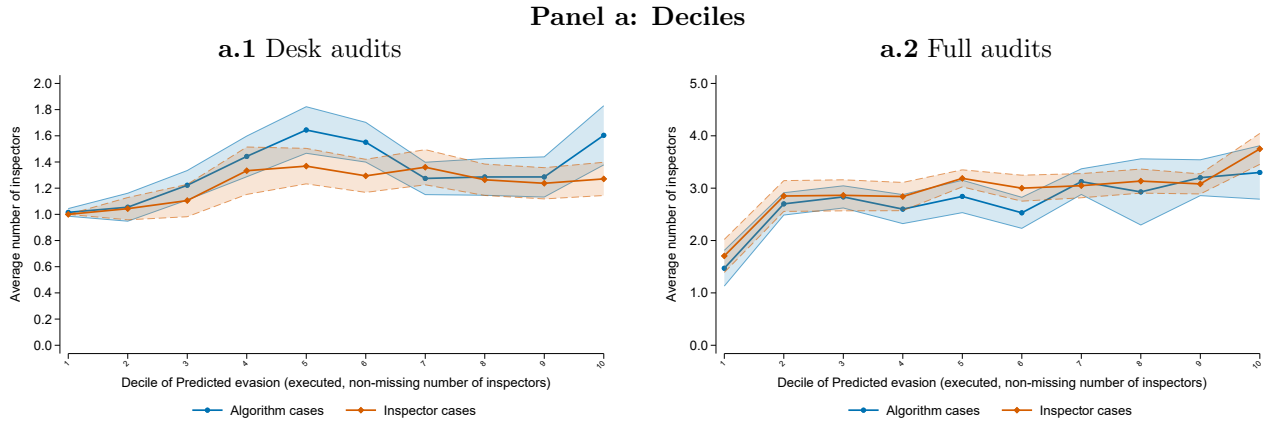
5.1 Whole Sample Bins

The bin-plot analysis uses the Random Forest predicted evasion score `yhatrf`, generated from the replication R pipeline and stored in `fullaudits_predicted.dta` / `deskaudits_predicted.dta`. The model is trained on executed audits with observed realized evasion (`y2==1` and non-missing `y4`) and then applied to the full selected sample.

- The two prediction files (`fullaudits_predicted.dta` and `deskaudits_predicted.dta`) are merged, and all steps are run separately by audit type (`x2==1` full, `x2==0` desk).
- Admin duration is cleaned by setting `y19=.` when `y19==0`.
- Firm size is constructed as mean turnover over 2014–2020 (`turnover_mean`).
- For each x-variable (`yhatrf`, `turnover_mean`), bins are created on the pooled selected sample using quantile bins (quintiles and deciles) and fixed-width bins (`w1/w2/w4/w5`). Bin assignment is done before `keep if y2==1`, so executed-only filtering is applied later for number of inspectors and duration outcomes.
- For number of inspectors and duration plots, only executed audits are kept (`y2==1`), data are collapsed by method and bin for `numberagents`, `y19`, `q30`, and `y8`, and 95% confidence intervals are computed using outcome-specific non-missing counts.
- Algorithm and inspector series are plotted with confidence bands, and one PDF is exported per outcome, x-variable, audit type, bin definition, and bin scope.

5.1.1 Predicted evasion

Figure 20: Average number of inspectors by predicted evasion bins (outcome-specific executed-sample quantile binning).



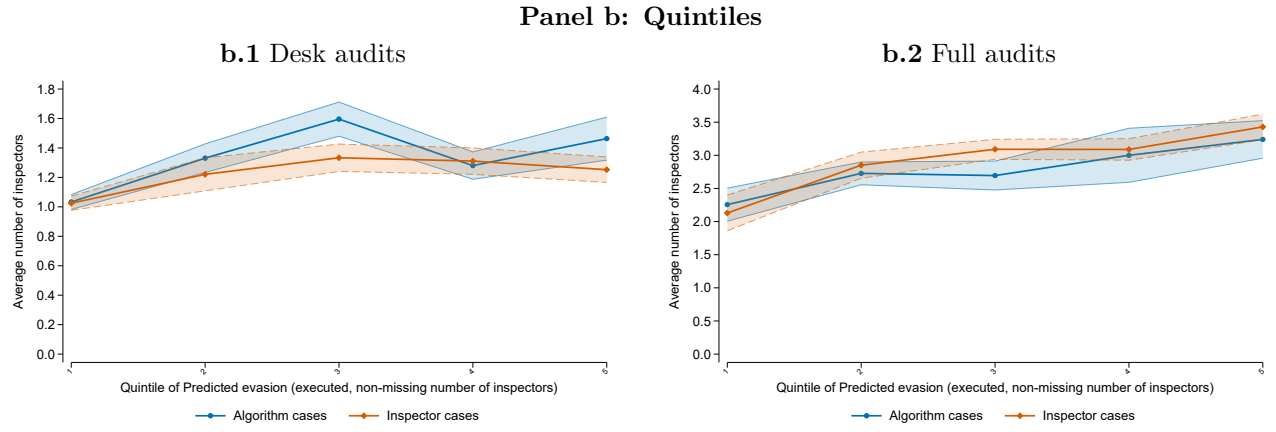


Figure 21: Average duration (admin data) by predicted evasion bins (outcome-specific executed-sample quantile binning).

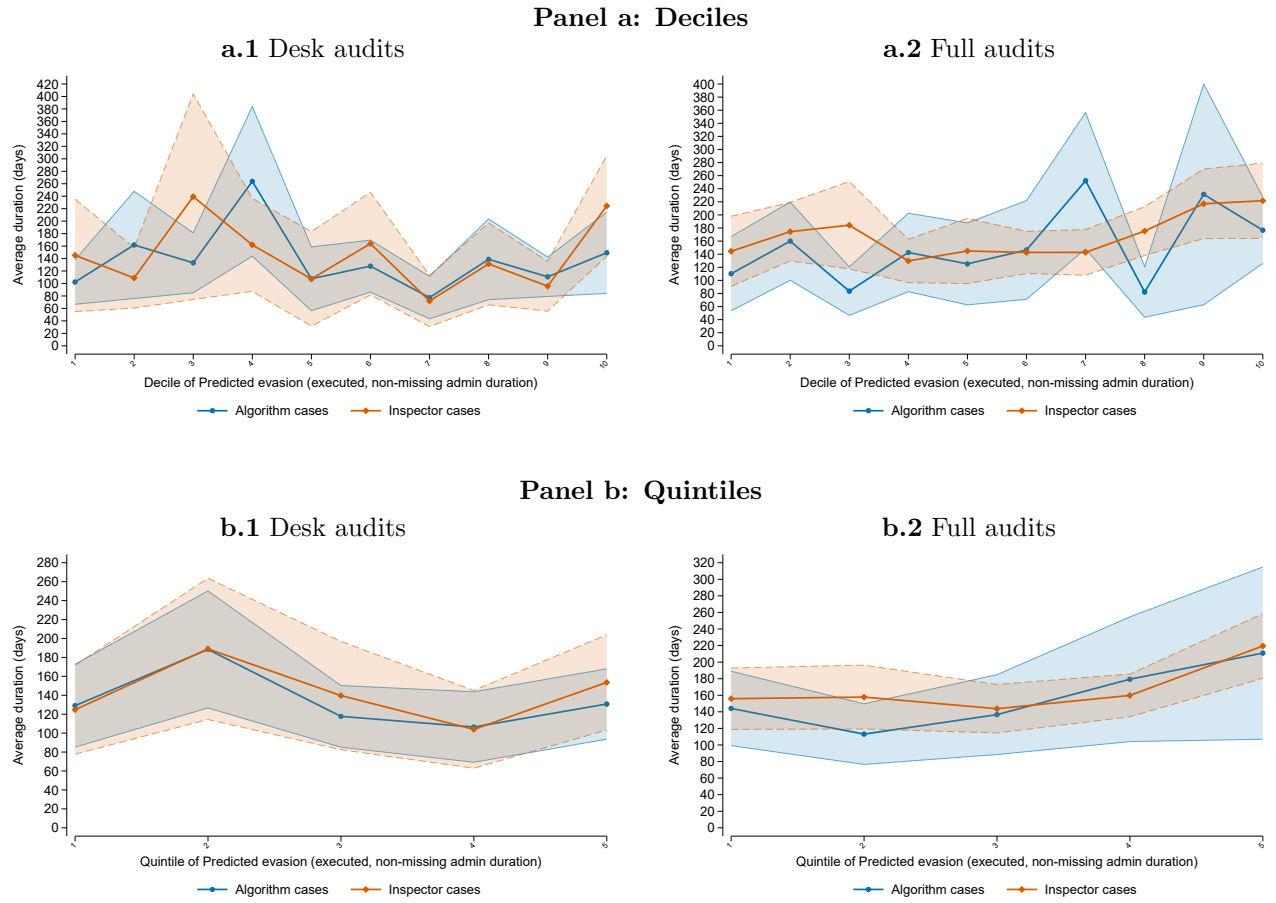
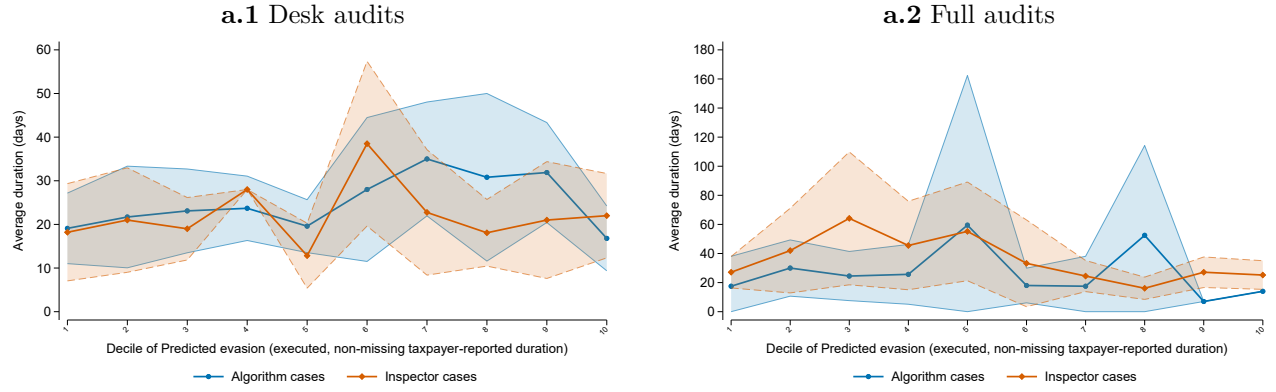


Figure 22: Average duration (taxpayer survey) by predicted evasion bins (outcome-specific executed-sample quantile binning).

Panel a: Deciles



Panel b: Quintiles

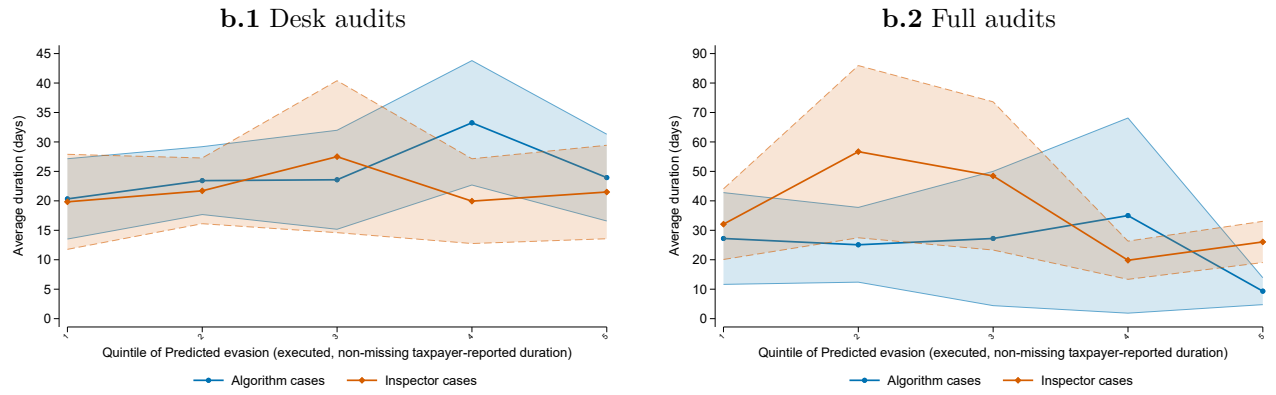
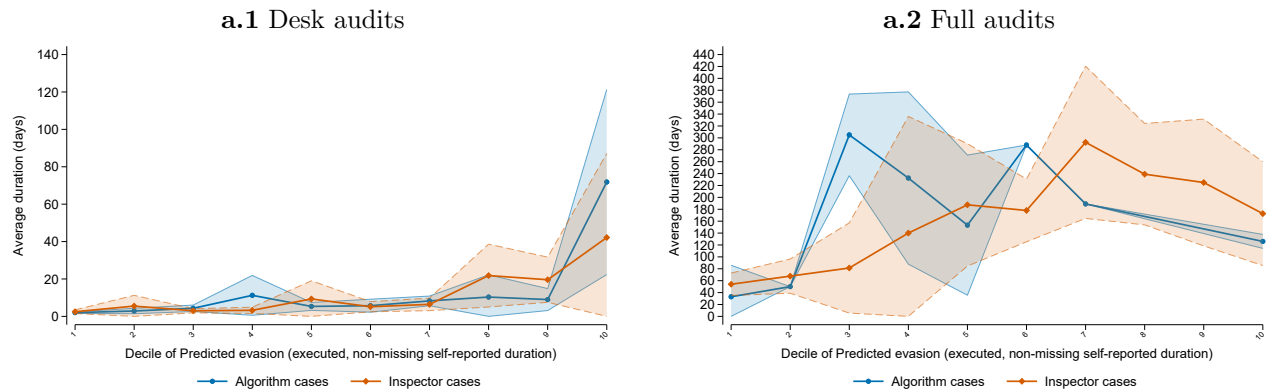


Figure 23: Average duration (self-reported) by predicted evasion bins (outcome-specific executed-sample quantile binning).

Panel a: Deciles



Panel b: Quintiles

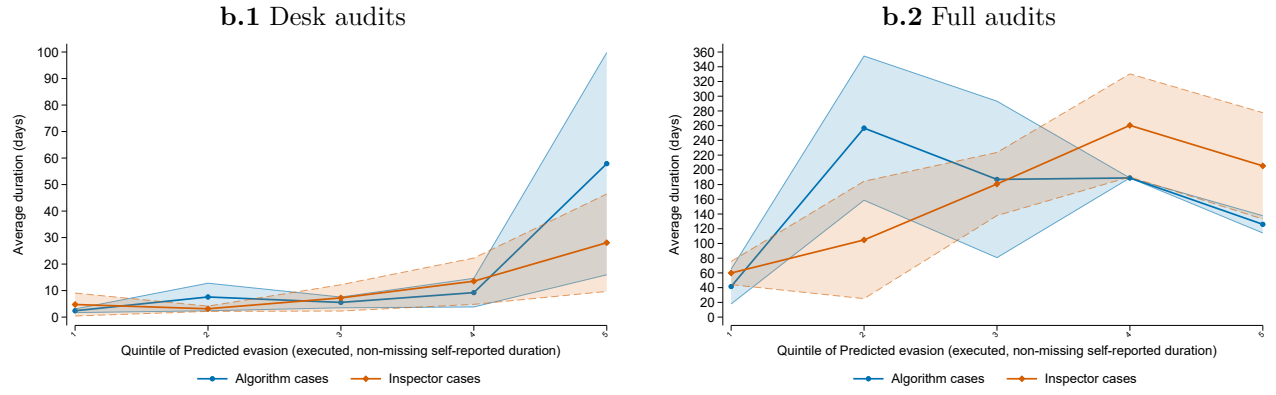
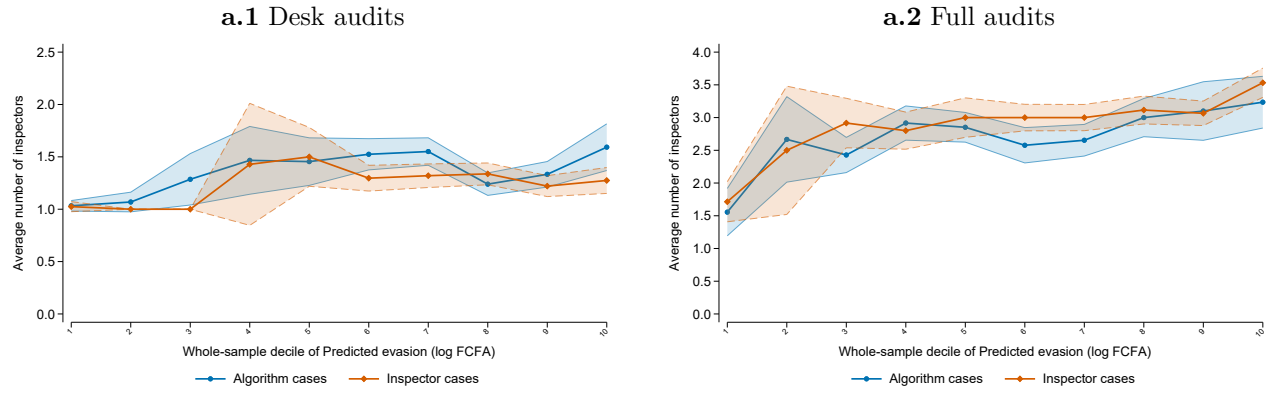
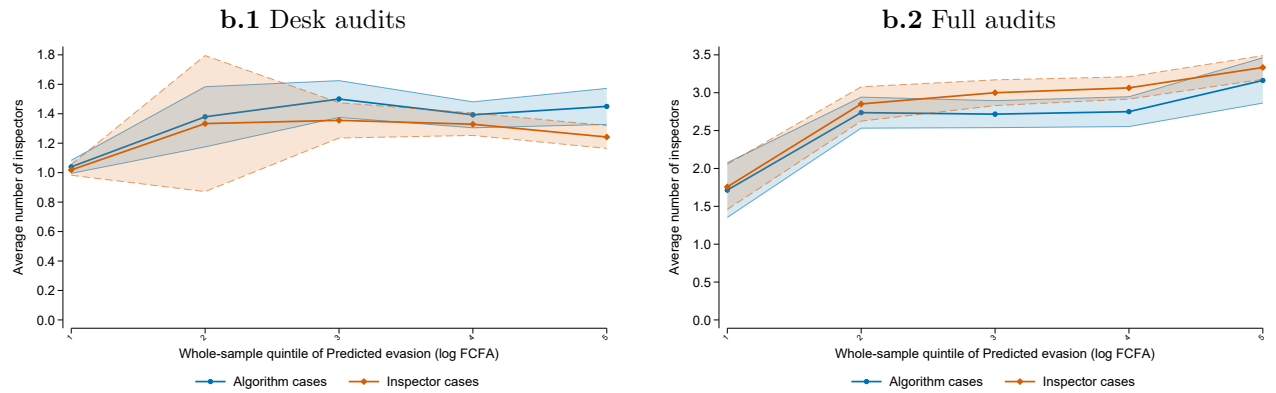


Figure 24: Average number of inspectors by predicted evasion bins (Executed-sample binning).

Panel a: Deciles

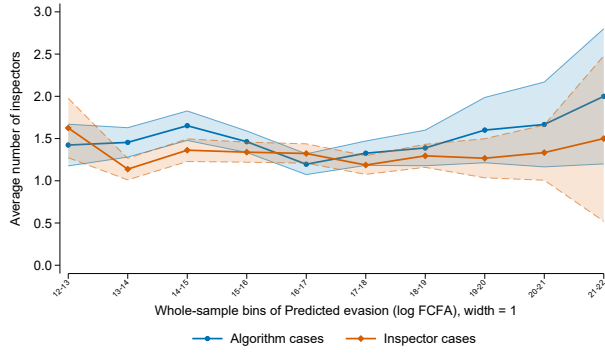


Panel b: Quintiles

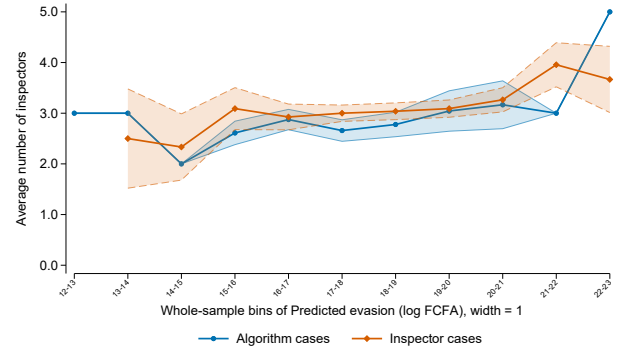


Panel c: Width 1

c.1 Desk audits

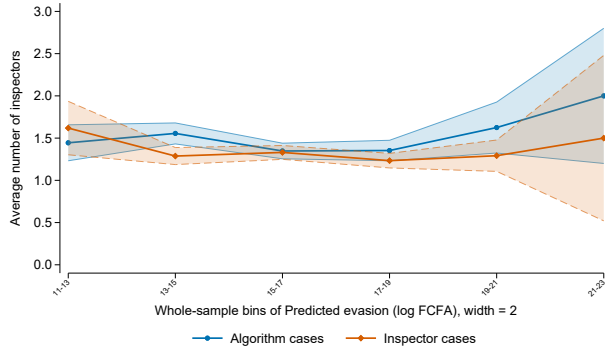


c.2 Full audits

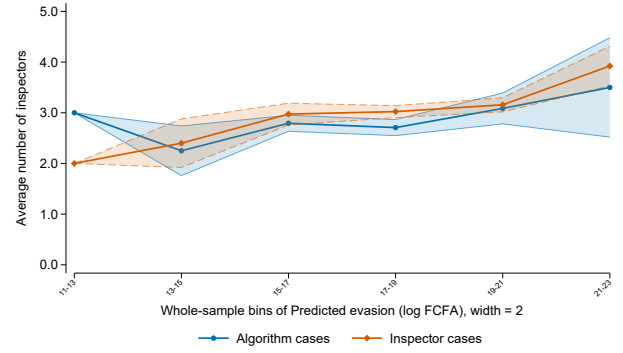


Panel d: Width 2

d.1 Desk audits

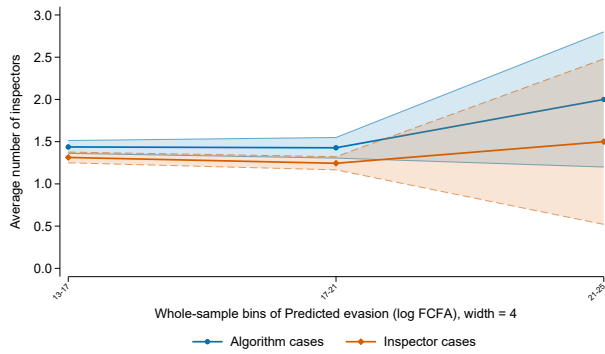


d.2 Full audits

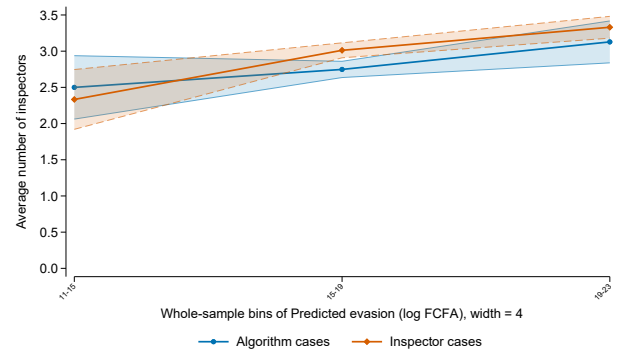


Panel e: Width 4

e.1 Desk audits



e.2 Full audits



Panel f: Width 5

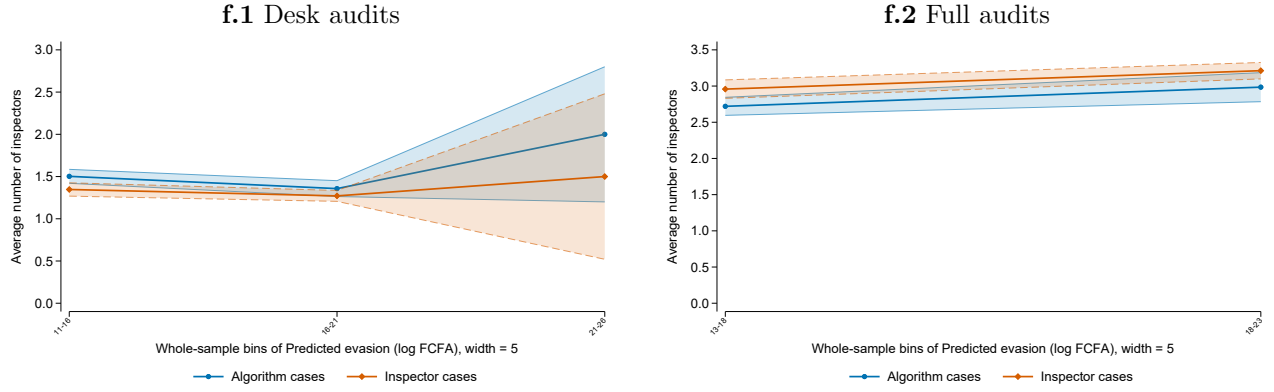
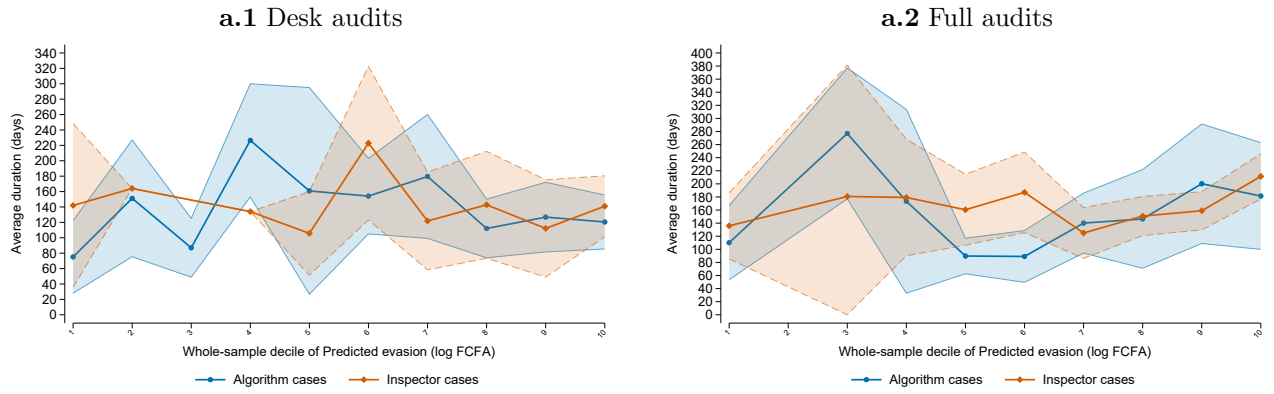
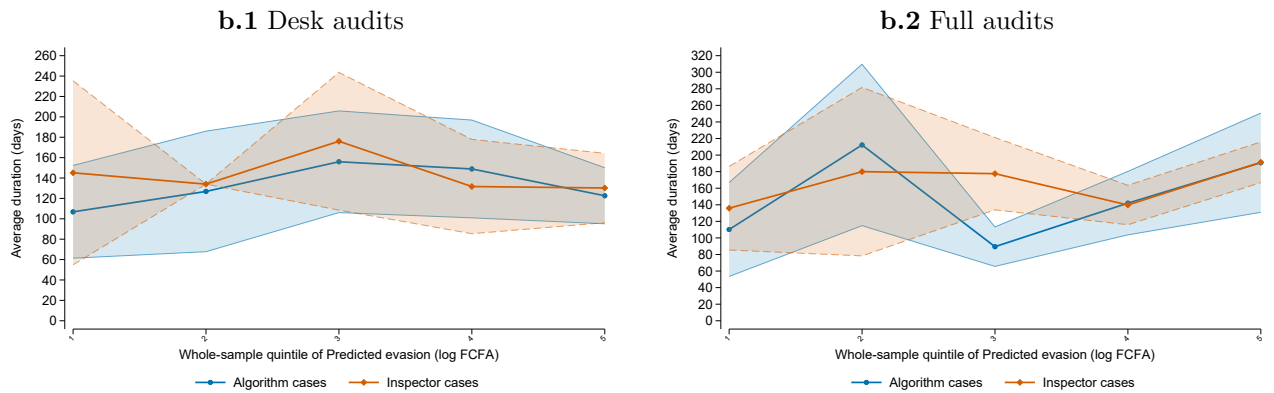


Figure 25: Average duration (admin data) by predicted evasion bins (Executed-sample binning).

Panel a: Deciles

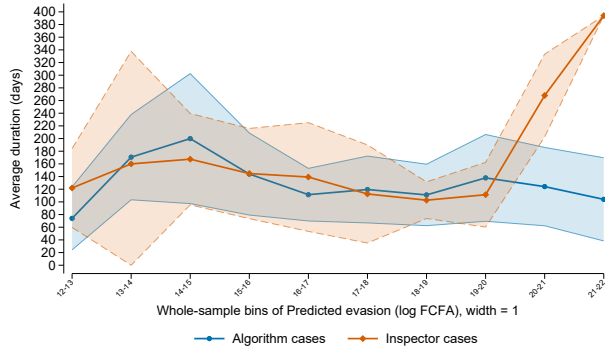


Panel b: Quintiles

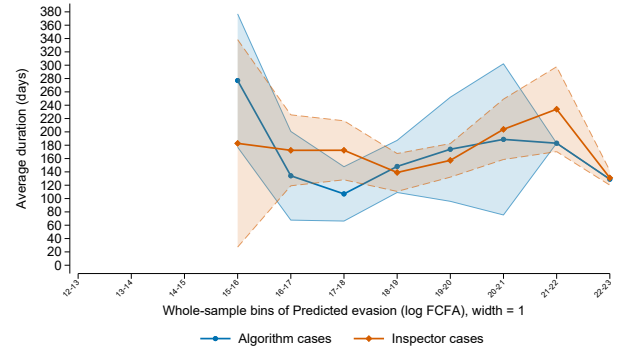


Panel c: Width 1

c.1 Desk audits

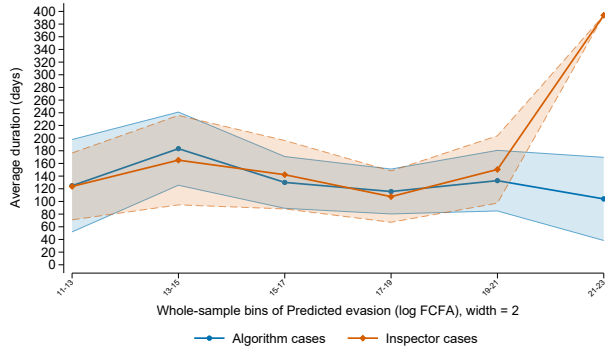


c.2 Full audits

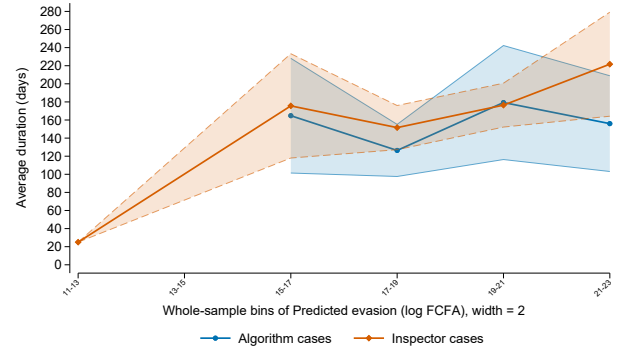


Panel d: Width 2

d.1 Desk audits

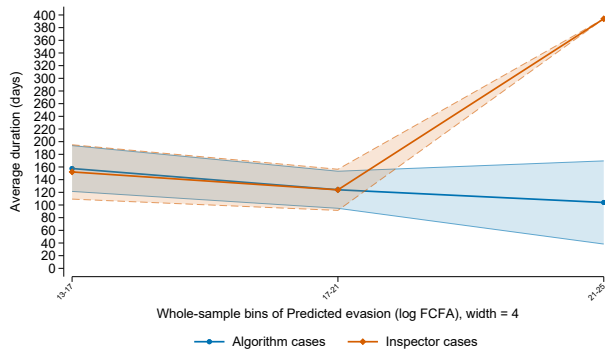


d.2 Full audits

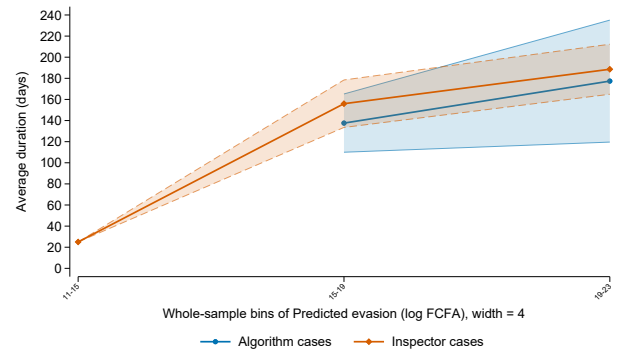


Panel e: Width 4

e.1 Desk audits



e.2 Full audits



Panel f: Width 5

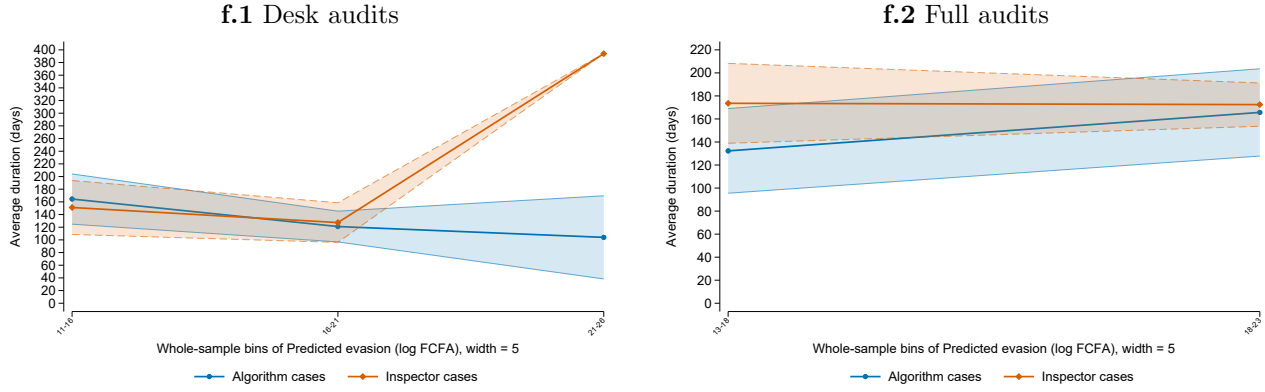
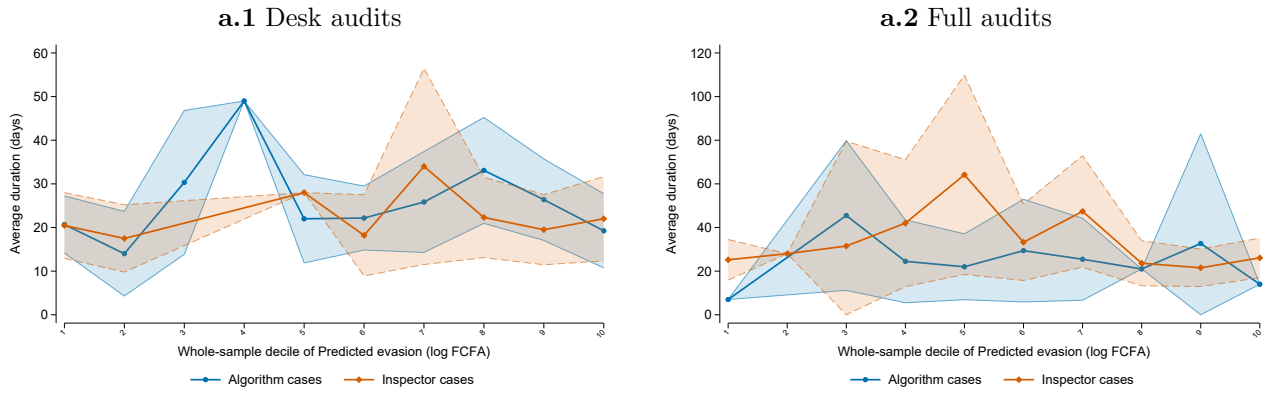
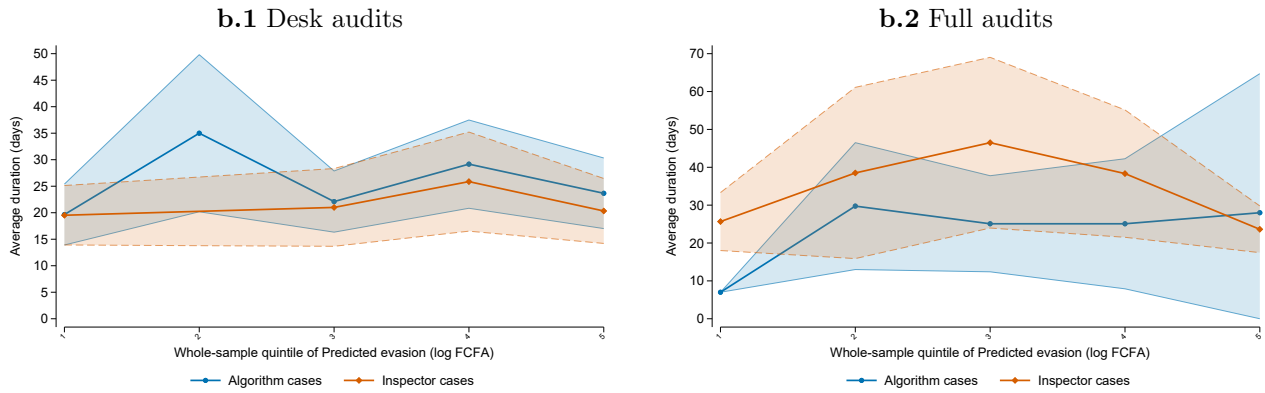


Figure 26: Average duration (taxpayer survey) by predicted evasion bins (Executed-sample binning).

Panel a: Deciles

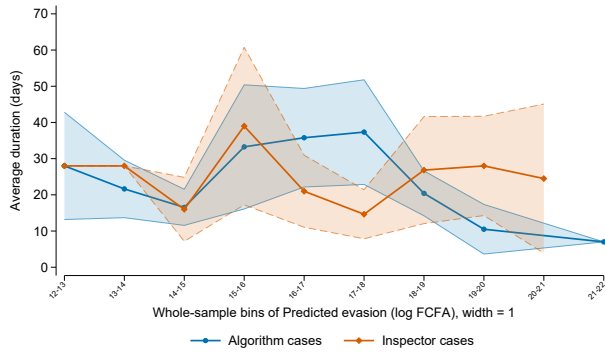


Panel b: Quintiles

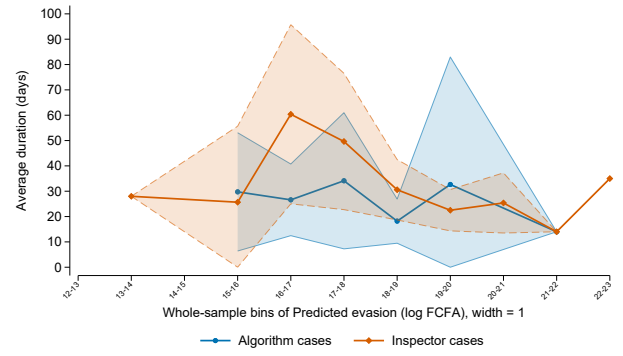


Panel c: Width 1

c.1 Desk audits

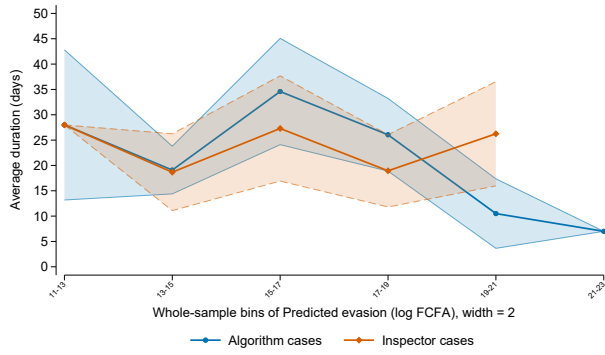


c.2 Full audits

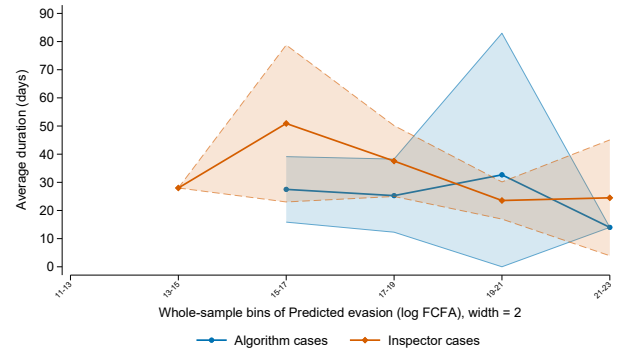


Panel d: Width 2

d.1 Desk audits

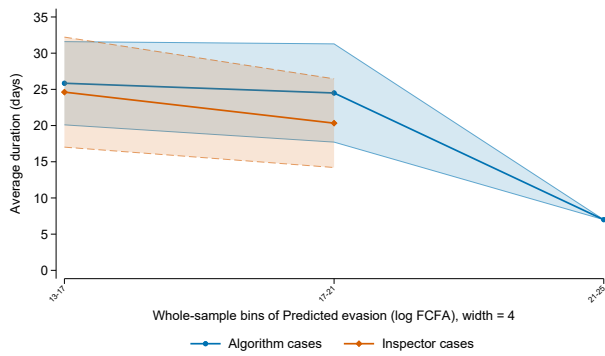


d.2 Full audits

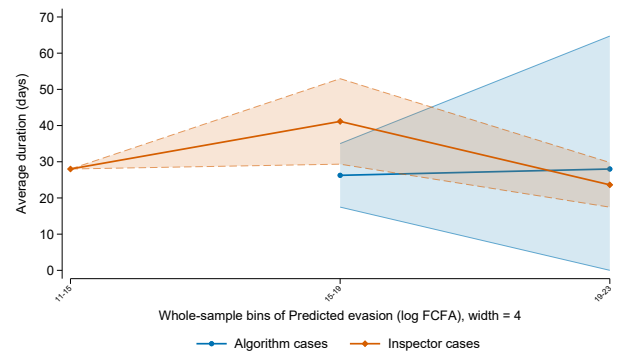


Panel e: Width 4

e.1 Desk audits



e.2 Full audits



Panel f: Width 5

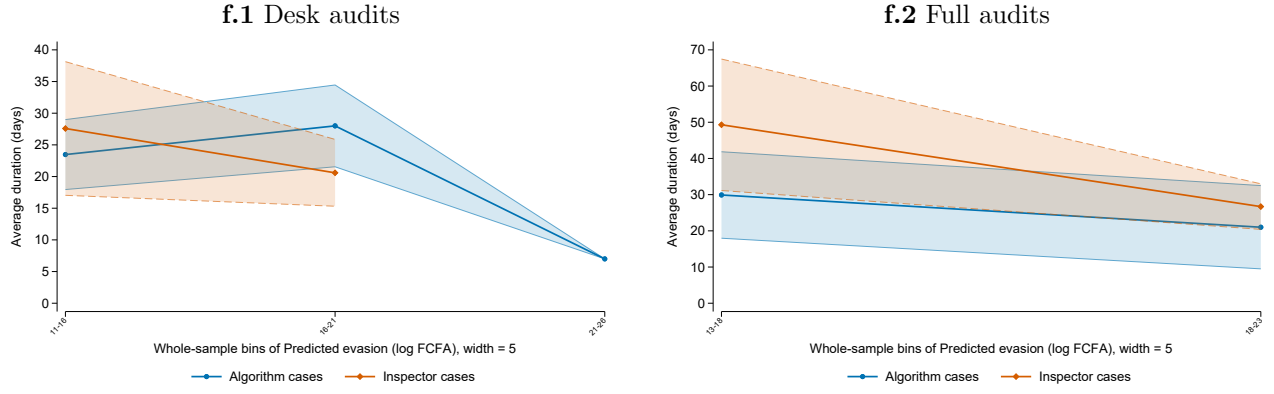
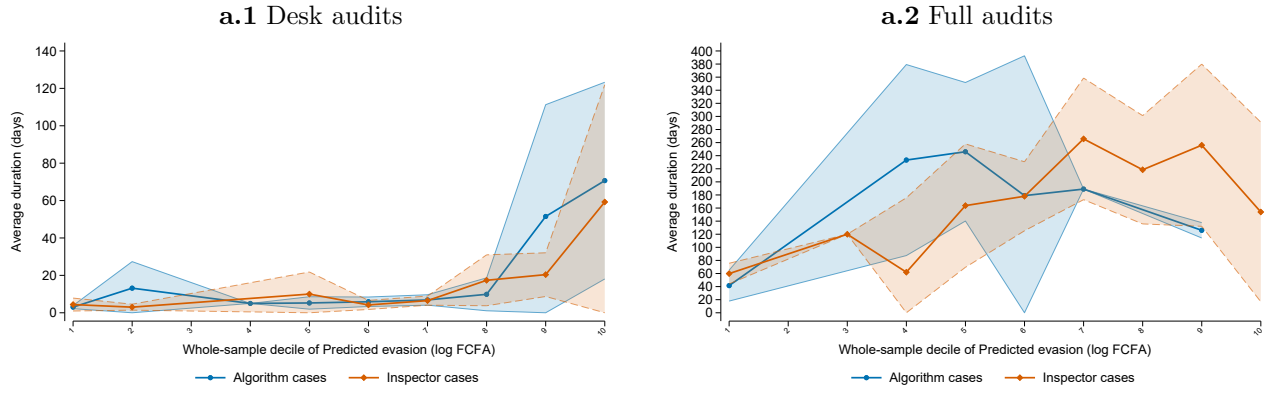
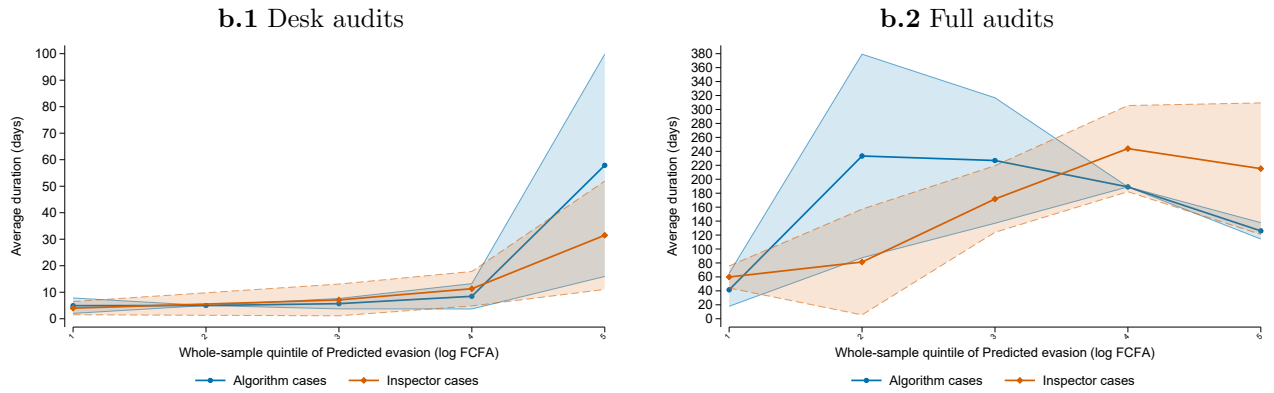


Figure 27: Average duration (self-reported) by predicted evasion bins (Executed-sample binning).

Panel a: Deciles

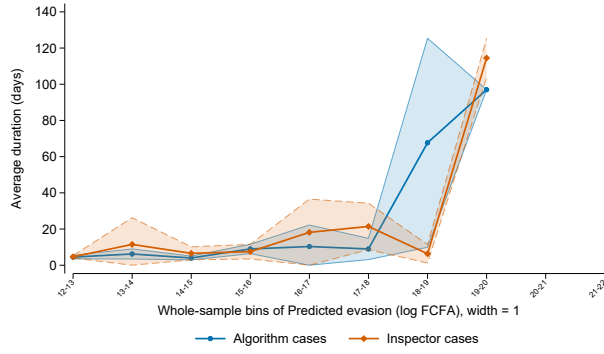


Panel b: Quintiles

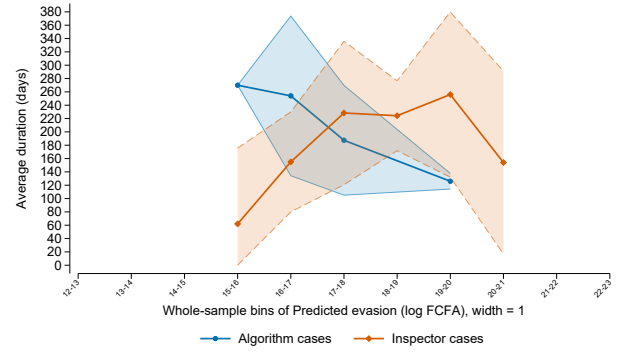


Panel c: Width 1

c.1 Desk audits

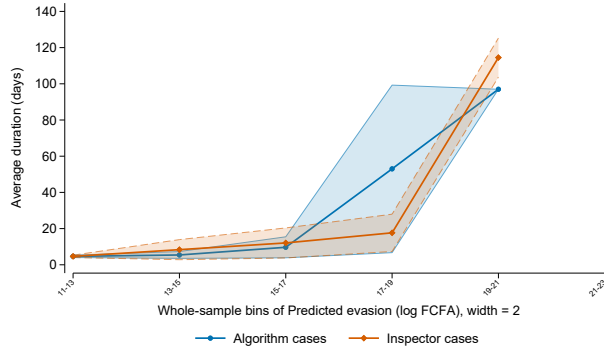


c.2 Full audits

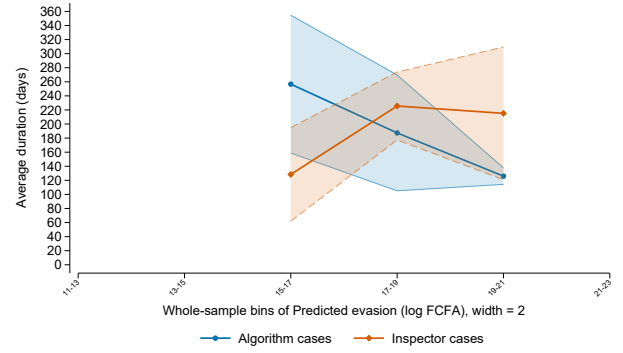


Panel d: Width 2

d.1 Desk audits

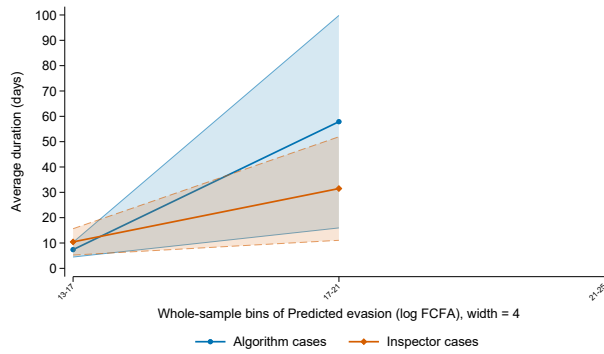


d.2 Full audits

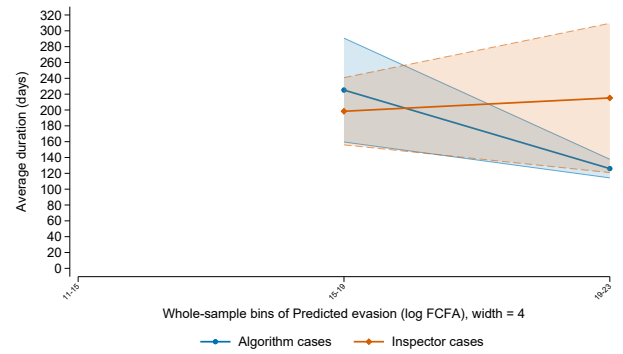


Panel e: Width 4

e.1 Desk audits

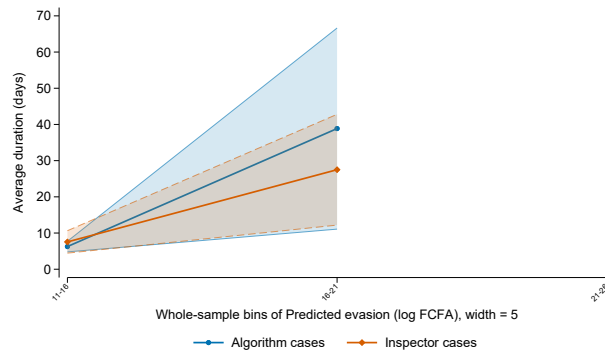


e.2 Full audits



Panel f: Width 5

f.1 Desk audits



f.2 Full audits

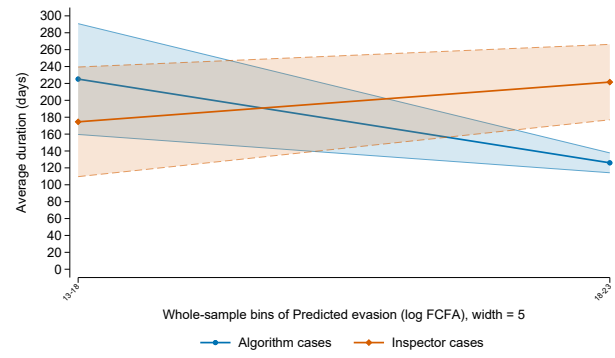


Table 35: Support for whole-sample predicted-evasion decile effort plots, full audits

Decile	Method	Selected N	Executed N	Number of inspectors						Admin duration						Taxpayer duration						Self-reported duration					
				N	Mean	SD	SE	CI low	CI high	N	Mean	SD	SE	CI low	CI high	N	Mean	SD	SE	CI low	CI high	N	Mean	SD	SE	CI low	CI high
1	Algorithm	40	18	18	1.56	0.78	0.18	1.19	1.92	7	110.14	76.52	28.92	53.45	166.83	1	7.00	0.00	0.00	7.00	7.00	4	41.50	24.13	12.07	17.85	65.15
1	Inspector	55	35	35	1.71	0.93	0.16	1.41	2.02	17	135.88	106.14	25.74	85.42	186.34	5	25.20	10.62	4.75	15.89	34.51	7	59.86	21.31	8.05	44.07	75.64
2	Algorithm	60	3	3	2.67	0.58	0.33	2.01	3.32	0						0						0					
2	Inspector	34	2	2	2.50	0.71	0.50	1.52	3.48	0						1	28.00	0.00	0.00	28.00	28.00	0					
3	Algorithm	51	14	14	2.43	0.51	0.14	2.16	2.70	3	277.00	88.36	51.01	177.01	376.99	2	45.50	24.75	17.50	11.20	79.80	0					
3	Inspector	44	12	12	2.92	0.67	0.19	2.54	3.29	4	180.75	204.94	102.47	0.00	381.59	2	31.50	34.65	24.50	0.00	79.52	1	120.00	0.00	0.00	120.00	120.00
4	Algorithm	54	24	24	2.92	0.65	0.13	2.66	3.18	5	173.40	160.22	71.65	32.96	313.84	6	24.50	23.74	9.69	5.51	43.49	3	233.33	128.97	74.46	87.39	379.28
4	Inspector	40	15	15	2.80	0.56	0.14	2.52	3.08	4	179.25	90.92	45.46	90.15	268.35	4	42.00	29.70	14.85	12.90	71.10	2	62.00	82.02	58.00	0.00	175.68
5	Algorithm	55	27	27	2.85	0.60	0.12	2.62	3.08	9	89.78	41.57	13.86	62.62	116.94	7	22.00	20.38	7.70	6.90	37.10	5	246.00	120.75	54.00	140.16	351.84
5	Inspector	39	23	23	3.00	0.74	0.15	2.70	3.30	10	160.50	87.47	27.66	106.28	214.72	6	64.17	57.11	23.32	18.47	109.87	4	163.75	96.21	48.10	69.46	258.04
6	Algorithm	39	26	26	2.58	0.70	0.14	2.31	2.85	10	89.10	63.86	20.19	49.52	128.68	5	29.40	26.84	12.00	5.88	52.92	2	179.00	154.15	109.00	0.00	392.64
6	Inspector	56	34	34	3.00	0.60	0.10	2.80	3.20	18	187.06	133.23	31.40	125.51	248.60	8	33.25	25.31	8.95	15.71	50.79	5	178.00	60.17	26.91	125.26	230.74
7	Algorithm	40	26	26	2.65	0.63	0.12	2.41	2.90	16	140.06	93.49	23.37	94.25	185.87	11	25.45	31.81	9.59	6.65	44.25	1	189.00	0.00	0.00	189.00	189.00
7	Inspector	54	42	42	3.00	0.66	0.10	2.80	3.20	24	124.96	96.97	19.79	86.16	163.75	13	47.38	47.06	13.05	21.80	72.97	7	265.71	125.41	47.40	172.81	358.62
8	Algorithm	18	10	10	3.00	0.47	0.15	2.71	3.29	7	146.43	101.76	38.46	71.04	221.82	1	21.00	0.00	0.00	21.00	21.00	0					
8	Inspector	77	52	52	3.12	0.78	0.11	2.90	3.33	32	150.81	86.03	15.21	121.00	180.62	8	23.62	14.94	5.28	13.27	33.98	6	218.50	103.43	42.22	135.74	301.26
9	Algorithm	30	20	20	3.10	1.02	0.23	2.65	3.55	11	200.18	154.50	46.58	108.88	291.49	3	32.67	44.46	25.67	0.00	82.97	2	126.00	8.49	6.00	114.24	137.76
9	Inspector	64	46	46	3.07	0.65	0.10	2.88	3.25	34	158.88	86.83	14.89	129.69	188.07	13	21.54	15.77	4.37	12.96	30.11	3	256.00	109.50	63.22	132.09	379.91
10	Algorithm	20	17	17	3.24	0.83	0.20	2.84	3.63	11	181.45	138.08	41.63	99.85	263.06	1	14.00	0.00	0.00	14.00	14.00	0					
10	Inspector	74	62	62	3.53	0.90	0.11	3.31	3.76	55	211.38	130.99	17.66	176.76	246.00	11	26.09	15.37	4.63	17.01	35.17	2	154.00	98.99	70.00	16.80	291.20

Notes: Predicted-evasion deciles are formed on the pooled sample of selected full audits. Outcome statistics are computed for executed cases only ($y2=1$). **Selected N** reports all selected cases in each method-by-decile cell, while **Executed N** and all outcome moments are based on executed cases; outcome-specific columns further restrict to non-missing observations for that outcome among executed cases only. Confidence intervals are 95% normal-approximation intervals constructed to match the corresponding bin plots.

5.1.2 Firm turnover

Figure 28: Average number of inspectors by firm turnover bins (outcome-specific executed-sample quantile binning).

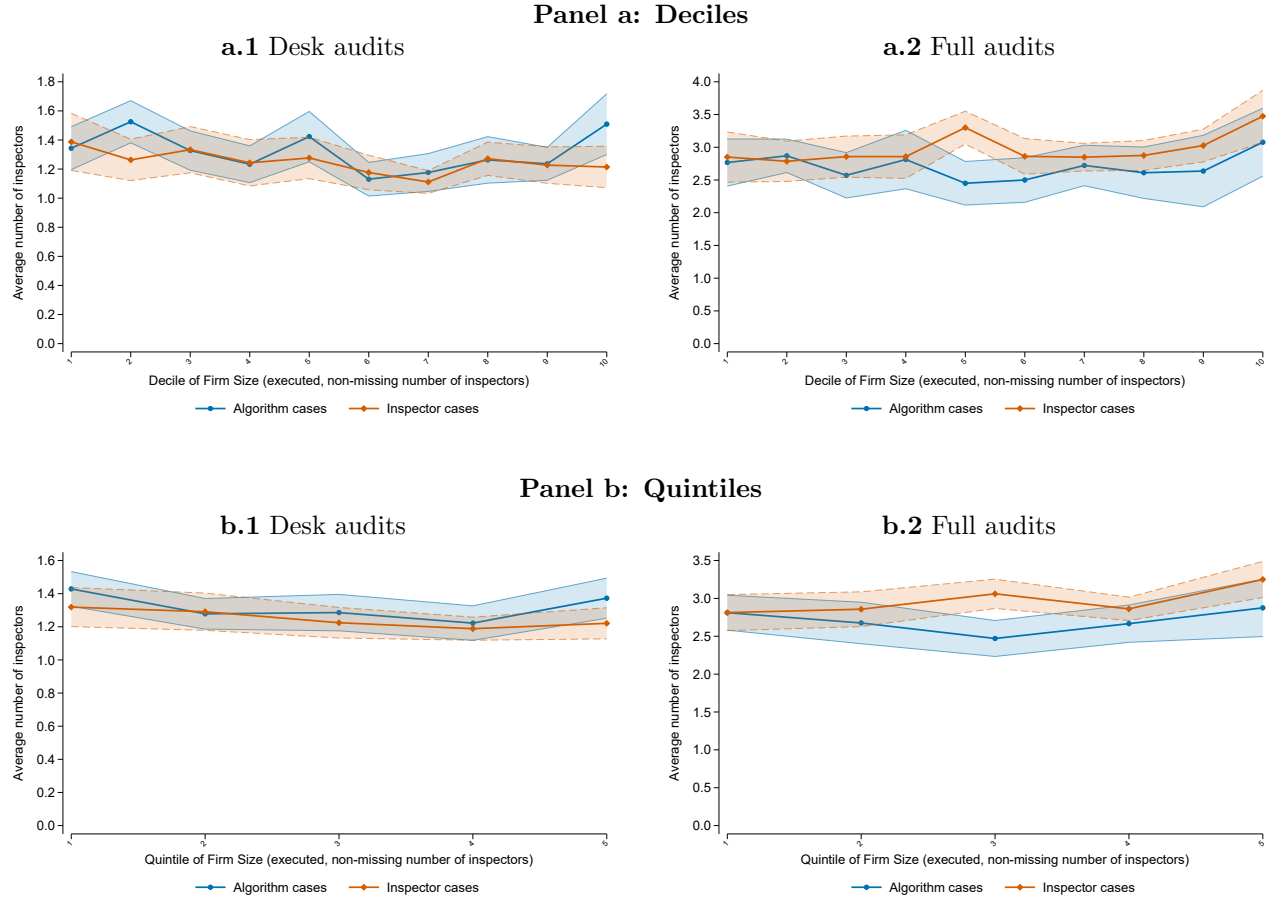
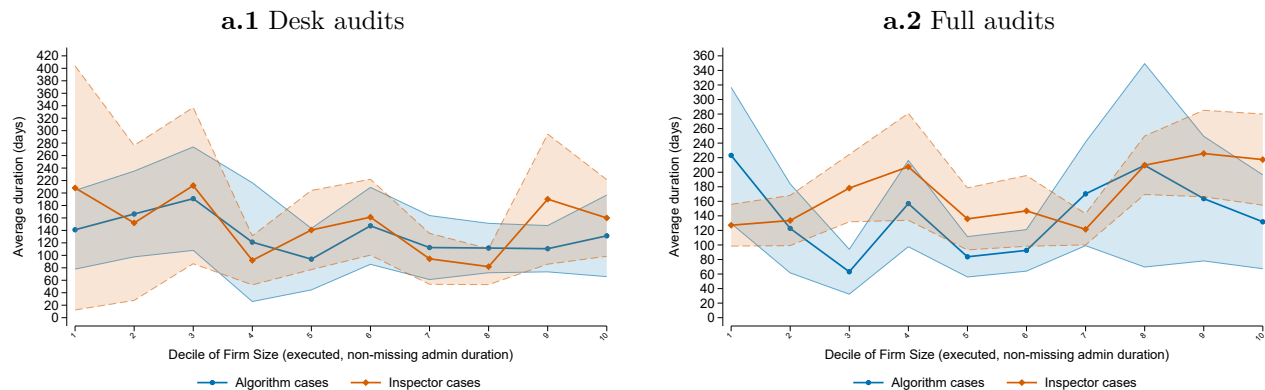


Figure 29: Average duration (admin data) by firm turnover bins (outcome-specific executed-sample quantile binning).

Panel a: Deciles



Panel b: Quintiles

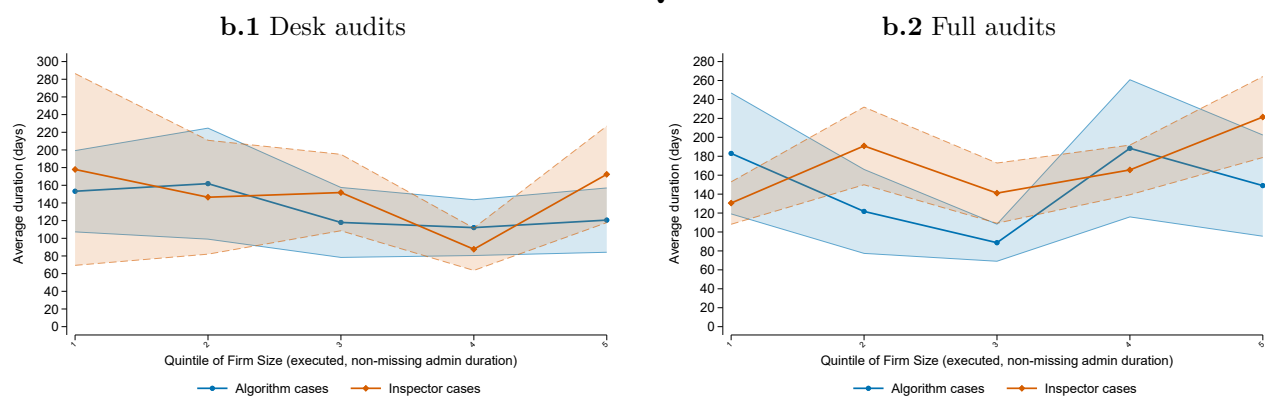
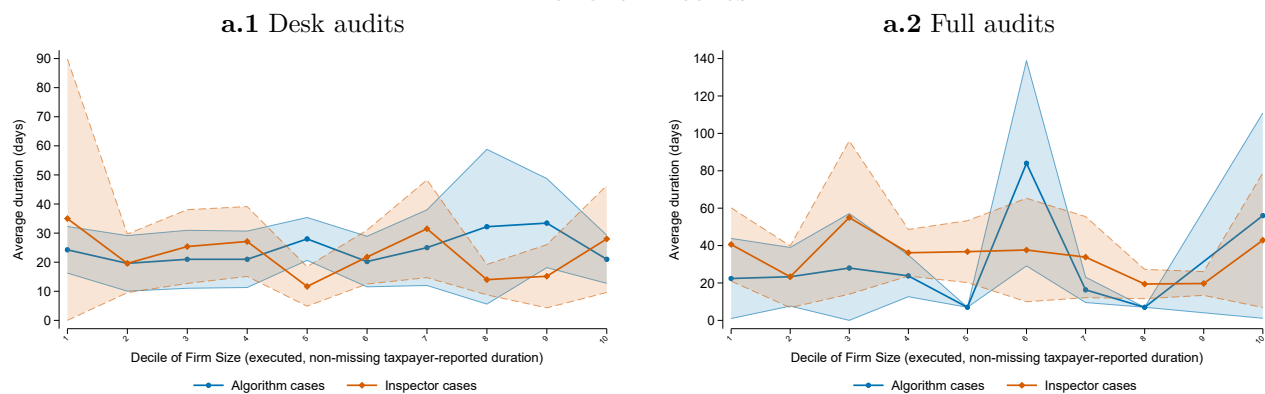


Figure 30: Average duration (taxpayer survey) by firm turnover bins (outcome-specific executed-sample quantile binning).

Panel a: Deciles



Panel b: Quintiles

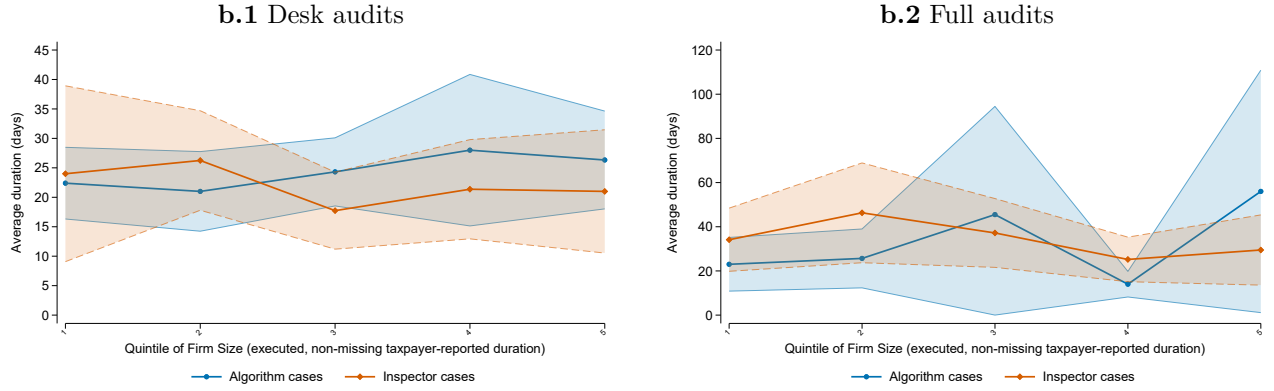
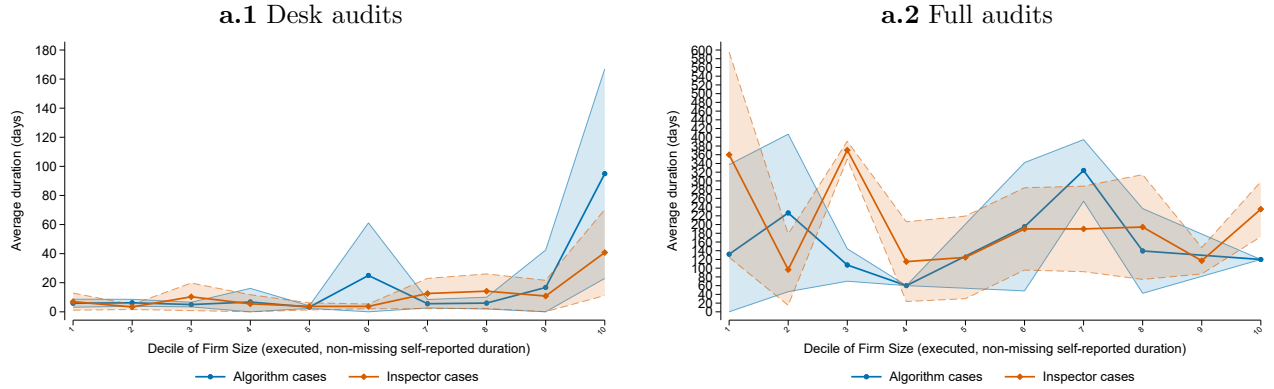


Figure 31: Average duration (self-reported) by firm turnover bins (outcome-specific executed-sample quantile binning).

Panel a: Deciles



Panel b: Quintiles

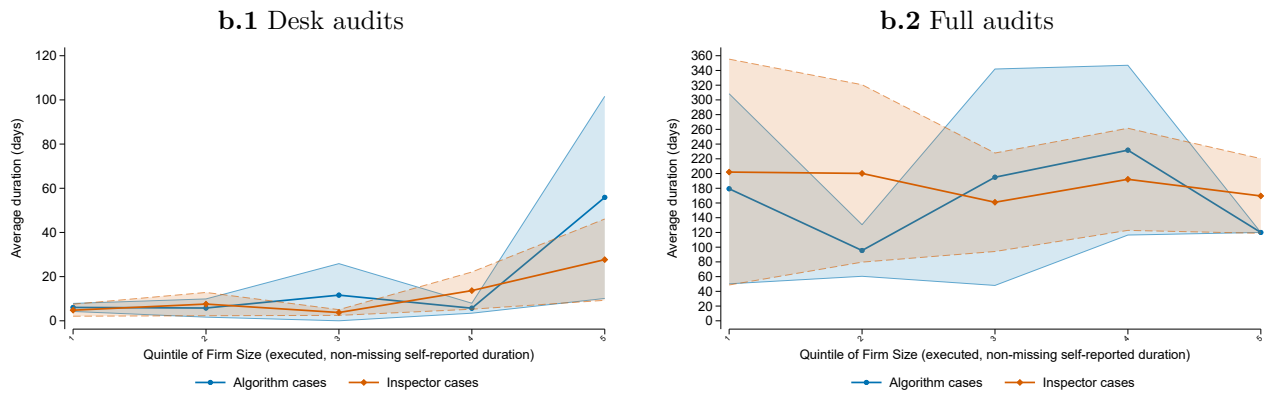
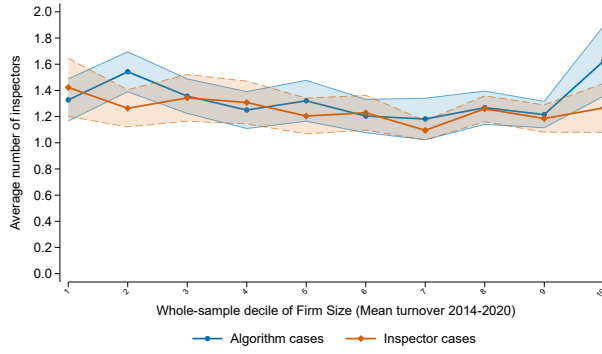


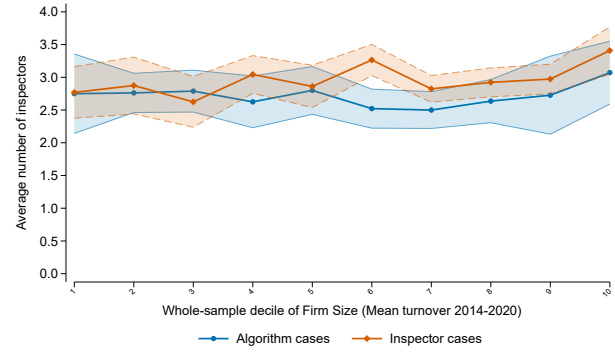
Figure 32: Average number of inspectors by firm turnover bins (Executed-sample binning).

Panel a: Deciles

a.1 Desk audits

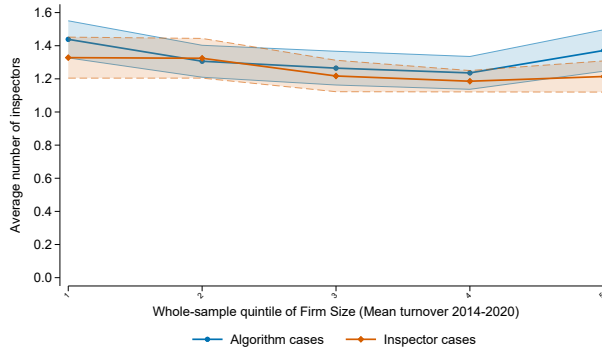


a.2 Full audits

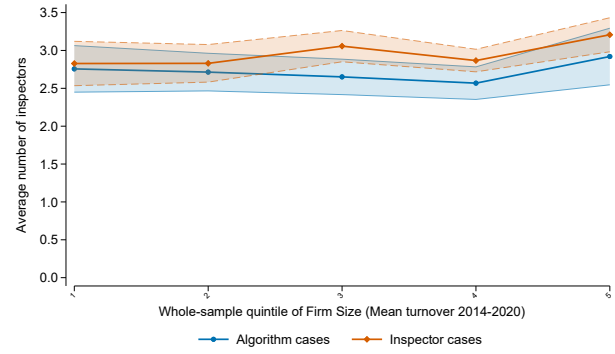


Panel b: Quintiles

b.1 Desk audits

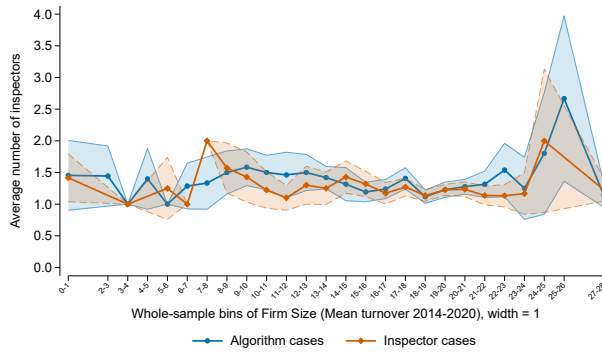


b.2 Full audits

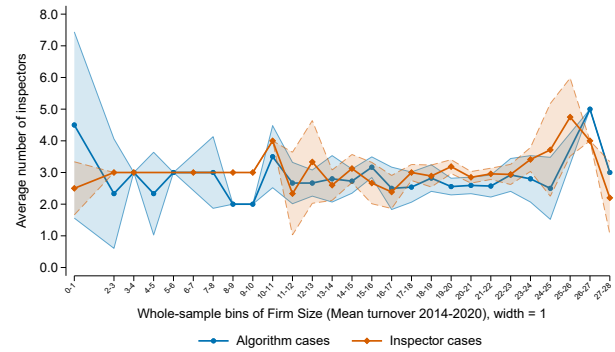


Panel c: Width 1

c.1 Desk audits

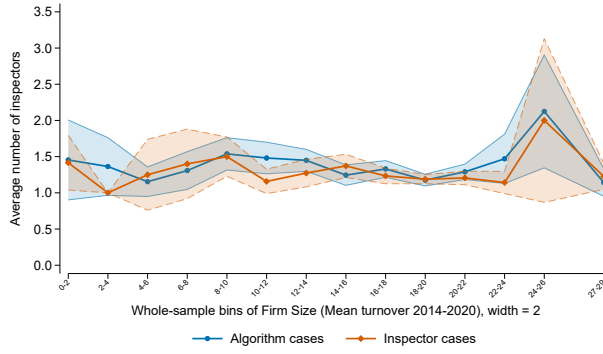


c.2 Full audits

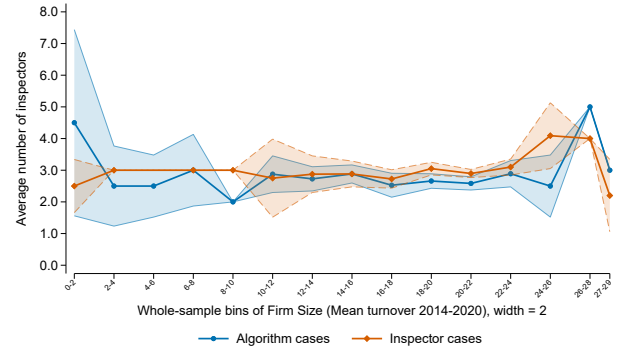


Panel d: Width 2

d.1 Desk audits

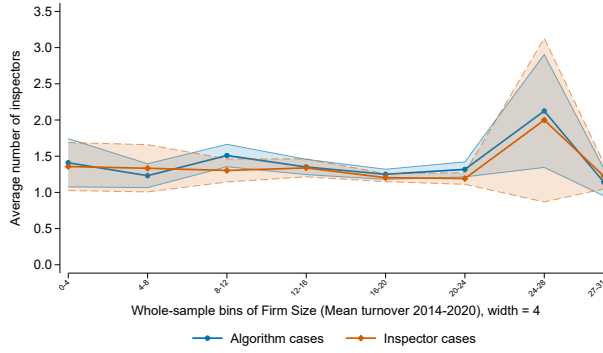


d.2 Full audits

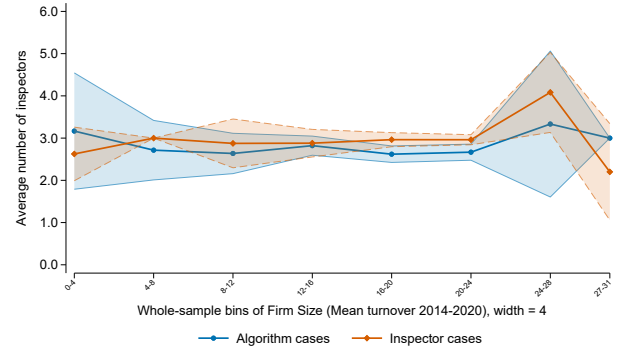


Panel e: Width 4

e.1 Desk audits

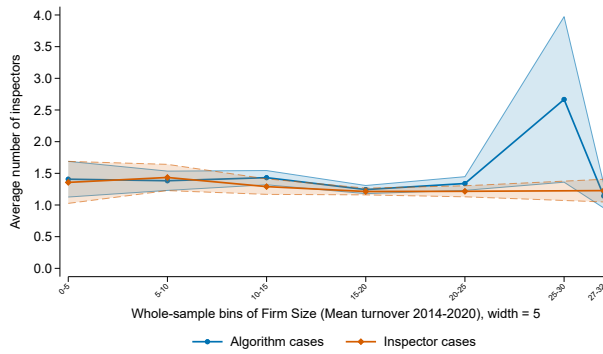


e.2 Full audits



Panel f: Width 5

f.1 Desk audits



f.2 Full audits

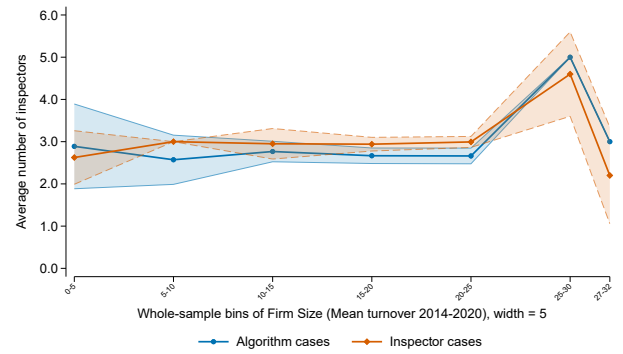
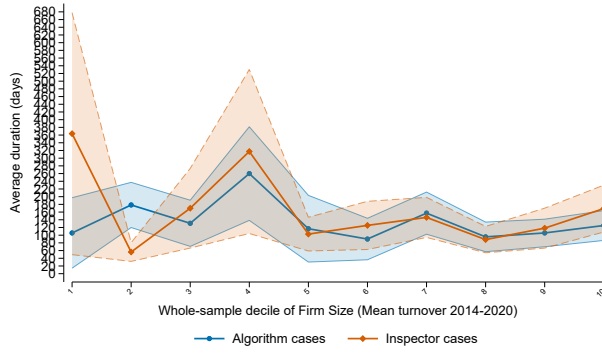


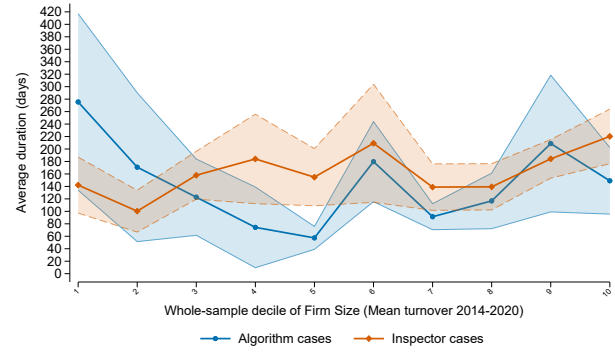
Figure 33: Average duration (admin data) by firm turnover bins (Executed-sample binning).

Panel a: Deciles

a.1 Desk audits

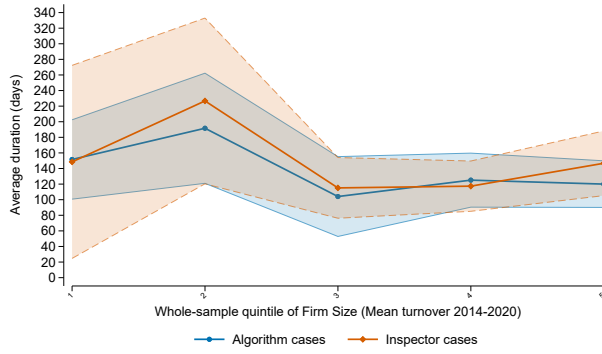


a.2 Full audits

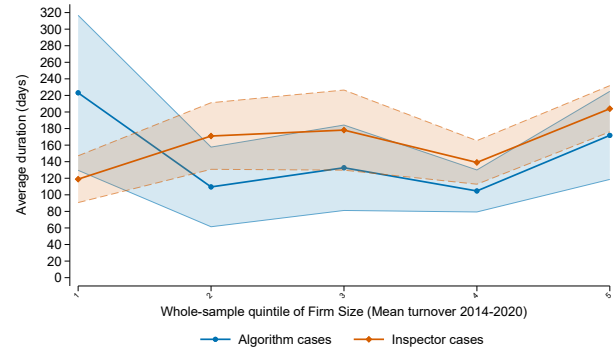


Panel b: Quintiles

b.1 Desk audits

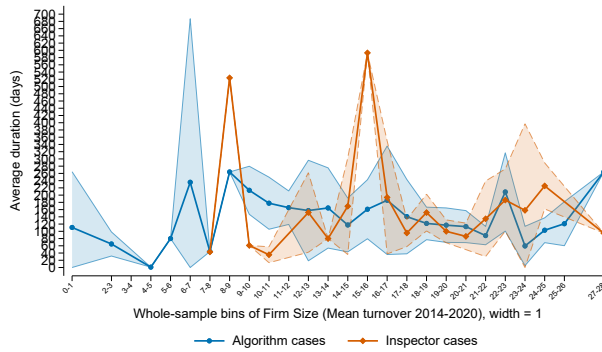


b.2 Full audits

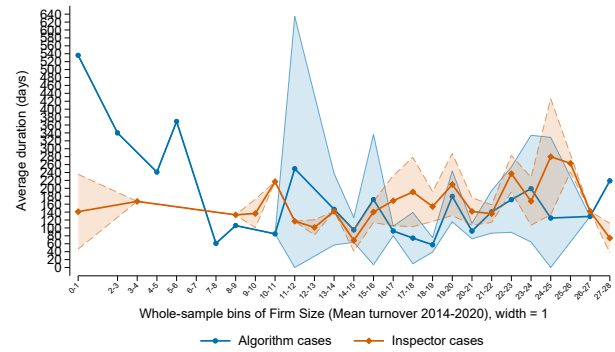


Panel c: Width 1

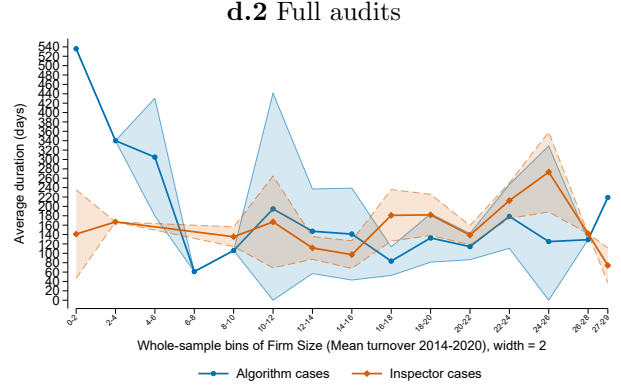
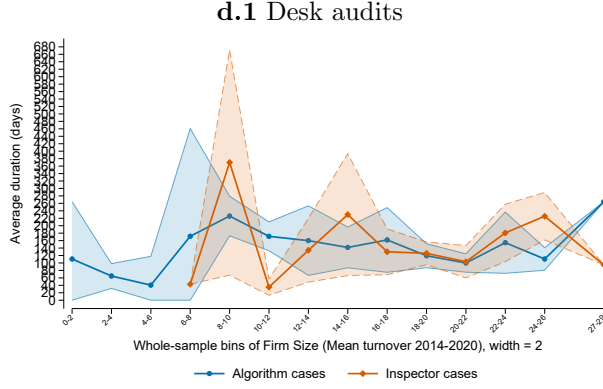
c.1 Desk audits



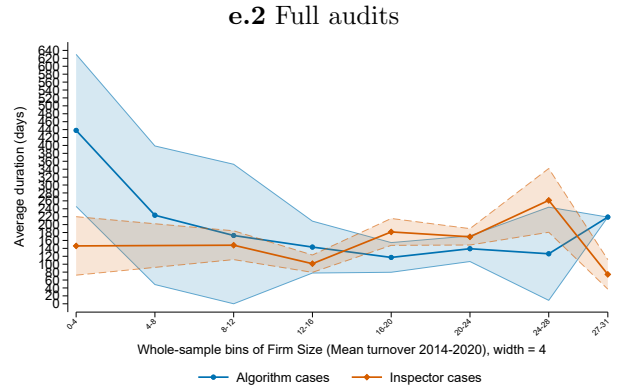
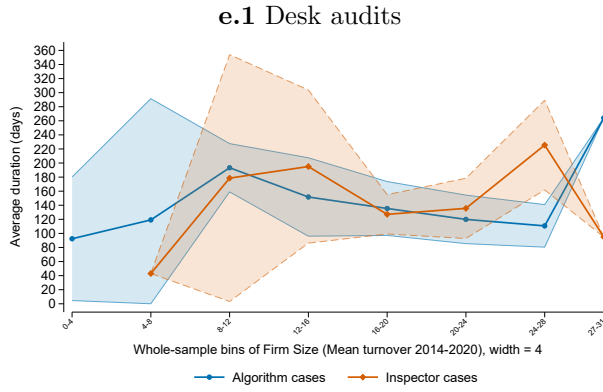
c.2 Full audits



Panel d: Width 2



Panel e: Width 4



Panel f: Width 5

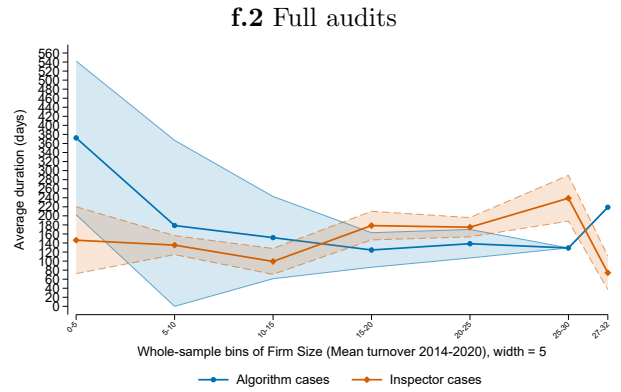
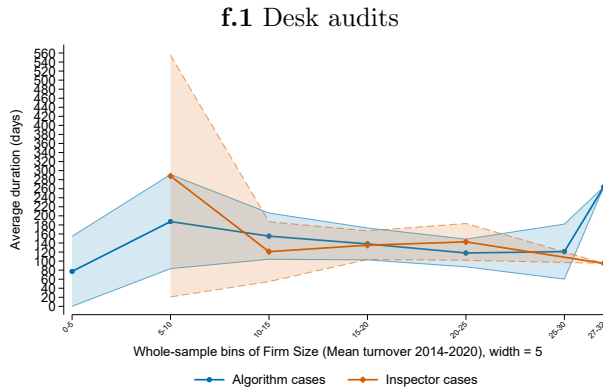
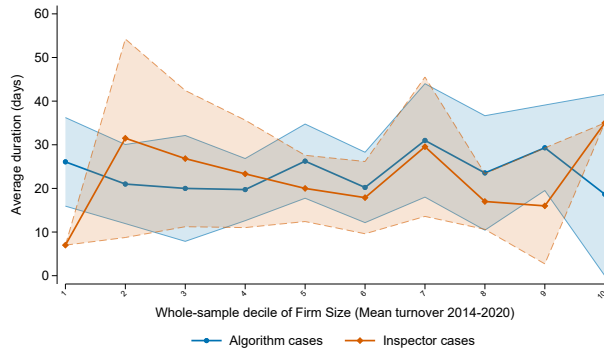


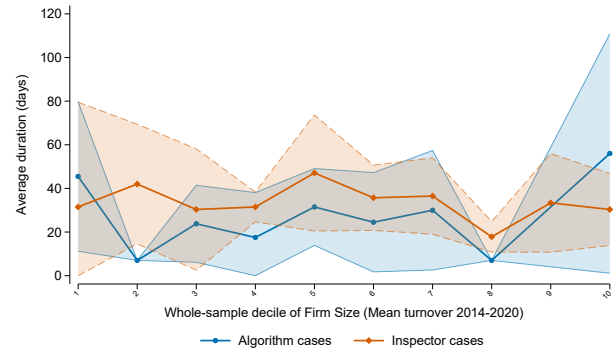
Figure 34: Average duration (taxpayer survey) by firm turnover bins (Executed-sample binning).

Panel a: Deciles

a.1 Desk audits

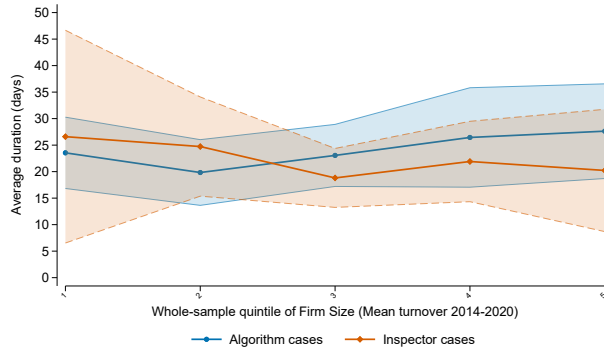


a.2 Full audits

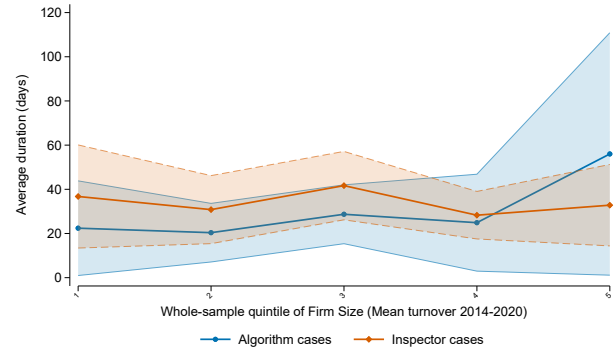


Panel b: Quintiles

b.1 Desk audits

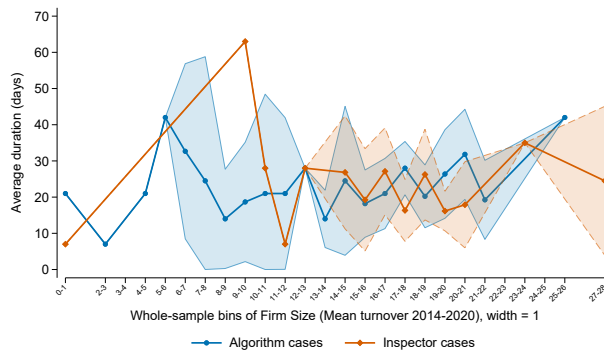


b.2 Full audits

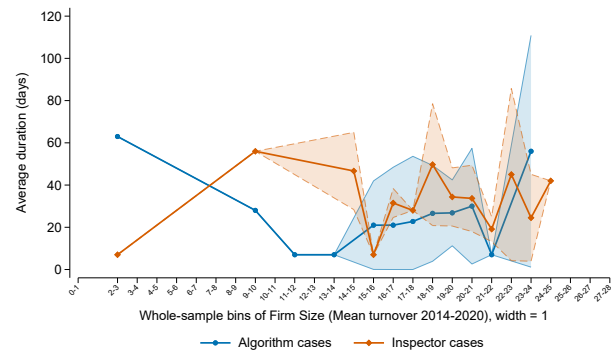


Panel c: Width 1

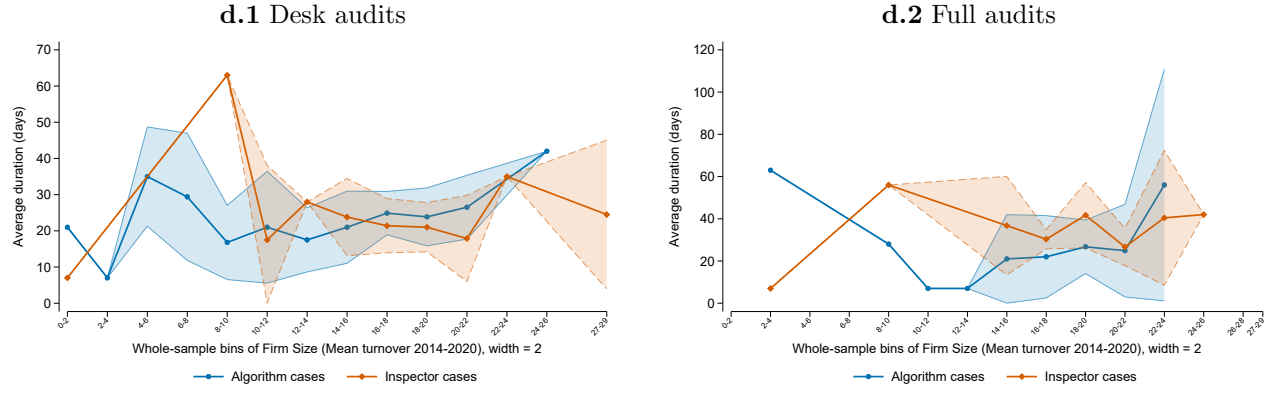
c.1 Desk audits



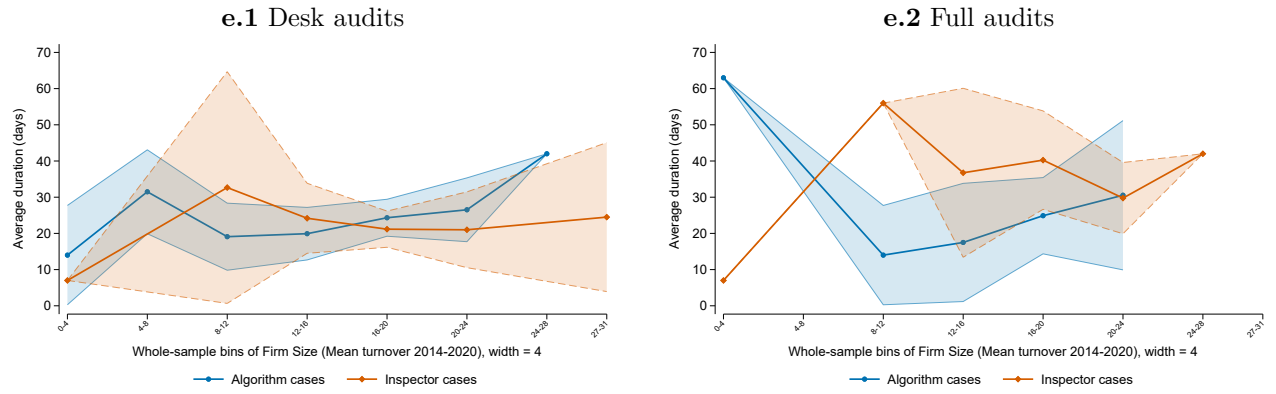
c.2 Full audits



Panel d: Width 2



Panel e: Width 4



Panel f: Width 5

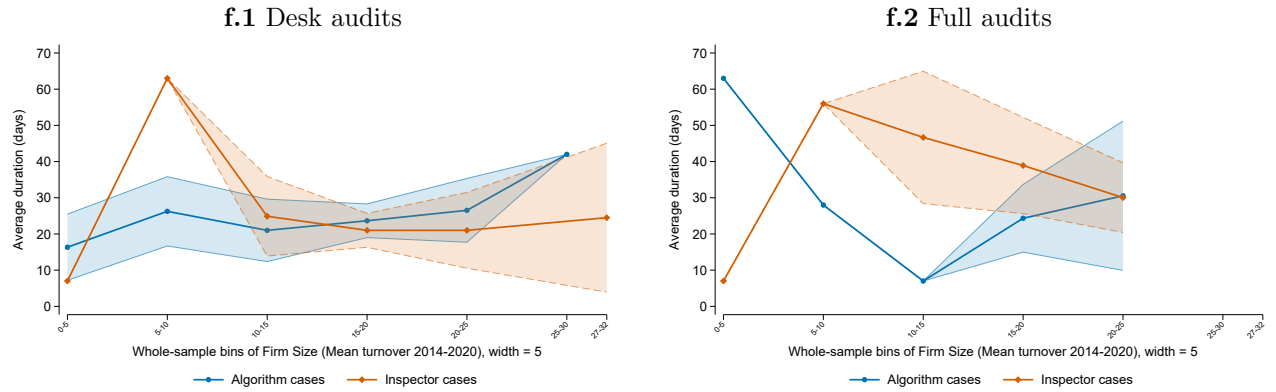
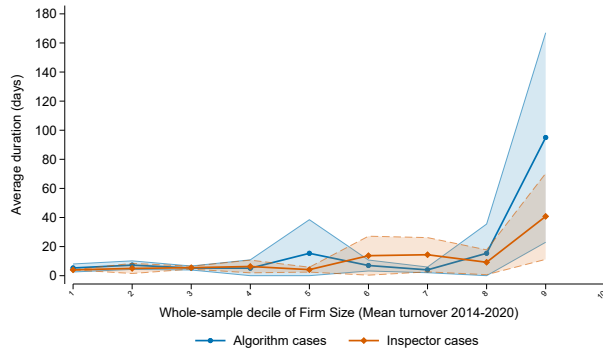


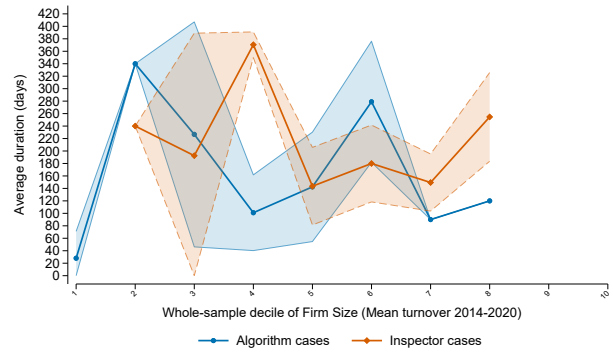
Figure 35: Average duration (self-reported) by firm turnover bins (Executed-sample binning).

Panel a: Deciles

a.1 Desk audits

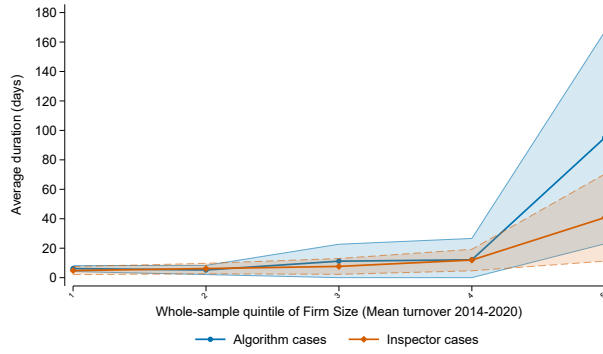


a.2 Full audits

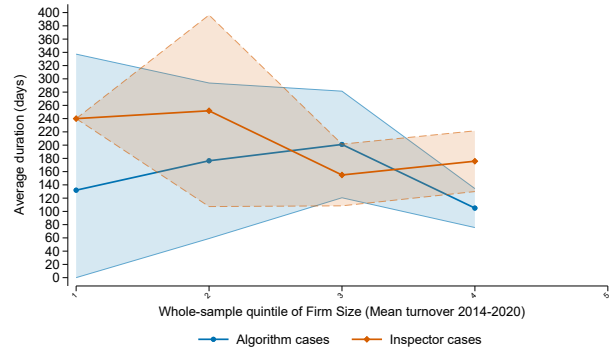


Panel b: Quintiles

b.1 Desk audits

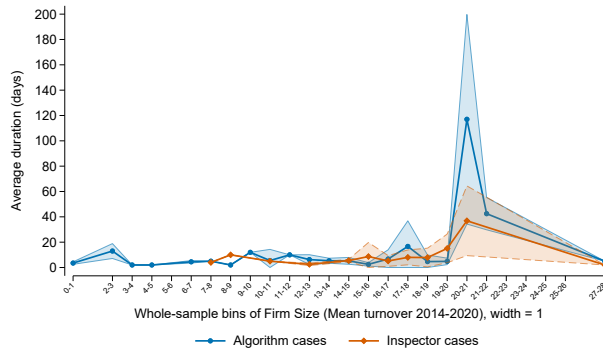


b.2 Full audits

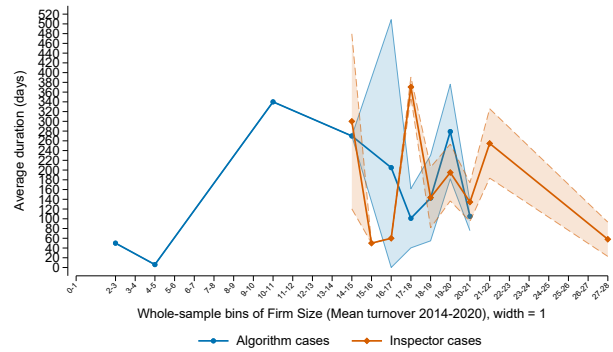


Panel c: Width 1

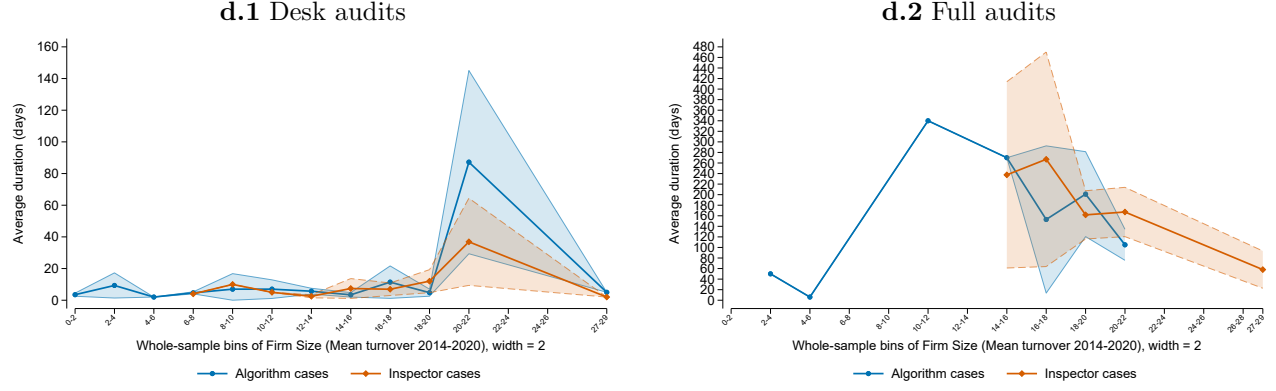
c.1 Desk audits



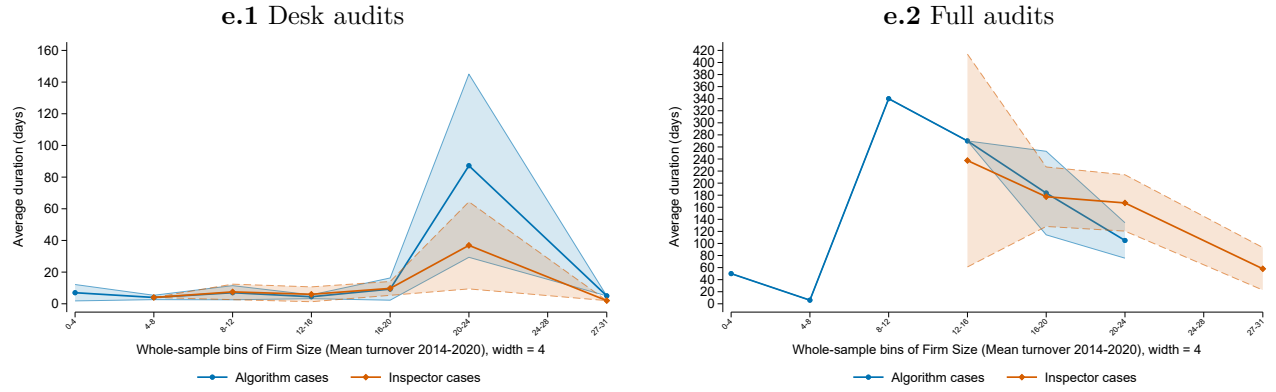
c.2 Full audits



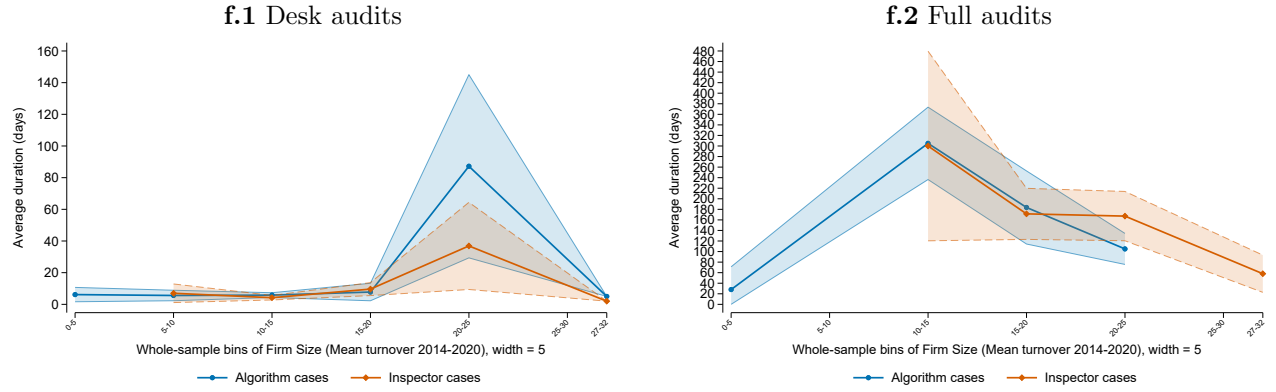
Panel d: Width 2



Panel e: Width 4



Panel f: Width 5

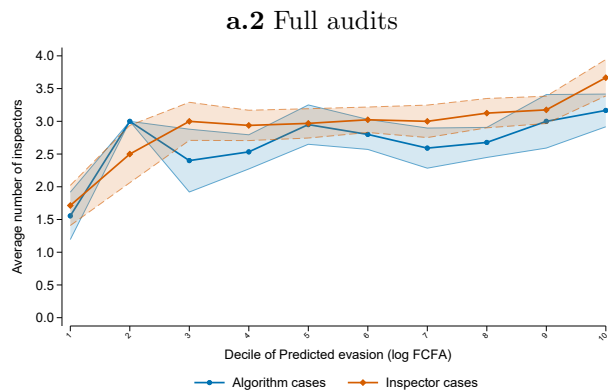
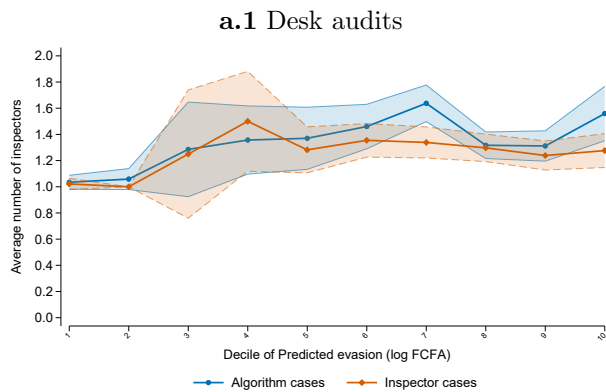


5.2 Within Algo/Inspector Estimated Bins

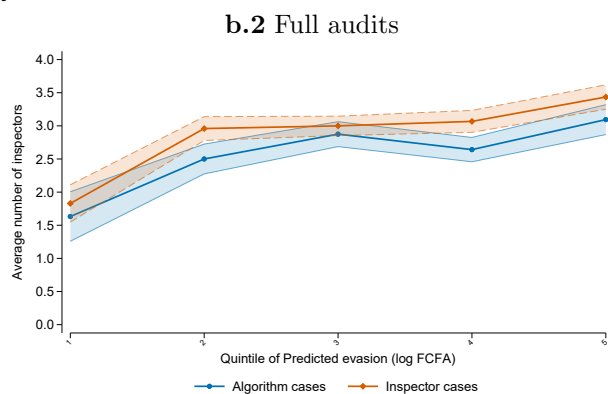
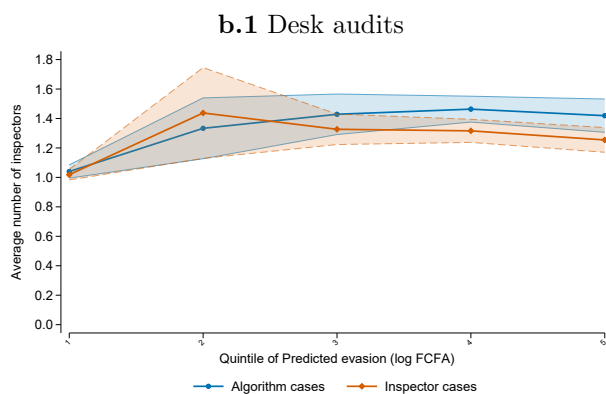
5.2.1 Predicted evasion

Figure 36: Average number of inspectors by predicted evasion bins (within algorithm/inspector estimated bins).

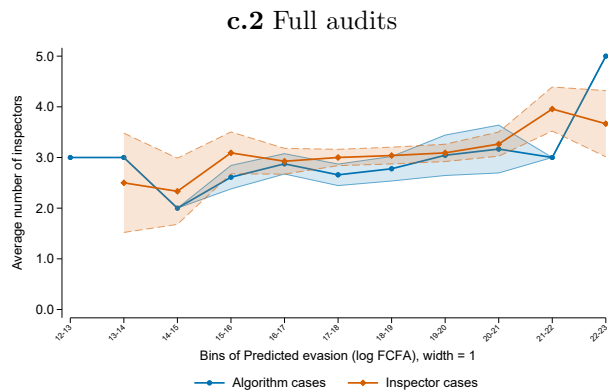
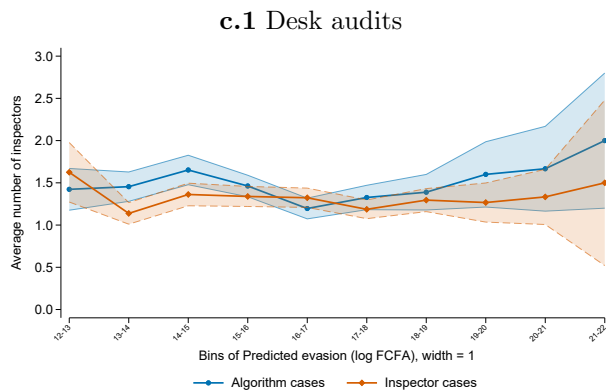
Panel a: Deciles



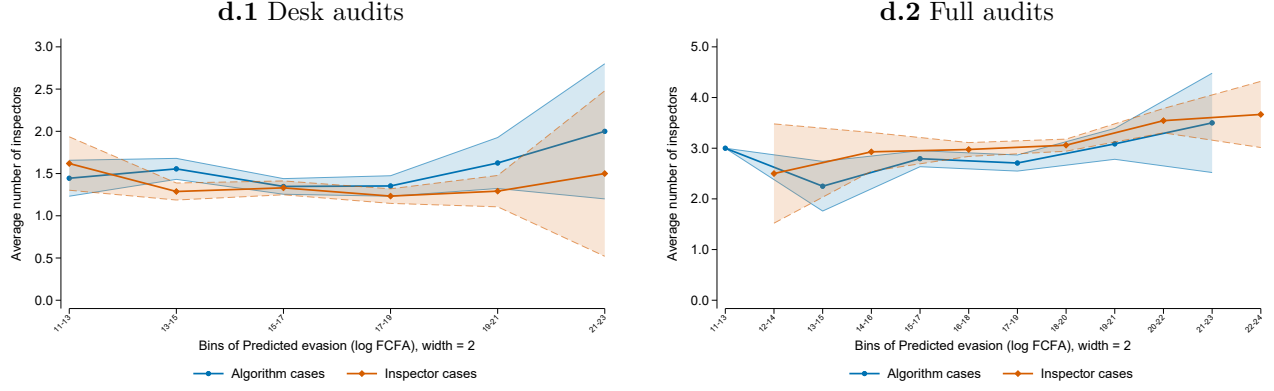
Panel b: Quintiles



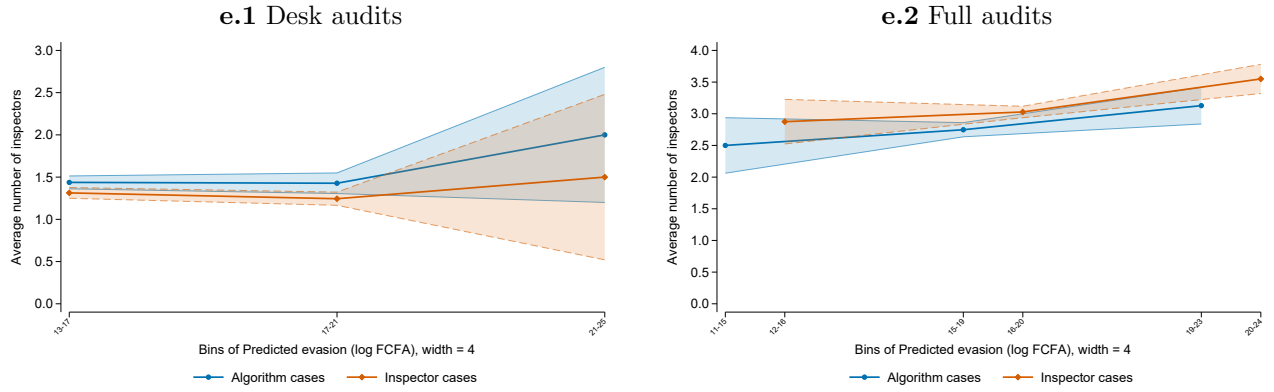
Panel c: Width 1



Panel d: Width 2



Panel e: Width 4



Panel f: Width 5

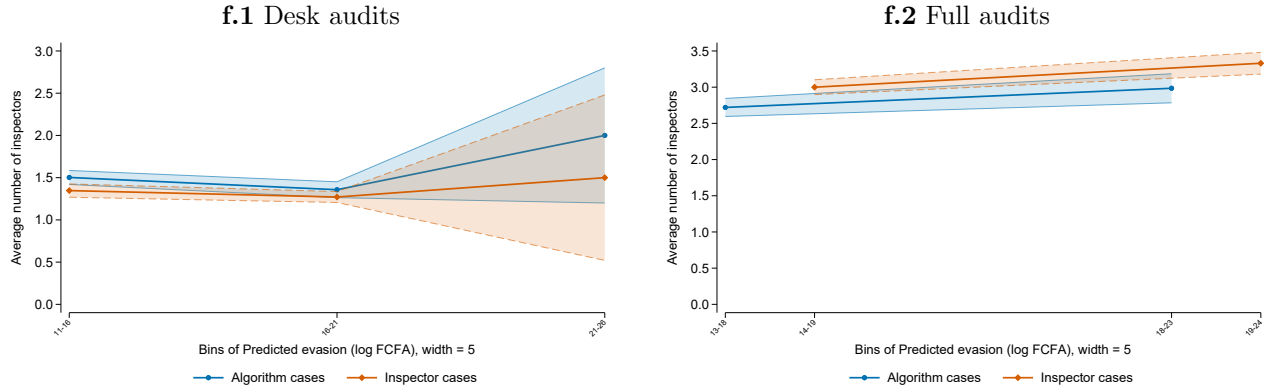
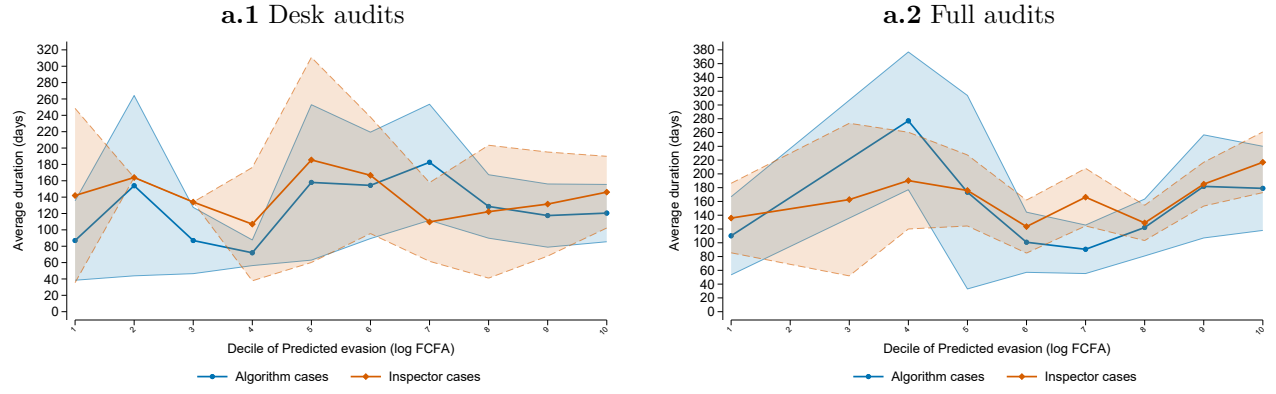
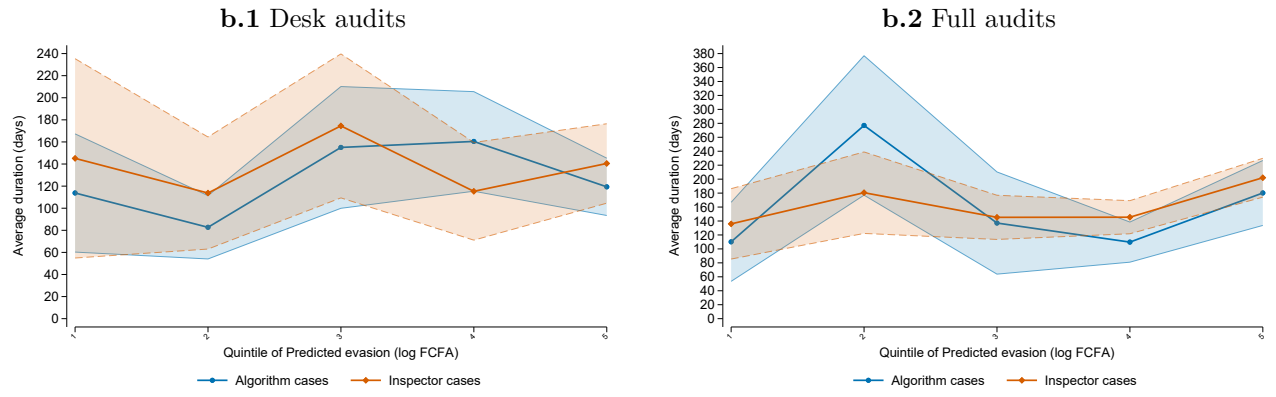


Figure 37: Average duration (admin data) by predicted evasion bins (within algorithm/inspector estimated bins).

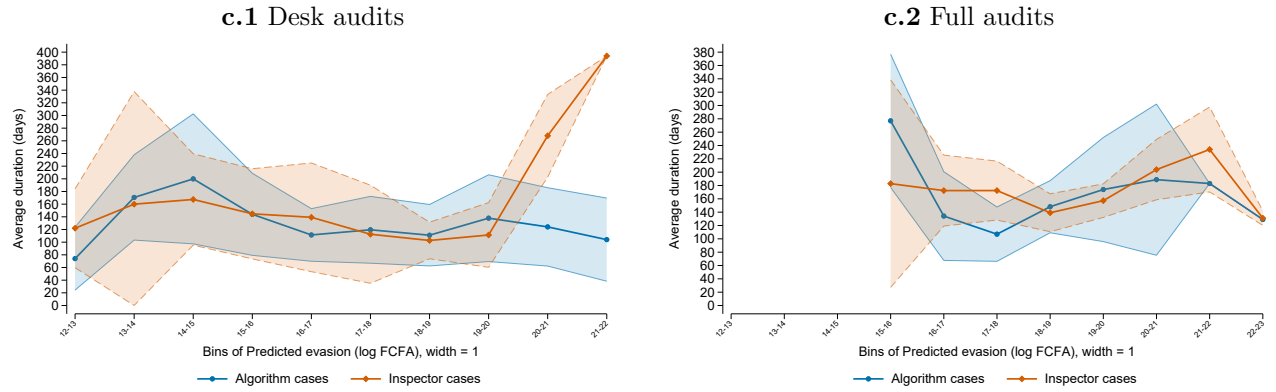
Panel a: Deciles



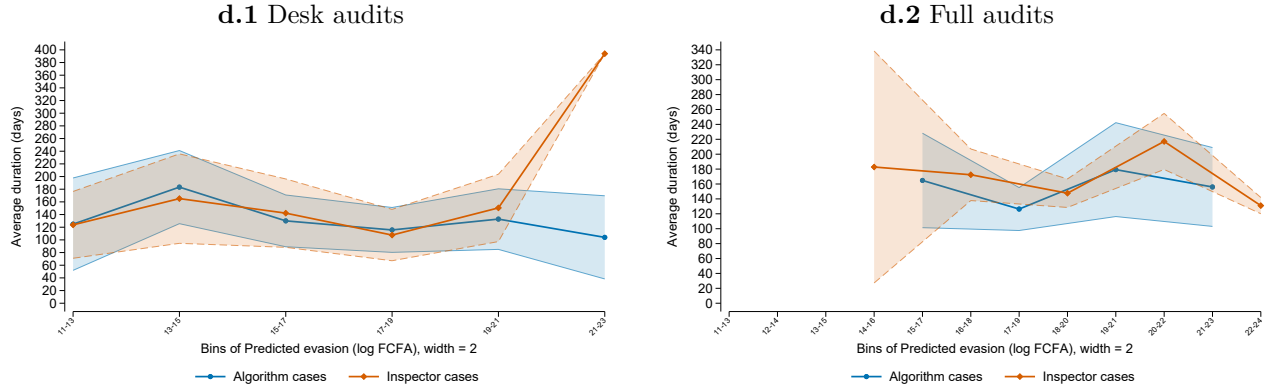
Panel b: Quintiles



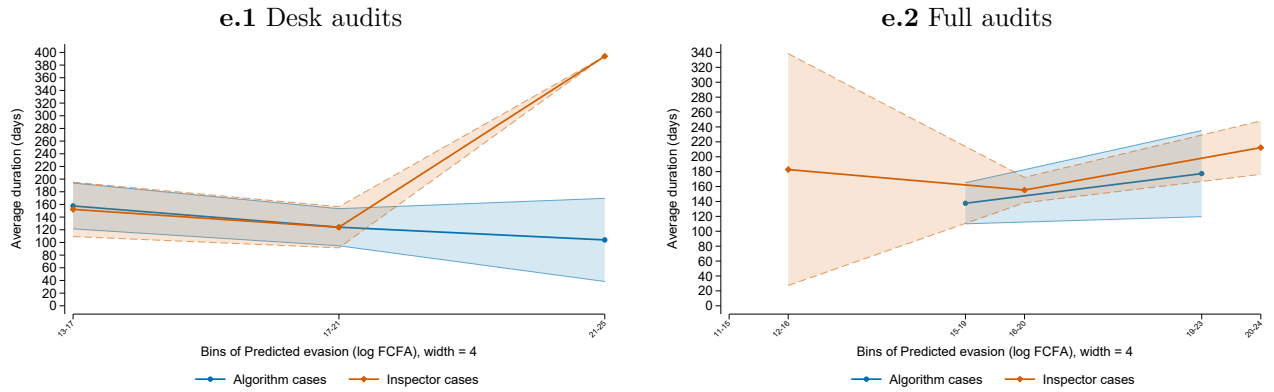
Panel c: Width 1



Panel d: Width 2



Panel e: Width 4



Panel f: Width 5

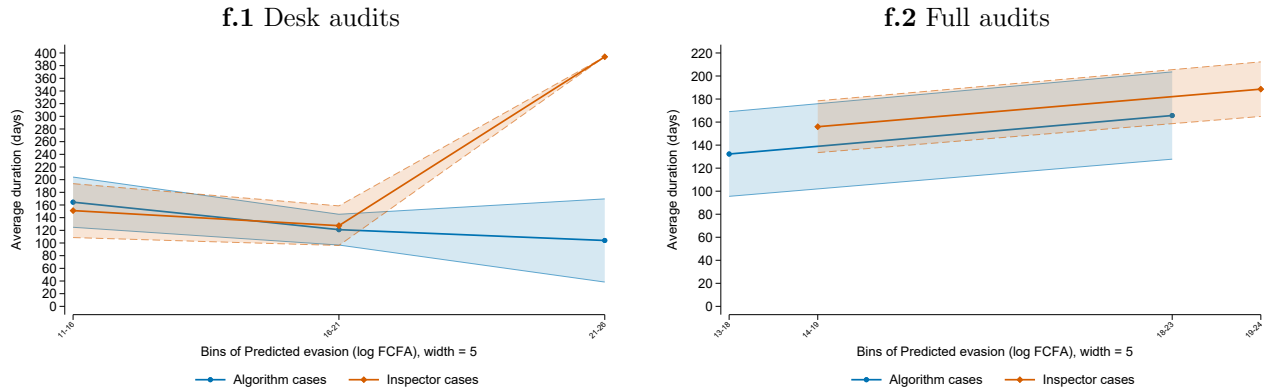
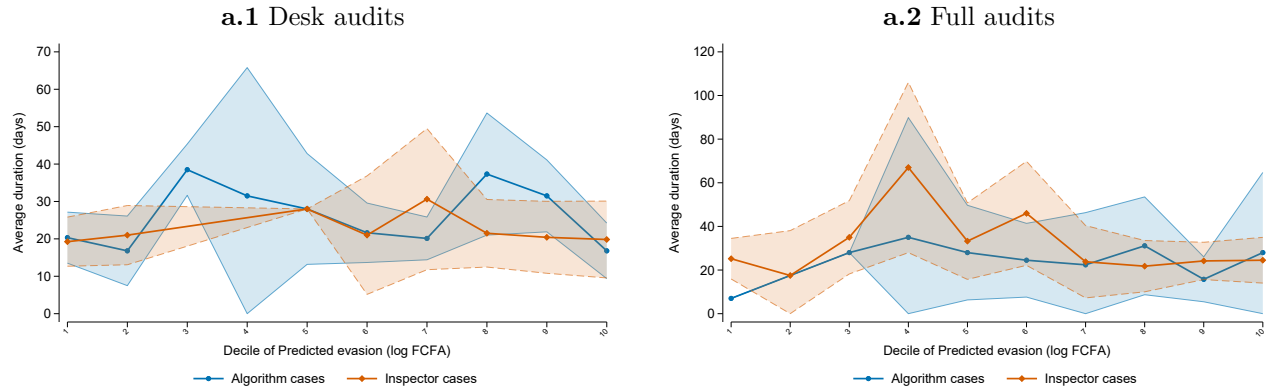
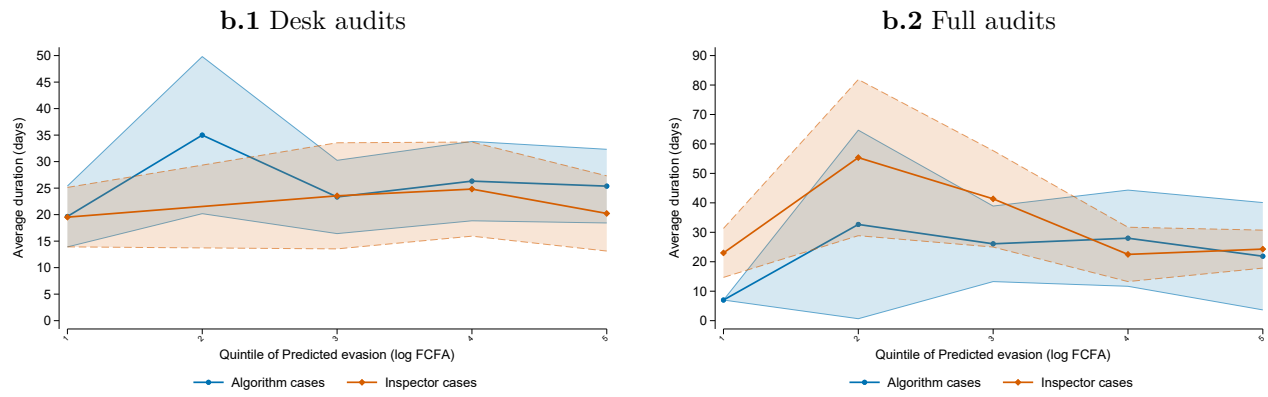


Figure 38: Average duration (taxpayer survey) by predicted evasion bins (within algorithm/inspector estimated bins).

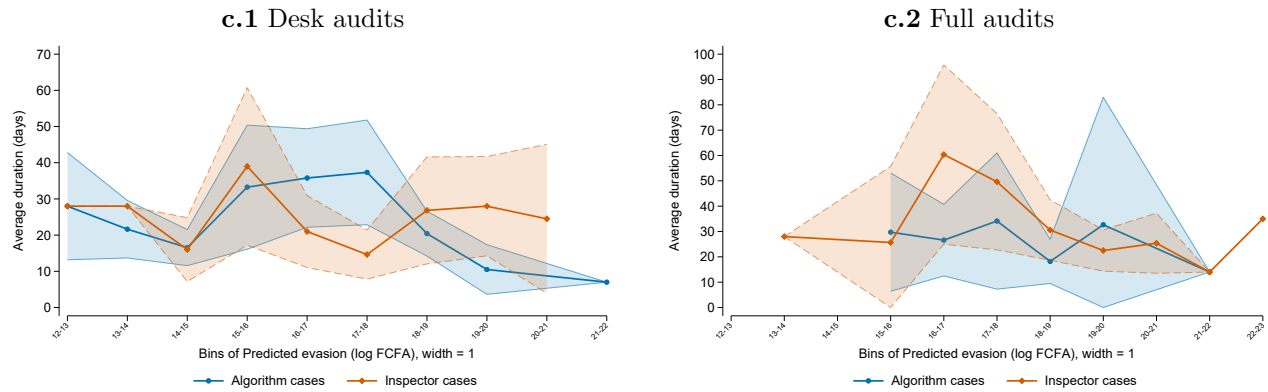
Panel a: Deciles



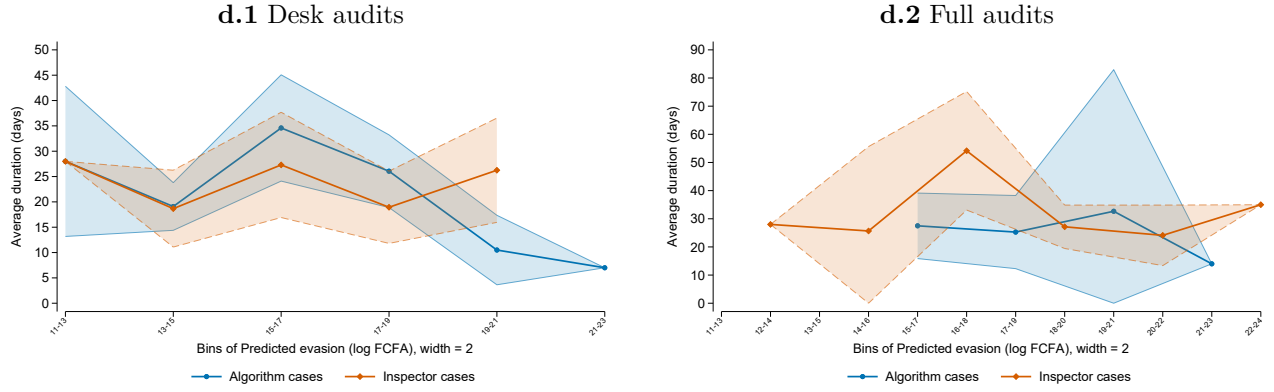
Panel b: Quintiles



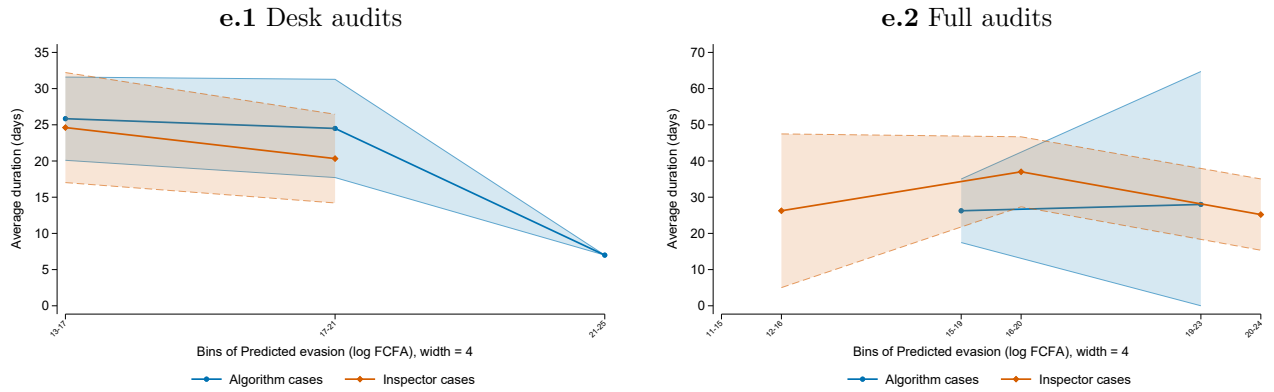
Panel c: Width 1



Panel d: Width 2



Panel e: Width 4



Panel f: Width 5

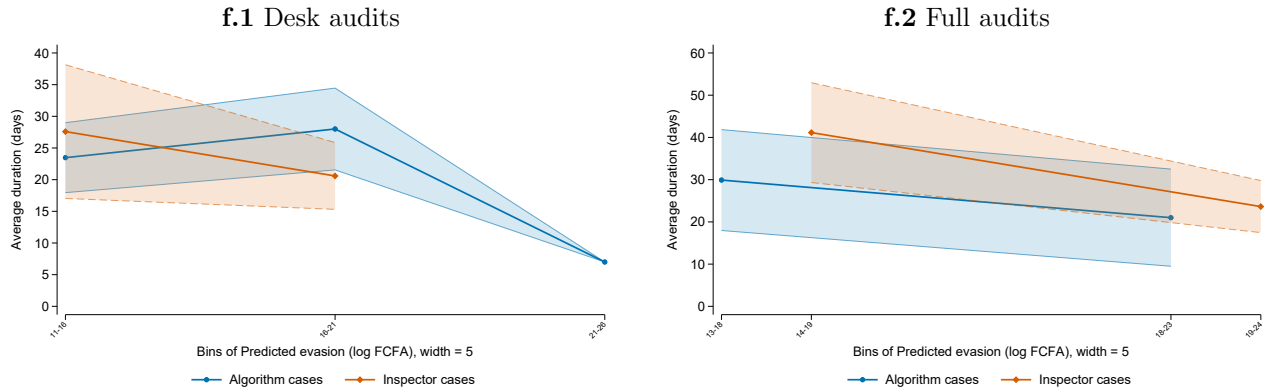
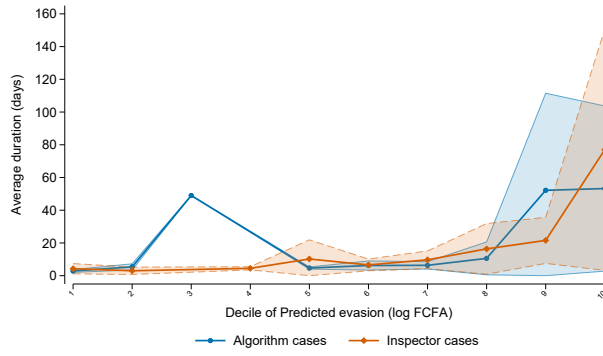


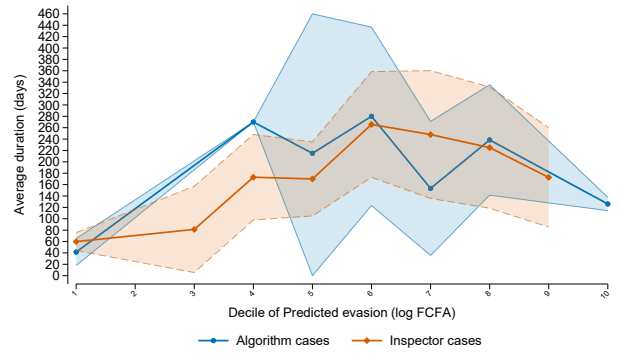
Figure 39: Average duration (self-reported) by predicted evasion bins (within algorithm/inspector estimated bins).

Panel a: Deciles

a.1 Desk audits

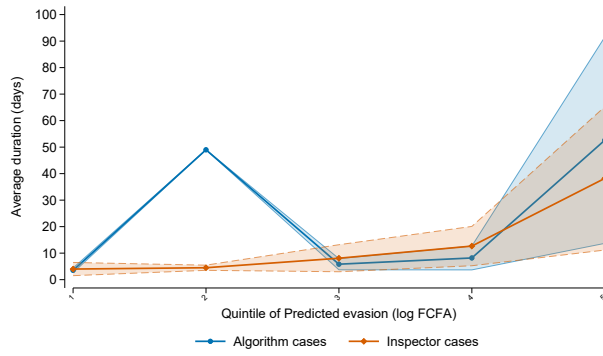


a.2 Full audits

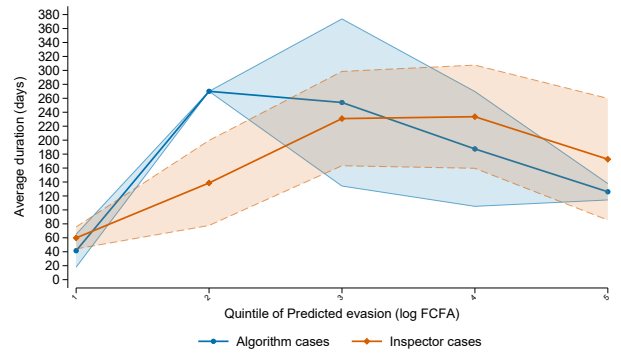


Panel b: Quintiles

b.1 Desk audits

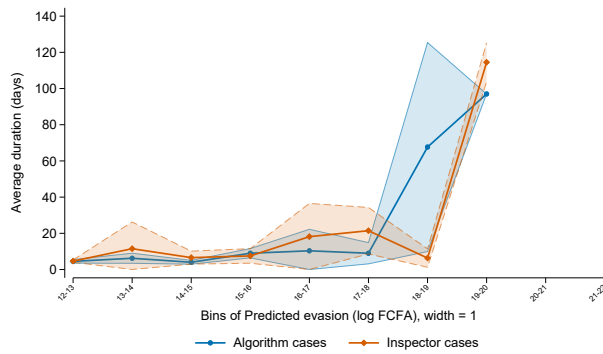


b.2 Full audits

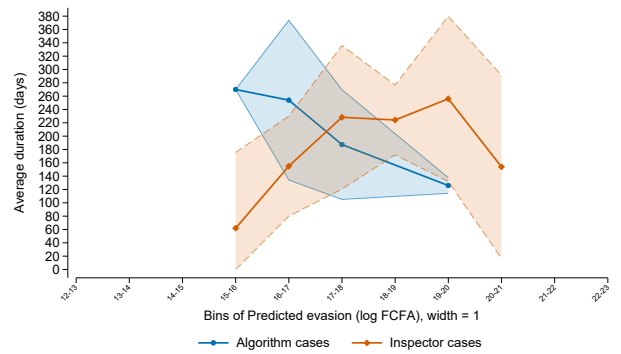


Panel c: Width 1

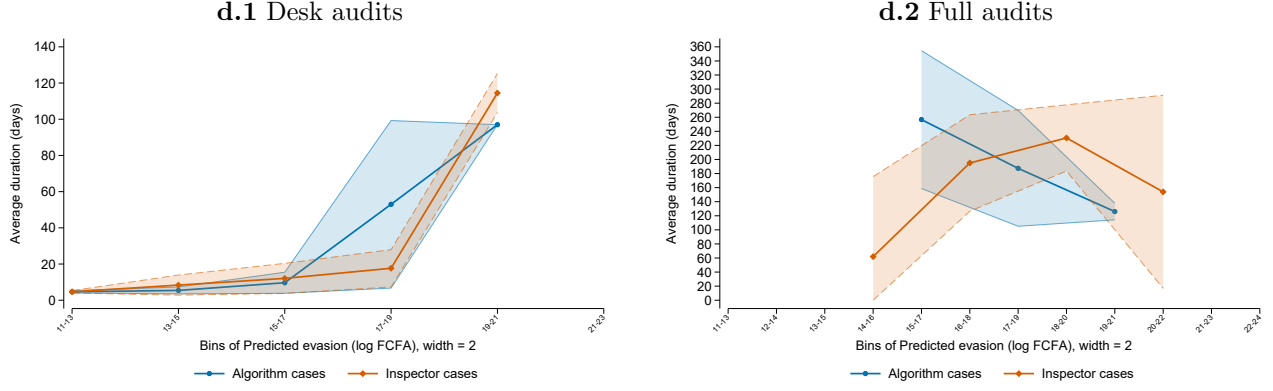
c.1 Desk audits



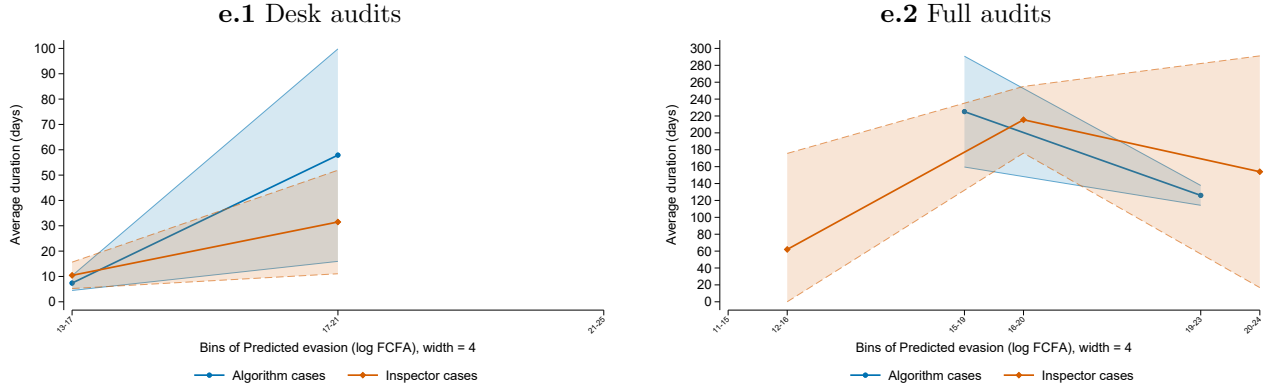
c.2 Full audits



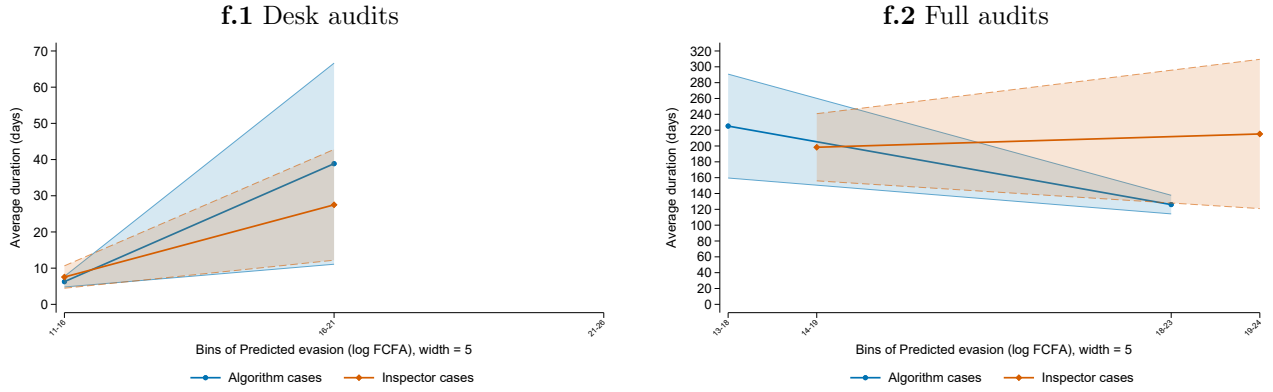
Panel d: Width 2



Panel e: Width 4



Panel f: Width 5

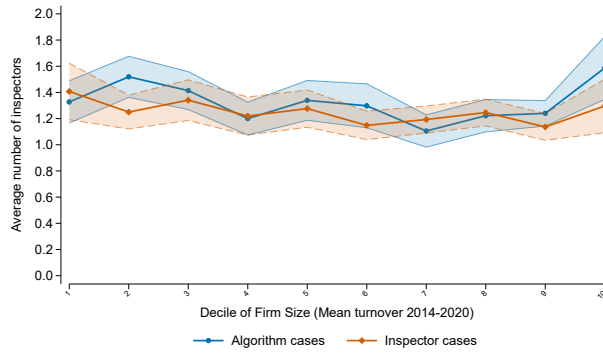


5.2.2 Firm turnover

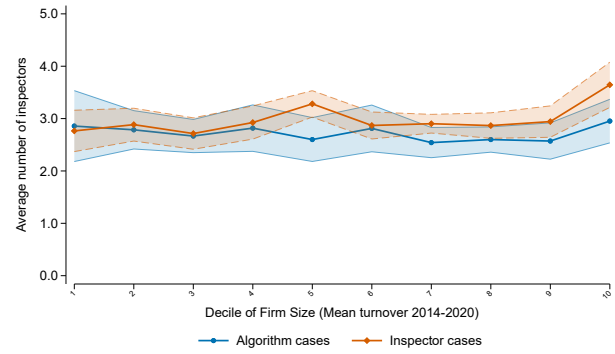
Figure 40: Average number of inspectors by firm turnover bins (within algorithm/inspector estimated bins).

Panel a: Deciles

a.1 Desk audits

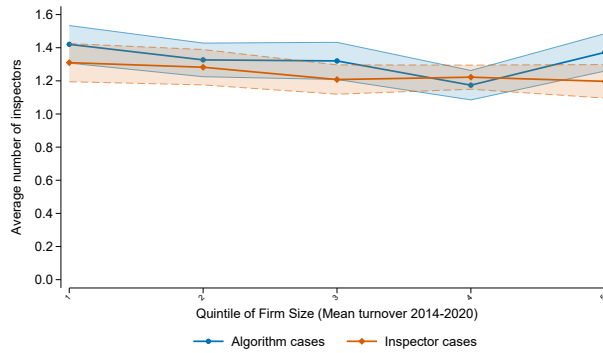


a.2 Full audits

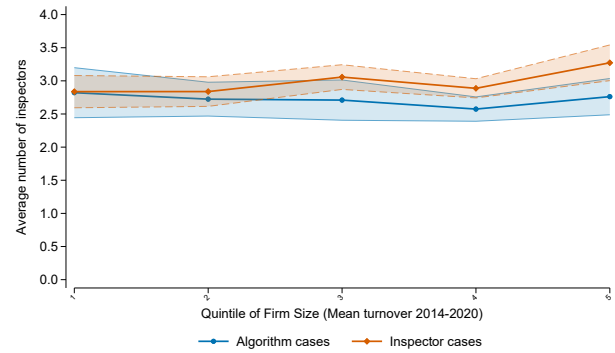


Panel b: Quintiles

b.1 Desk audits

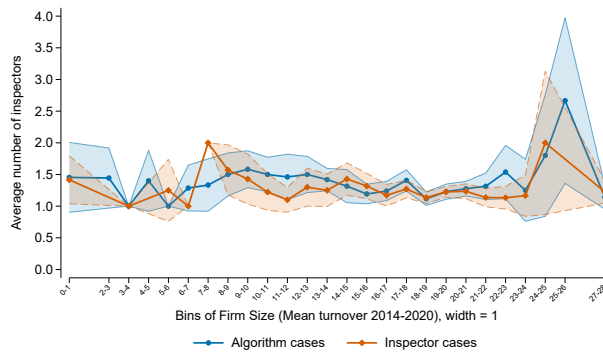


b.2 Full audits

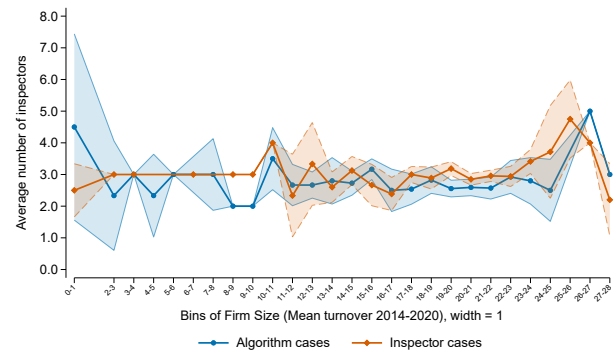


Panel c: Width 1

c.1 Desk audits



c.2 Full audits



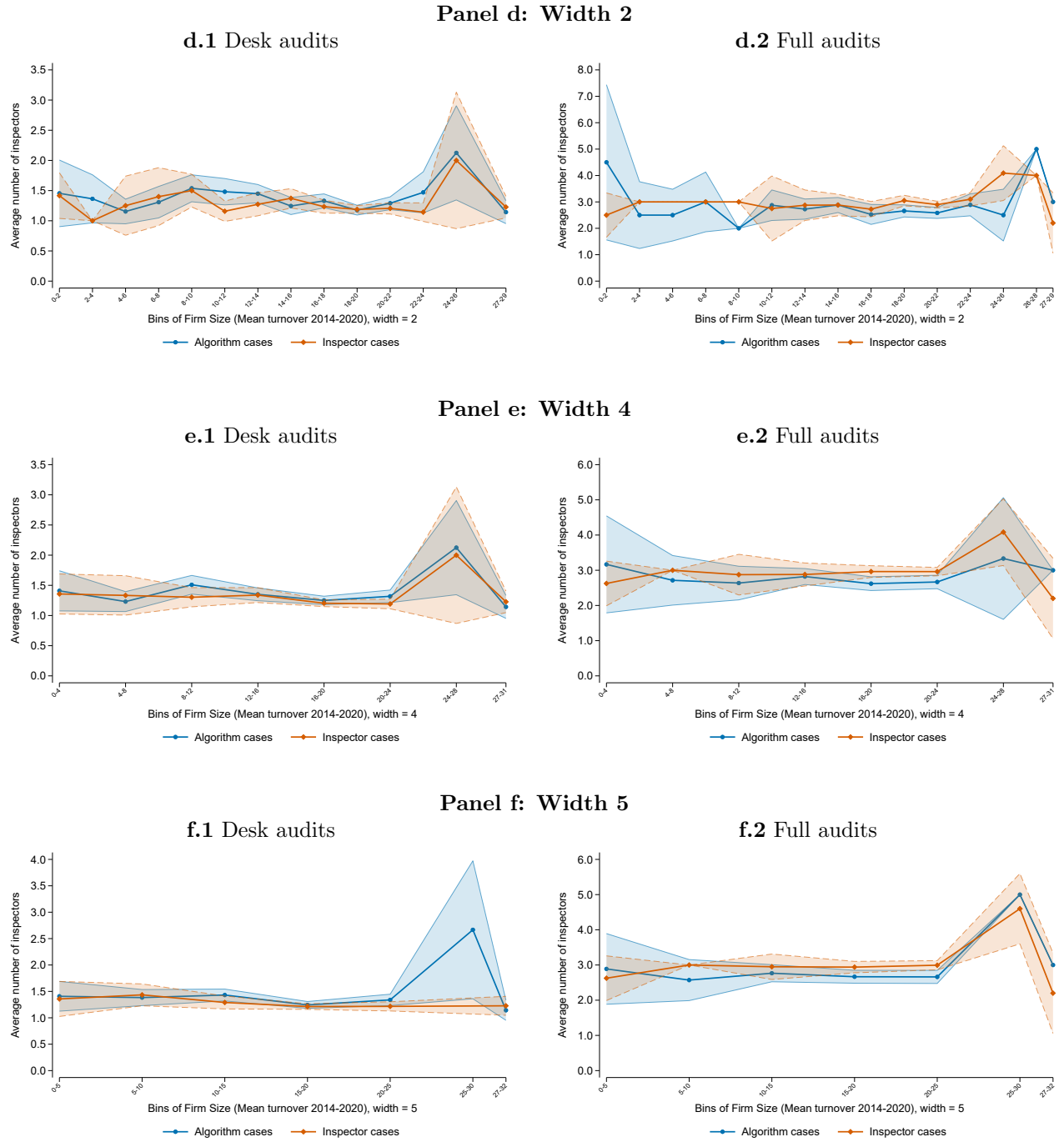
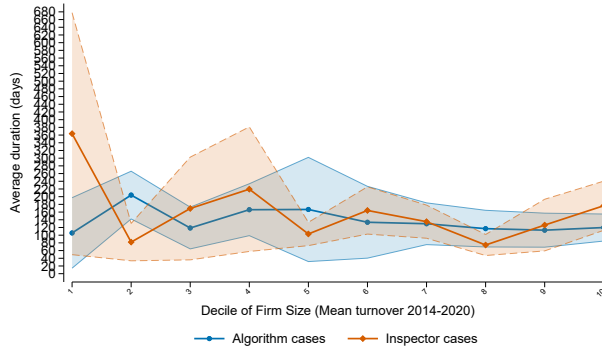


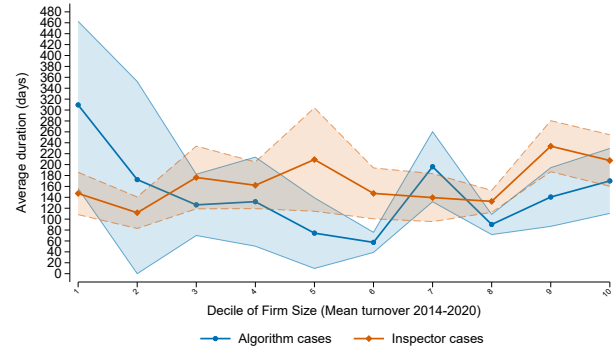
Figure 41: Average duration (admin data) by firm turnover bins (within algorithm/inspector estimated bins).

Panel a: Deciles

a.1 Desk audits

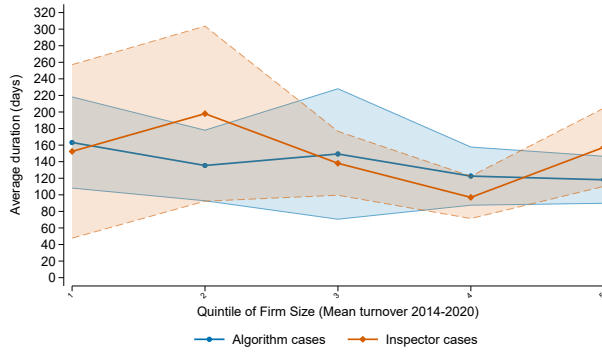


a.2 Full audits

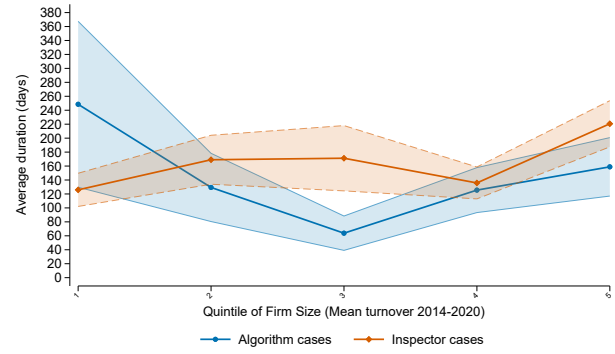


Panel b: Quintiles

b.1 Desk audits

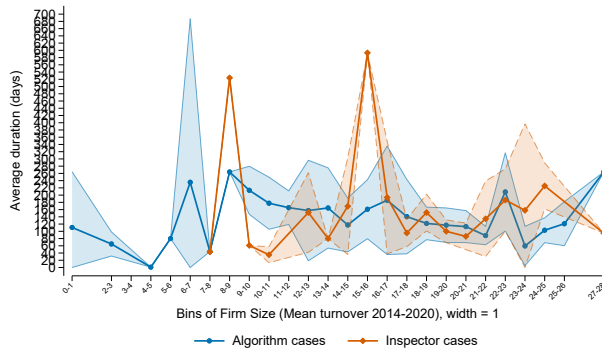


b.2 Full audits

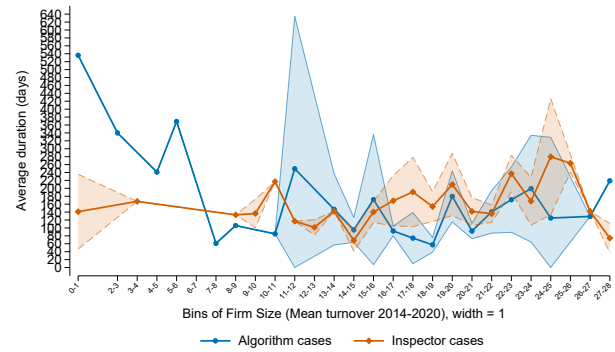


Panel c: Width 1

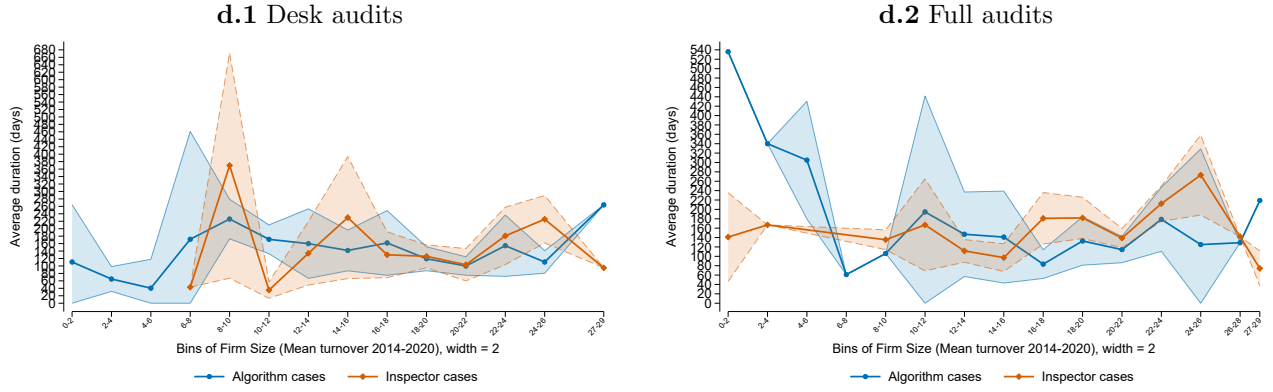
c.1 Desk audits



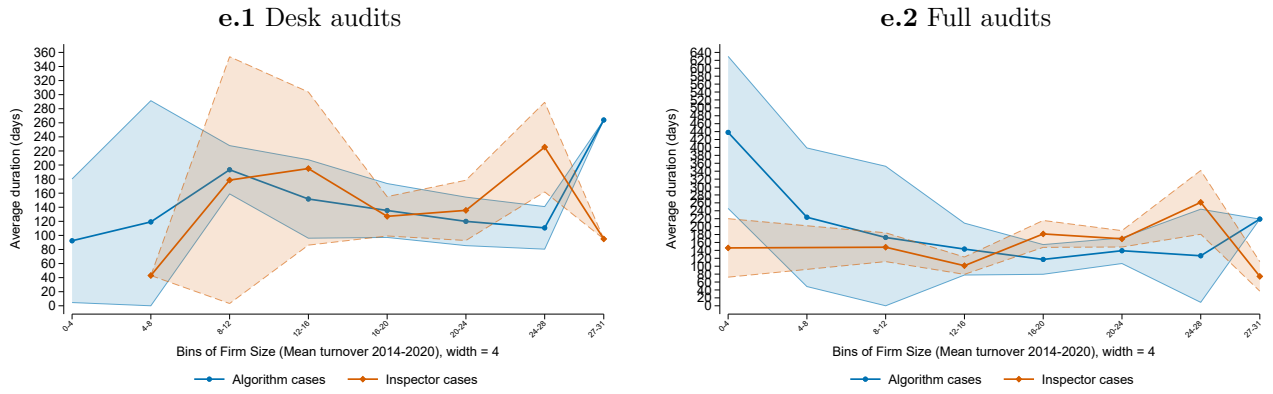
c.2 Full audits



Panel d: Width 2



Panel e: Width 4



Panel f: Width 5

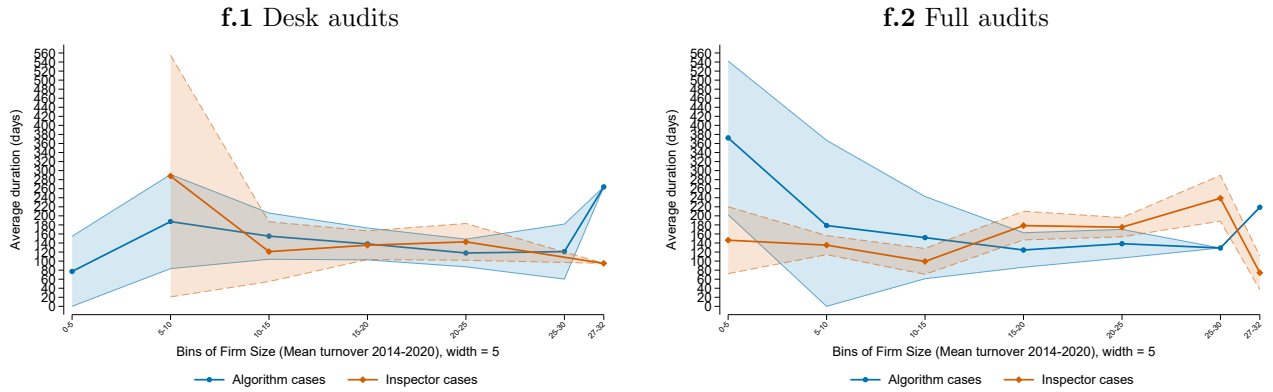
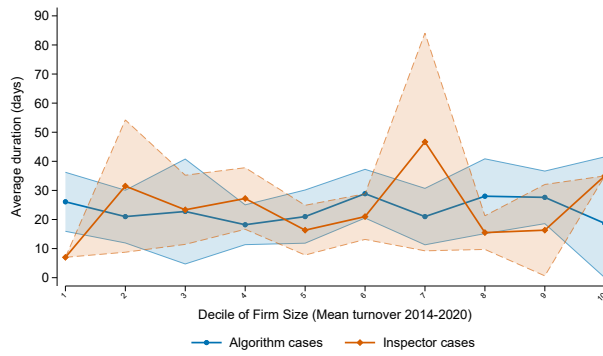


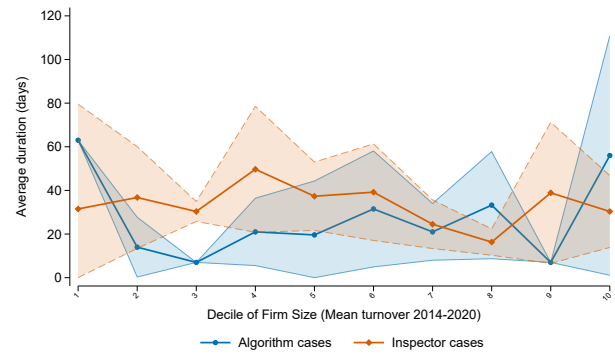
Figure 42: Average duration (taxpayer survey) by firm turnover bins (within algorithm/inspector estimated bins).

Panel a: Deciles

a.1 Desk audits

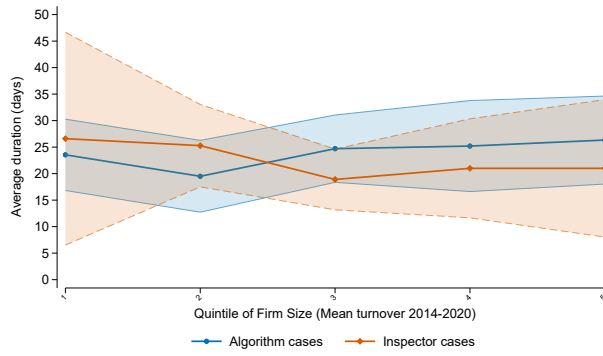


a.2 Full audits

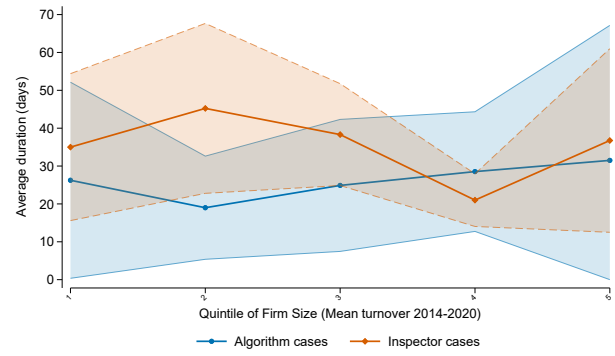


Panel b: Quintiles

b.1 Desk audits

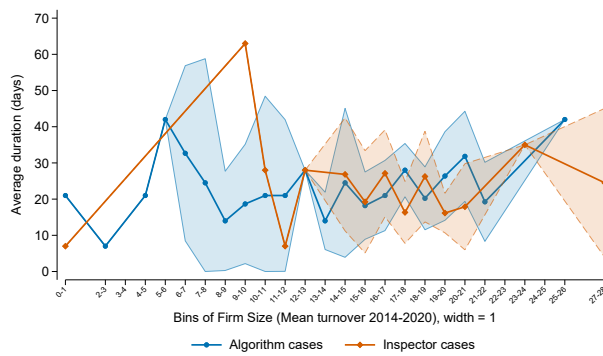


b.2 Full audits

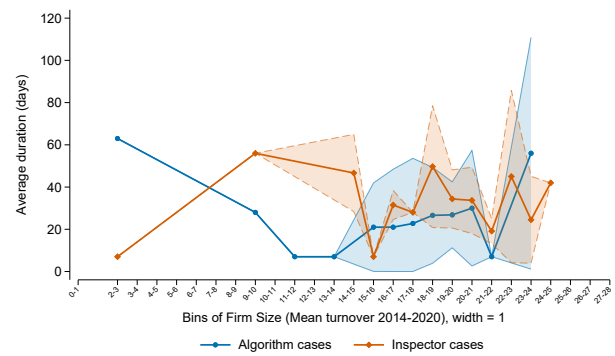


Panel c: Width 1

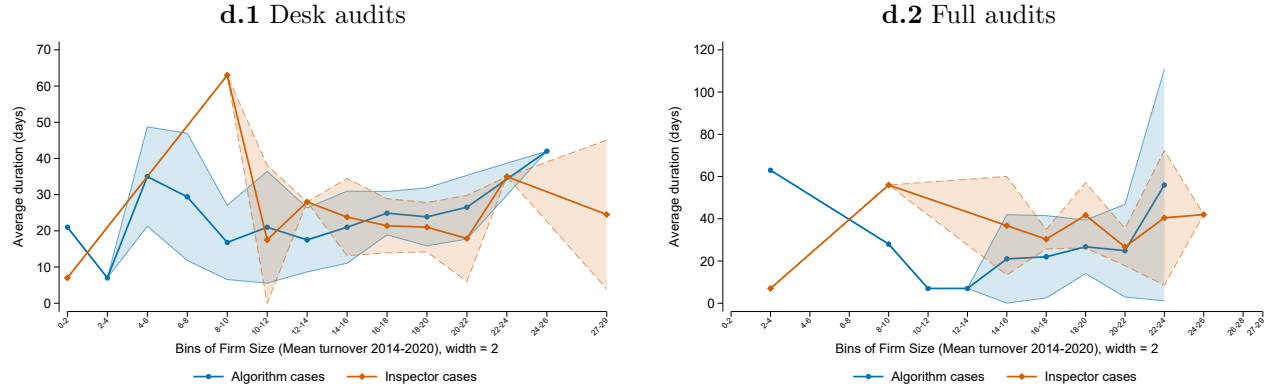
c.1 Desk audits



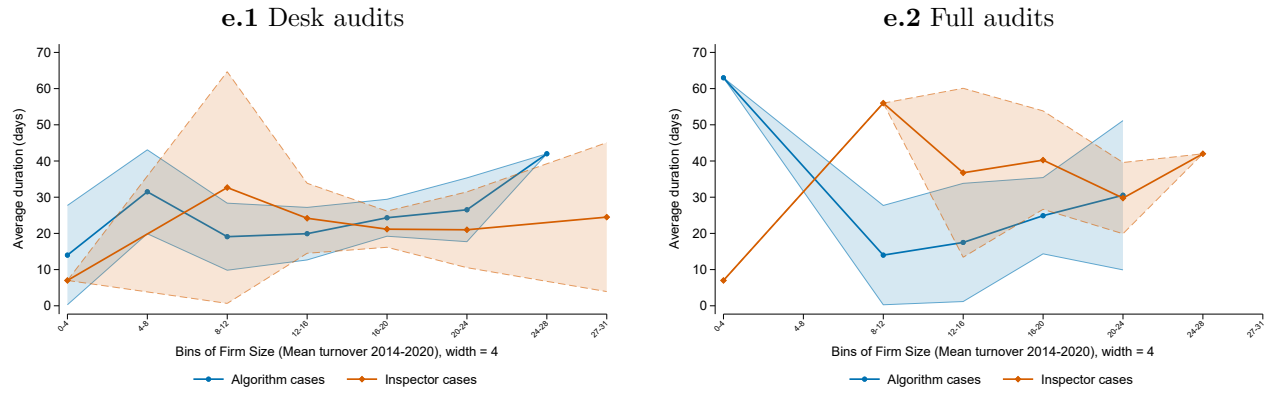
c.2 Full audits



Panel d: Width 2



Panel e: Width 4



Panel f: Width 5

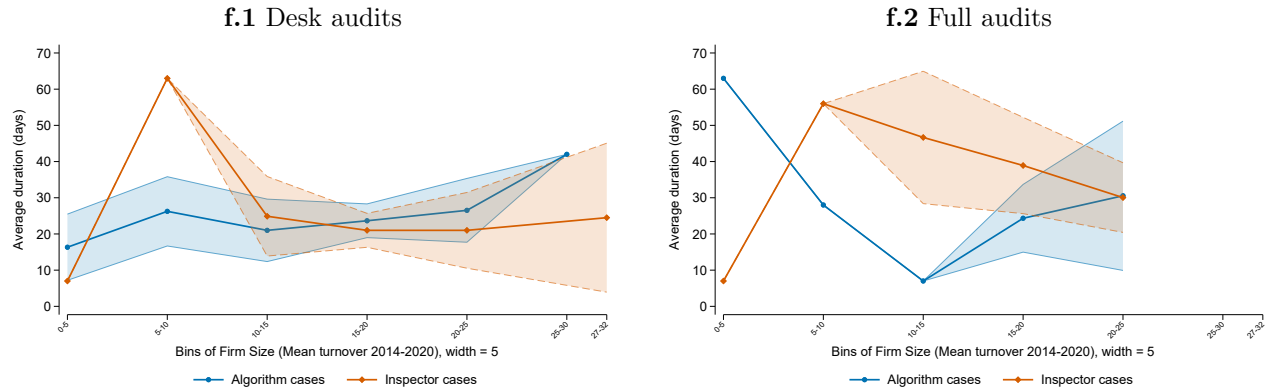
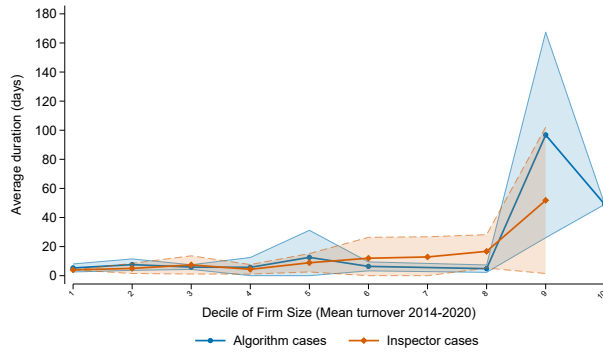


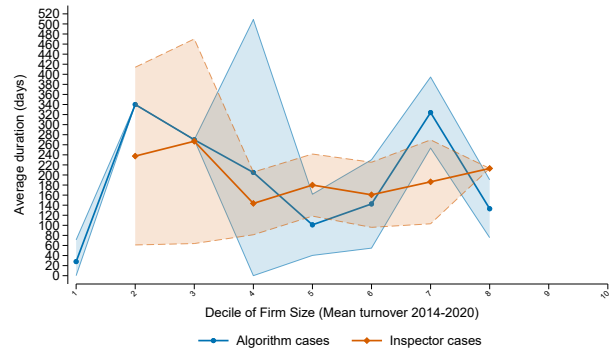
Figure 43: Average duration (self-reported) by firm turnover bins (within algorithm/inspector estimated bins).

Panel a: Deciles

a.1 Desk audits

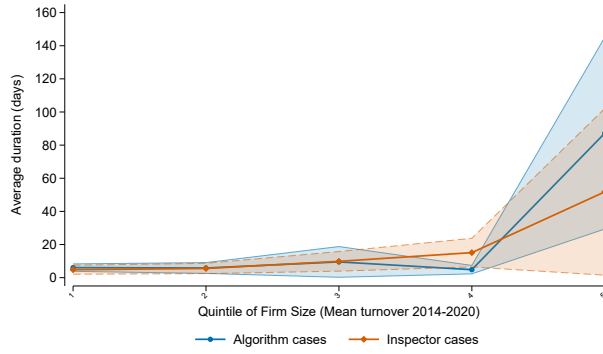


a.2 Full audits

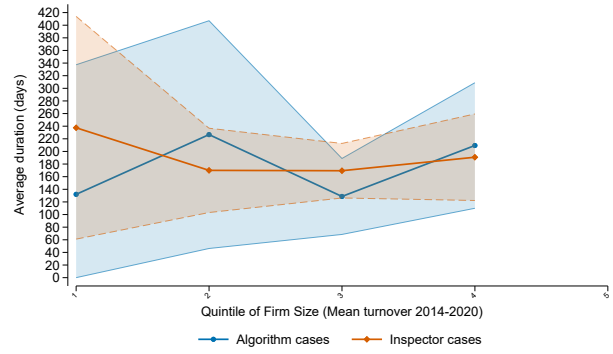


Panel b: Quintiles

b.1 Desk audits

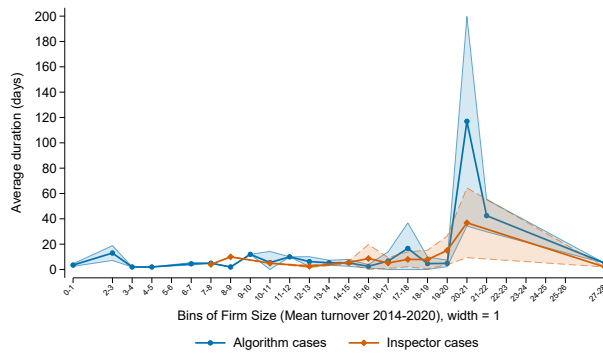


b.2 Full audits

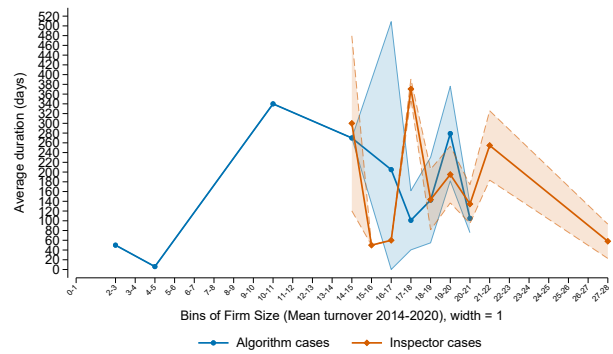


Panel c: Width 1

c.1 Desk audits

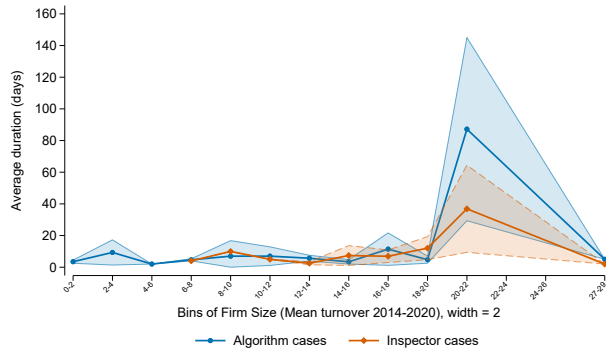


c.2 Full audits

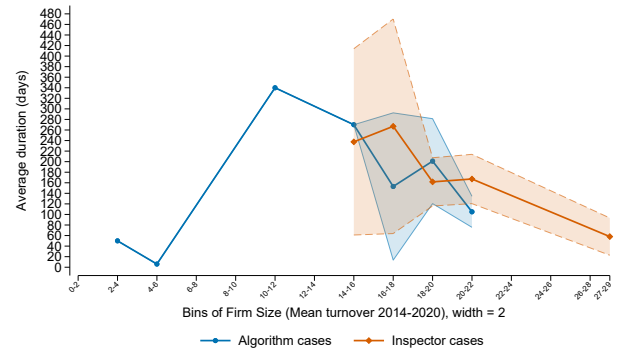


Panel d: Width 2

d.1 Desk audits

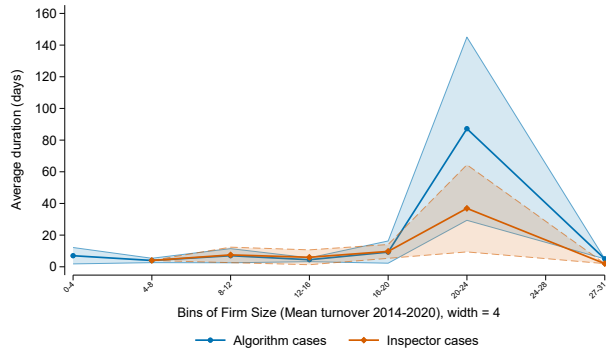


d.2 Full audits

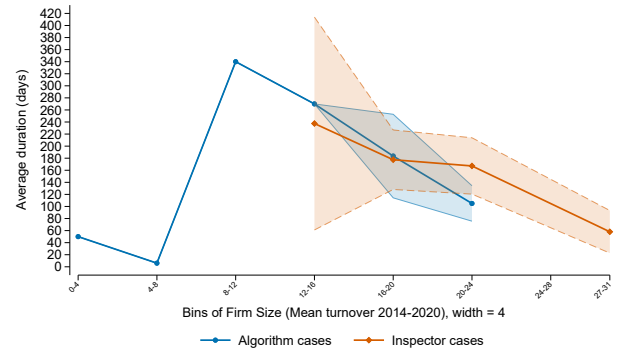


Panel e: Width 4

e.1 Desk audits

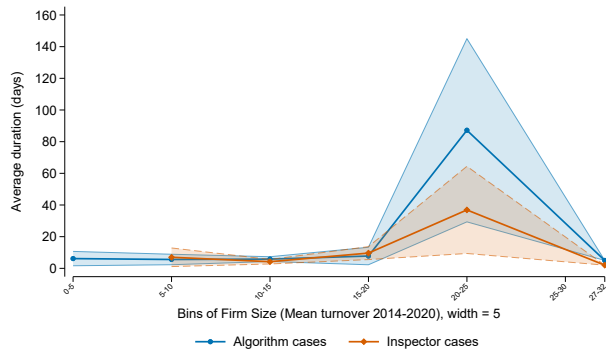


e.2 Full audits

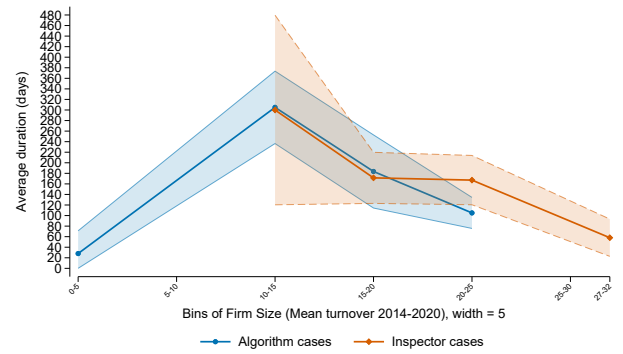


Panel f: Width 5

f.1 Desk audits



f.2 Full audits



5.3 More on inspector characteristics performance and firm size

Table 36: Inspector Characteristics and firm size, Desk Audits

	All				Algorithm cases				Inspector cases			
	log(Turnover) (Mean)	log(N. Employees) (Mean)	log(Firm age) (Mean)	Risk score (Mean)	log(Turnover) (Mean)	log(N. Employees) (Mean)	log(Firm age) (Mean)	Risk score (Mean)	log(Turnover) (Mean)	log(N. Employees) (Mean)	log(Firm age) (Mean)	Risk score (Mean)
Masters/PhD	-0.06 (0.29)	0.14 (0.28)	0.03 (0.04)	0.05 (0.08)	-0.37 (0.28)	0.27 (0.31)	0.10 (0.09)	-0.01 (0.12)	0.23 (0.65)	-0.11 (0.45)	-0.01 (0.06)	0.15 (0.09)
Above median age	-0.49 (0.35)	0.03 (0.21)	0.02 (0.04)	-0.02 (0.07)	-0.96* (0.49)	-0.09 (0.32)	0.01 (0.06)	0.09 (0.13)	0.04 (0.49)	0.48 (0.31)	0.06 (0.07)	-0.08 (0.09)
Above median experience	0.40 (0.41)	0.15 (0.19)	0.02 (0.04)	-0.06 (0.06)	0.48 (0.54)	0.25 (0.31)	0.12** (0.06)	-0.07 (0.13)	0.09 (0.51)	0.01 (0.28)	-0.07 (0.06)	-0.10 (0.09)
N	86.00	86.00	86.00	86.00	86.00	86.00	86.00	86.00	86.00	86.00	86.00	86.00
R2	0.63	0.61	0.05	0.78	0.43	0.50	0.11	0.70	0.66	0.42	0.10	0.54
Mean outcome	20.07	4.43	1.09	-0.28	20.38	4.43	1.10	0.19	19.83	4.61	1.07	-0.64

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: * 0.10 ** 0.05 *** 0.01 significance levels. This table shows the estimation of the OLS regressions of outcomes of inspectors on inspector characteristics. The analysis is done at the list (inspector x year) level and only includes the lists of desk audits. These lists were assigned individually to inspectors. The regressors are three individual characteristics of the inspectors collected using the inspector survey (whether the inspector had a Masters/PhD and whether they had higher than median experience working at the tax authority) and administrative data (age). Column 1's outcome is the total started cases for each list, Columns 2-4 are averages within list of the main outcomes, Column 5 uses the number of started algorithm cases, Column 6 the share of started algorithm cases among algorithm-selected cases, Column 7 uses an indicator of whether the number of started algorithm cases was larger than the number of started inspector cases in the list, and Column 8 uses the response of the inspector to whether they would favor the automation of case selection using an algorithm. The regressions control for fixed effects at the tax office-year level. Robust standard errors are shown in parentheses (Huber-White formula).

Table 37: Inspector Characteristics and Performance, Full Audits [Executed]

	All					Algorithm cases					Inspector cases				
	N Cases (Unw. Avg.)	log(Turnover) (W. Avg.)	log(Employees) (W. Avg.)	log(Firm age) (W. Avg.)	Risk score (W. Avg.)	N Cases (Unw. Avg.)	log(Turnover) (W. Avg.)	log(Employees) (W. Avg.)	log(Firm age) (W. Avg.)	Risk score (W. Avg.)	N Cases (Unw. Avg.)	log(Turnover) (W. Avg.)	log(Employees) (W. Avg.)	log(Firm age) (W. Avg.)	Risk score (W. Avg.)
Masters/PhD	-0.67 (1.51)	0.46 (0.49)	0.10 (0.31)	-0.12* (0.07)	0.04 (0.28)	0.37 (0.83)	0.08 (0.51)	-0.23 (0.42)	-0.07 (0.13)	0.03 (0.28)	-0.37 (0.99)	0.41 (0.57)	0.32 (0.35)	-0.11* (0.07)	0.07 (0.17)
Above median age	3.51*** (1.17)	0.57 (0.39)	0.06 (0.22)	0.03 (0.06)	0.05 (0.18)	1.71** (0.66)	0.02 (0.58)	-0.02 (0.37)	-0.15 (0.12)	0.14 (0.19)	2.20*** (0.77)	0.66 (0.47)	0.01 (0.27)	0.07 (0.06)	0.08 (0.16)
Above median experience	0.94 (0.80)	0.33 (0.50)	0.34 (0.21)	-0.07 (0.05)	-0.00 (0.18)	0.42 (0.47)	0.40 (0.57)	0.31 (0.36)	-0.07 (0.11)	-0.26 (0.21)	0.65 (0.57)	0.39 (0.54)	0.37 (0.24)	-0.04 (0.05)	-0.02 (0.15)
N	152.00	150.00	150.00	152.00	152.00	99.00	98.00	97.00	99.00	99.00	140.00	138.00	138.00	140.00	140.00
R2	0.26	0.28	0.42	0.12	0.46	0.28	0.28	0.48	0.15	0.64	0.24	0.27	0.25	0.11	0.53
Mean outcome	5.53	20.84	5.11	1.12	0.21	2.71	20.78	4.61	1.03	1.92	4.09	21.01	5.35	1.14	-0.59

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: * 0.10 ** 0.05 *** 0.01 significance levels. This table shows the estimation of OLS regressions of outcomes on inspector characteristics. The analysis is done at the inspector-year level and focuses on full audits. Unlike desk audits, full audits may involve teams: for team cases, we allocate each executed case equally across participating inspectors, assigning weight $1/k$ to each inspector, where k is the number of inspectors recorded on the case. Column 1's outcome is the team-adjusted number of executed cases for the inspector-year, Column 2 is the team-adjusted number of executed algorithm cases, Column 3 is the share of executed algorithm cases among executed cases (constructed from the team-adjusted counts), Column 4 is an indicator for whether the share of executed algorithm cases exceeds 0.5, Columns 5-6 report averages among executed cases of the main outcomes (detection and log evasion), and Columns 7-10 report averages among executed cases of firm characteristics (log turnover, log number of employees, log firm age, and the risk score). The regressors are individual characteristics of inspectors collected using the inspector survey (e.g., whether the inspector had a Masters/PhD and whether they had higher than median experience working at the tax authority) and administrative data (age). The regressions control for fixed effects at the tax office-year level. Robust standard errors are shown in parentheses (Huber-White formula).

6 Audit Discrepancies and taxpayer dissatisfaction with the audit process.

This section studies whether larger discrepancies between notified and confirmed audit amounts are associated with taxpayer-reported efficiency and corruption perceptions.

6.1 Current empirical steps (updated main specification)

1. Start from `datasetforanalysis.dta`.
2. Keep the broad analysis sample by keeping safeties and non-selected, and account for them in the regression controls.
3. Recreate cluster groups used in fixed effects:

```
capture drop clusterid
egen clusterid = group(inspectionyear)
```

4. Recreate Table C5 dispute variables (`d1–d21`) exactly as in the main analysis script.
5. Construct the baseline discrepancy regressor:

$$W_{\text{main}} = \frac{\text{notificationvalue} - \text{confirmationvalue}}{\text{notificationvalue}}$$

6. Cap negative baseline values to zero:

```
replace W_main = 0 if W_main < 0 & W_main != .
```

7. Build robustness discrepancy regressors:

```
W_nonzero = 1{W_main != 0}
W_above_med = 1{W_main > median(W_main)}
W_topquart = 1{W_main >= p75(W_main)}
```

where $\text{median}(W_{\text{main}})$ and $p75(W_{\text{main}})$ are single pooled moments from non-missing W_{main} in the regression-eligible sample.

8. In exported regression tables, coefficient labels are: *Downward revision (% of notification)*, *Downward revision (% of notification) $\neq 0$* , *Downward revision (% of notification) $> \text{median}$* , and *Downward revision (% of notification) in top quartile*.
9. Build panel indicators:

```
selfreported_audit = 1 if q28a in [2018,2021] or q29a in [2018,2021]
```

```
Panel A uses q1  $\neq$  . AND selfreported = 1; Panel B uses q1  $\neq$  ..
```

10. Build index outcomes as in taxpayer-survey code:

```
index_efficiency from q31 q33 q41
index_corruption from q32_inverted q42 q34 (swindex)
```

11. Estimate Table 6 / D7-style columns by audit type (full, desk, all), using selection method and the following controls:

```
W_main algorithm overlap random safeties horsprogramme
```

Index outcomes use `a(selectionyear center)`; question outcomes (`q34 q35 q32 q42`) use `a(inspectionyear)`. The same backbone is used for the three robustness W variants.

6.2 Regression equation and notation

Our empirical exercise studies whether larger discrepancies between the notified and confirmed audit amounts are associated with taxpayer-reported perceptions of the audit process. Concretely, for each taxpayer-survey outcome, we estimate the following model standard OLS regression:

$$y_{ig} = \beta_0 + \beta_1 W_i + \beta_2 \text{Algorithm}_i + \beta_3 \text{Overlap}_i + \beta_4 \text{Random}_i + \beta_5 \text{Safeties}_i + \beta_6 \text{Horsprogramme}_i + \gamma_g + \varepsilon_{ig},$$

where y_{ig} is the taxpayer-survey outcome for firm-audit observation i , and γ_g denotes the relevant fixed effects. The unit of observation is a firm-audit observation in the merged administrative audit and taxpayer-survey data, so a given firm may appear more than once if it is associated with multiple audit cases. Our main discrepancy regressor is

$$W_i = \max \left\{ \frac{\text{Notification}_i - \text{Confirmation}_i}{\text{Notification}_i}, 0 \right\},$$

defined for observations with strictly positive notified and confirmed amounts and an executed audit. Thus, $W_i = 0$ both when there is no positive discrepancy and when the confirmed amount is at least as large as the notified amount.

The indicators Algorithm_i , Overlap_i , Random_i , Safeties_i , and Horsprogramme_i capture the audit-case classification. The dummy for random selection is applicable only to desk-audit specifications. The coefficient of interest is β_1 , which captures the association between the capped discrepancy measure and the taxpayer-reported outcome.

We estimate this equation separately by panel and audit type. Panel A restricts the sample to firms that report having been audited recently,

$$\text{q1} != . \quad \& \quad \text{selfreported_audit} == 1,$$

while Panel B includes all surveyed firms with non-missing q1 ,

$$\text{q1} != ..$$

For audit type, we estimate the specification separately for full audits ($\text{x2} == 1$), desk audits ($\text{x2} == 0$), and the pooled sample with no restriction on x2 .

The outcomes are the taxpayer-survey indices

$$\text{index_efficiency}, \quad \text{index_corruption},$$

and the individual survey questions

$$\text{q34}, \quad \text{q35}, \quad \text{q32}, \quad \text{q42}.$$

For index outcomes, the fixed effects γ_g correspond to $\text{selectionyear} \times \text{center}$. For question-level outcomes, γ_g corresponds to $\text{inspectorclusteryear}$. We use robust standard errors to perform inference.

In robustness checks, we also replace W_i with indicator versions:

$$W_i^{\neq 0} = \mathbf{1}\{W_i \neq 0\}, \quad W_i^{>\text{med}} = \mathbf{1}\{W_i > \widetilde{W}\}, \quad W_i^{\geq q75} = \mathbf{1}\{W_i \geq \widetilde{W}_{75}\},$$

where $\widetilde{W}$ is the pooled median and $\widetilde{W}_{75}$ is the pooled 75th percentile of non-missing W_i in the relevant regression sample.

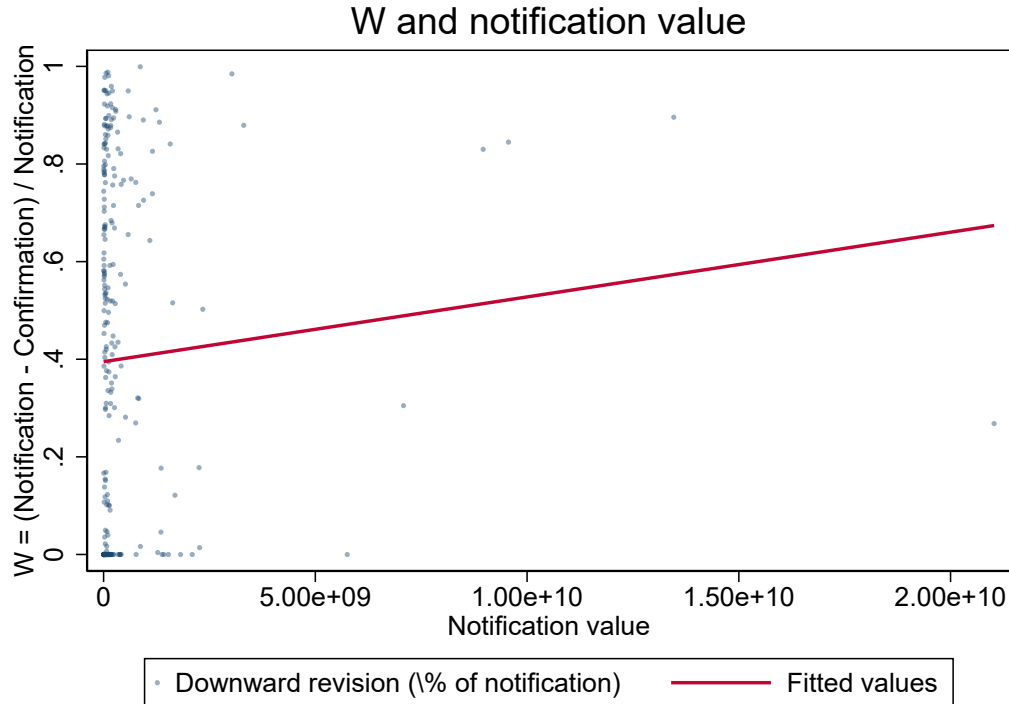
6.3 Coverage and W diagnostics

6.4 Main regression tables (current outputs)

Table 38: Informative regression-sample coverage by panel and audit type (counts and % of base)

	Panel A: Recent audit			Panel B: All surveyed		
	Full audits	Desk audits	All audits	Full audits	Desk audits	All audits
Eligible surveyed firms (q1 observed)	257	696	953	299	859	1158
Executed cases (y2=1)	136	394	530	156	477	633
W-eligible by construction (d4=1,d5=1,y2=1)	73	153	226	83	182	265
Non-missing capped W	73	153	226	83	182	265
Non-missing capped W + method controls	73	153	226	83	182	265
Downward revision (% of notification) $\neq 0$	52	106	158	59	124	183
Downward revision (% of notification) > pooled median	38	76	114	43	89	132
Downward revision (% of notification) in top quartile	16	43	59	17	50	67
Regression sample: Efficiency index	73	149	222	82	174	256
Regression sample: Corruption index	72	149	221	81	175	256
Regression sample: Corruption in General (q34)	53	99	152	58	116	174
Regression sample: Corruption Experience (q35)	73	152	225	83	180	263
Regression sample: Grade on Honesty (q32)	68	127	195	75	141	216
Regression sample: Friend at tax Auth. (q42)	65	133	198	70	154	224
Eligible surveyed firms (q1 observed), % of base	100.0	100.0	100.0	100.0	100.0	100.0
Executed cases (y2=1), % of base	52.9	56.6	55.6	52.2	55.5	54.7
W-eligible by construction, % of base	28.4	22.0	23.7	27.8	21.2	22.9
Non-missing capped W, % of base	28.4	22.0	23.7	27.8	21.2	22.9
Non-missing capped W + controls, % of base	28.4	22.0	23.7	27.8	21.2	22.9
Downward revision (% of notification) $\neq 0$, % of base	20.2	15.2	16.6	19.7	14.4	15.8
Downward revision (% of notification) > pooled median, % of base	14.8	10.9	12.0	14.4	10.4	11.4
Downward revision (% of notification) in top quartile, % of base	6.2	6.2	6.2	5.7	5.8	5.8
Regression sample: Efficiency index, % of base	28.4	21.4	23.3	27.4	20.3	22.1
Regression sample: Corruption index, % of base	28.0	21.4	23.2	27.1	20.4	22.1
Regression sample: Corruption in General (q34), % of base	20.6	14.2	15.9	19.4	13.5	15.0
Regression sample: Corruption Experience (q35), % of base	28.4	21.8	23.6	27.8	21.0	22.7
Regression sample: Grade on Honesty (q32), % of base	26.5	18.2	20.5	25.1	16.4	18.7
Regression sample: Friend at tax Auth. (q42), % of base	25.3	19.1	20.8	23.4	17.9	19.3

Notes: Coverage rows report counts and percentages relative to the base surveyed sample in each panel-column cell (base = firms with q1 observed). Panels are defined as in the regression tables.

Figure 44: Scatter and linear fit: W_{main} vs notification value

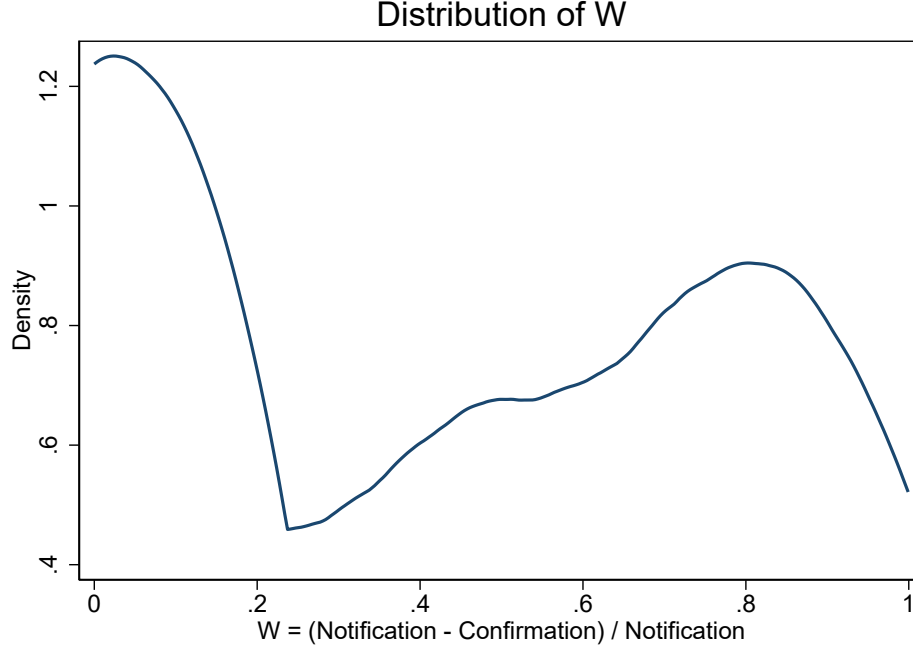


Figure 45: Density of W_{main}

Table 39: Selection method, capped W , and survey indexes: two panels (Recent audit / All surveyed firms)

A: Surveyed Firms (Self-Reporting a Recent Audit)	Efficiency Index			Corruption Index		
	(1) Full audits	(2) Desk audits	(3) All audits	(4) Full audits	(5) Desk audits	(6) All audits
Downward revision (% of notification)	0.51 (0.54)	-0.06 (0.34)	0.19 (0.27)	0.15 (0.46)	0.24 (0.29)	0.12 (0.23)
Algorithm	-0.53 (0.36)	0.43 (0.32)	-0.12 (0.24)	0.30 (0.28)	-0.40 (0.29)	-0.03 (0.21)
Inspectors x Overlap	0.11 (0.40)	-1.12* (0.63)	-0.75 (0.54)	0.61 (0.37)	-0.42 (0.37)	0.26 (0.25)
Algorithm x Random		-0.01 (0.33)	0.03 (0.30)		-0.16 (0.33)	-0.11 (0.25)
N	55	81	138	54	81	137
R2	0.14	0.25	0.08	0.27	0.33	0.18
Mean outcome	-0.01	-0.01	0.00	-0.13	-0.10	-0.11
B: All Surveyed Firms	(1)	(2)	(3)	(4)	(5)	(6)
Downward revision (% of notification)	0.63 (0.47)	0.00 (0.29)	0.25 (0.24)	-0.04 (0.41)	0.31 (0.28)	0.09 (0.22)
Algorithm	-0.53 (0.35)	0.18 (0.28)	-0.20 (0.22)	0.37 (0.28)	-0.38 (0.29)	-0.03 (0.21)
Inspectors x Overlap	0.10 (0.37)	-1.43** (0.58)	-0.82 (0.53)	0.60* (0.35)	-0.25 (0.37)	0.38* (0.23)
Algorithm x Random		0.03 (0.30)	0.03 (0.27)		-0.22 (0.33)	-0.16 (0.25)
N	61	96	159	60	97	159
R2	0.18	0.20	0.09	0.30	0.22	0.12
Mean outcome	0.01	0.03	0.03	-0.22	-0.05	-0.11

Notes: OLS results for the Efficiency Index and Corruption Index. The specification is $y_{ig} = \beta_0 + \beta_1 W_i + \beta_2 \text{Algorithm}_i + \beta_3 \text{Overlap}_i + \beta_4 \text{Random}_i + \beta_5 \text{Safeties}_i + \beta_6 \text{Horsprogramme}_i + \gamma_g + \varepsilon_{ig}$. Here $W_i = W_{\text{main},i} = \max\{(\text{notificationvalue}_i - \text{confirmationvalue}_i) / \text{notificationvalue}_i, 0\}$. The control set follows Table D7 (algorithm-selection, overlap, random-selection, replacement, and ad hoc indicators), with the addition of W_i . Panel A includes surveyed firms that report having been audited recently; Panel B includes all surveyed firms. Within each panel, columns report full audits, desk audits, and all audits. Fixed effects are at the selection-year $\times$ tax-office level. Robust standard errors (Huber-White) are in parentheses. Stars denote significance at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 40: Selection method, capped W , and survey-question outcomes: two panels (Recent audit / All surveyed firms)

A: Surveyed Firms (Self-Reporting a Recent Audit)	Corruption in General			Corruption Experience			Grade on Honesty			Friend at tax Auth.		
	(1) Full audits	(2) Desk audits	(3) All audits	(4) Full audits	(5) Desk audits	(6) All audits	(7) Full audits	(8) Desk audits	(9) All audits	(10) Full audits	(11) Desk audits	(12) All audits
Downward revision (% of notification)	21.050* (10.949)	5.150 (16.810)	12.704 (9.462)	-0.016 (0.099)	0.034 (0.092)	-0.006 (0.060)	-0.301 (0.903)	-0.622 (0.765)	-0.358 (0.643)	0.188 (0.576)	0.914 (0.554)	0.438 (0.413)
Algorithm	-4.177 (6.638)	-28.556 (21.741)	-7.982 (7.555)	0.025 (0.088)	-0.045 (0.070)	0.007 (0.075)	-0.875 (0.862)	-1.005 (1.214)	-0.884 (0.766)	0.172 (0.497)	0.440 (0.872)	0.241 (0.471)
Inspectors x Overlap	33.008*** (4.789)		33.960*** (5.046)	0.010 (0.037)		0.004 (0.026)	-0.879 (1.569)		-0.864 (1.664)	-0.016 (0.560)		-0.091 (0.564)
Algorithm x Random					-0.280 (0.226)	-0.264 (0.197)		0.136 (1.798)	-1.296 (0.988)		-1.664 (1.025)	-0.733 (0.666)
N	49	45	94	65	89	154	61	67	128	57	76	133
R2	0.324	0.429	0.314	0.135	0.369	0.237	0.137	0.544	0.325	0.078	0.417	0.238
Mean outcome	12.63	17.33	14.88	0.07	0.04	0.05	6.68	6.44	6.56	3.45	3.35	3.39
B: All Surveyed Firms	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Downward revision (% of notification)	14.436 (11.523)	-4.864 (16.297)	6.499 (9.495)	-0.099 (0.117)	-0.107 (0.085)	-0.106 (0.067)	-0.343 (0.832)	-0.258 (0.770)	-0.234 (0.594)	0.097 (0.551)	0.519 (0.517)	0.304 (0.381)
Algorithm	-5.729 (6.325)	-19.904 (15.256)	-8.736 (6.289)	0.020 (0.087)	0.032 (0.071)	0.025 (0.063)	-0.851 (0.841)	-0.315 (0.854)	-0.712 (0.680)	0.228 (0.488)	0.412 (0.713)	0.303 (0.419)
Inspectors x Overlap	17.455 (16.372)		20.064 (16.305)	-0.120 (0.145)		-0.116 (0.166)	-0.427 (1.347)		-0.445 (1.392)	0.007 (0.486)		-0.082 (0.441)
Algorithm x Random					-0.298* (0.172)	-0.279* (0.161)		-1.434 (0.908)	-1.690** (0.700)		-1.011 (0.977)	-0.654 (0.600)
N	54	54	108	73	111	184	67	78	145	61	88	149
R2	0.208	0.409	0.285	0.109	0.420	0.269	0.145	0.528	0.336	0.079	0.335	0.222
Mean outcome	12.85	15.64	14.25	0.09	0.06	0.07	6.74	6.50	6.61	3.39	3.40	3.40

Notes: OLS results for four taxpayer-survey outcomes: Corruption in general (Columns 1 to 3), Corruption experience (Columns 4 to 6), Grade on honesty (Columns 7 to 9), and Friend at tax authority (Columns 10 to 12). The specification is $y_{ig} = \beta_0 + \beta_1 W_i + \beta_2 \text{Algorithm}_i + \beta_3 \text{Overlap}_i + \beta_4 \text{Random}_i + \beta_5 \text{Safeties}_i + \beta_6 \text{Horsprogramme}_i + \gamma_g + \varepsilon_{ig}$. Here $W_i = W_{\text{main},i} = \max\{(\text{notificationvalue}_i - \text{confirmationvalue}_i)/\text{notificationvalue}_i, 0\}$. The control set follows Table D7 (algorithm-selection, overlap, random-selection, replacement, and ad hoc indicators), with the addition of W_i . Panel A includes surveyed firms that report having been audited recently; Panel B includes all surveyed firms. Within each panel, columns report full audits, desk audits, and all audits. Corruption in general is the declared belief about the percentage of audits in Senegal affected by corruption; Corruption experience indicates that the respondent declared an instance of bribery to obtain tax favors; Grade on honesty is a 0-to-10 grade for the honesty of the inspector with whom the respondent last interacted during a full audit; and Friend at tax authority captures agreement with the statement that if the firm's owner has a friend at the tax authority, the firm will rarely be audited. Fixed effects are at the inspector-cluster-year level. Robust standard errors (Huber-White) are in parentheses. Stars denote significance at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.5 Robustness regression tables

Table 41: Robustness: $W_{i \neq 0}$ and survey indexes (Recent audit / All surveyed firms)

A: Surveyed Firms (Self-Reporting a Recent Audit)	Efficiency Index			Corruption Index		
	(1) Full audits	(2) Desk audits	(3) All audits	(4) Full audits	(5) Desk audits	(6) All audits
Downward revision (% of notification) $\neq 0$	0.43 (0.41)	-0.07 (0.27)	0.21 (0.22)	0.29 (0.33)	0.21 (0.22)	0.15 (0.17)
Algorithm	-0.52 (0.36)	0.43 (0.32)	-0.12 (0.24)	0.31 (0.28)	-0.42 (0.29)	-0.03 (0.21)
Inspectors x Overlap	0.29 (0.38)	-1.13* (0.63)	-0.68 (0.54)	0.68* (0.36)	-0.36 (0.35)	0.31 (0.24)
Algorithm x Random		-0.01 (0.31)	0.04 (0.30)		-0.14 (0.32)	-0.10 (0.25)
N	55	81	138	54	81	137
R2	0.15	0.25	0.09	0.28	0.33	0.19
Mean outcome	-0.01	-0.01	0.00	-0.13	-0.10	-0.11
B: All Surveyed Firms	(1)	(2)	(3)	(4)	(5)	(6)
Downward revision (% of notification) $\neq 0$	0.46 (0.35)	-0.08 (0.23)	0.18 (0.19)	0.08 (0.31)	0.21 (0.21)	0.09 (0.16)
Algorithm	-0.52 (0.34)	0.17 (0.28)	-0.20 (0.22)	0.37 (0.27)	-0.38 (0.29)	-0.03 (0.21)
Inspectors x Overlap	0.30 (0.33)	-1.44** (0.60)	-0.76 (0.54)	0.58 (0.35)	-0.17 (0.37)	0.41* (0.22)
Algorithm x Random		0.04 (0.29)	0.05 (0.27)		-0.20 (0.32)	-0.16 (0.25)
N	61	96	159	60	97	159
R2	0.18	0.20	0.09	0.30	0.22	0.12
Mean outcome	0.01	0.03	0.03	-0.22	-0.05	-0.11

Notes: OLS results for the Efficiency Index and Corruption Index. The specification is $y_{ig} = \beta_0 + \beta_1 W_i^{\neq 0} + \beta_2 \text{Algorithm}_i + \beta_3 \text{Overlap}_i + \beta_4 \text{Random}_i + \beta_5 \text{Safeties}_i + \beta_6 \text{Horsprogramme}_i + \gamma_g + \varepsilon_{ig}$. Here $W_i^{\neq 0} = \mathbf{1}\{W_{\text{main},i} \neq 0\}$. The control set follows Table D7 (algorithm-selection, overlap, random-selection, replacement, and ad hoc indicators), with the addition of the discrepancy term $W_i^{\neq 0}$. Panel A includes surveyed firms that report having been audited recently; Panel B includes all surveyed firms. Within each panel, columns report full audits, desk audits, and all audits. Fixed effects are at the selection-year $\times$ tax-office level. Robust standard errors (Huber-White) are in parentheses. Stars denote significance at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 42: Robustness: $W_{\neq 0}$ and survey questions (Recent audit / All surveyed firms)

A: Surveyed Firms (Self-Reporting a Recent Audit)	Corruption in General			Corruption Experience			Grade on Honesty			Friend at tax Auth.		
	(1) Full audits	(2) Desk audits	(3) All audits	(4) Full audits	(5) Desk audits	(6) All audits	(7) Full audits	(8) Desk audits	(9) All audits	(10) Full audits	(11) Desk audits	(12) All audits
Downward revision (% of notification) $\neq 0$	17.226* (9.243)	3.684 (10.664)	10.975 (7.172)	-0.014 (0.075)	0.099 (0.069)	0.038 (0.050)	0.419 (0.768)	0.044 (0.631)	0.281 (0.532)	0.655 (0.458)	0.656 (0.422)	0.629** (0.315)
Algorithm	-4.663 (6.720)	-28.728 (21.678)	-8.520 (7.599)	0.025 (0.089)	-0.062 (0.071)	0.005 (0.076)	-0.907 (0.873)	-1.165 (1.234)	-0.942 (0.774)	0.200 (0.481)	0.509 (0.782)	0.245 (0.456)
Inspectors x Overlap	38.446*** (2.649)		37.160*** (3.725)	0.005 (0.018)		0.001 (0.015)	-0.977 (1.569)		-0.986 (1.673)	0.067 (0.499)		0.082 (0.528)
Algorithm x Random					-0.287 (0.223)	-0.273 (0.195)		0.111 (1.806)	-1.226 (1.010)		-1.650* (0.950)	-0.835 (0.691)
N	49	45	94	65	89	154	61	67	128	57	76	133
R2	0.319	0.429	0.319	0.136	0.395	0.240	0.141	0.537	0.325	0.108	0.418	0.259
Mean outcome	12.63	17.33	14.88	0.07	0.04	0.05	6.68	6.44	6.56	3.45	3.35	3.39
B: All Surveyed Firms	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Downward revision (% of notification) $\neq 0$	9.151 (9.581)	-1.612 (10.466)	4.676 (7.148)	-0.116 (0.094)	0.016 (0.072)	-0.049 (0.059)	0.228 (0.692)	0.047 (0.578)	0.192 (0.482)	0.527 (0.424)	0.498 (0.362)	0.505* (0.280)
Algorithm	-6.042 (6.344)	-20.476 (14.568)	-8.787 (6.301)	0.015 (0.088)	0.016 (0.069)	0.024 (0.063)	-0.874 (0.848)	-0.380 (0.866)	-0.747 (0.681)	0.250 (0.475)	0.402 (0.686)	0.301 (0.410)
Inspectors x Overlap	21.202 (16.389)		21.634 (16.345)	-0.145 (0.144)		-0.154 (0.172)	-0.620 (1.305)		-0.588 (1.366)	-0.069 (0.425)		-0.051 (0.423)
Algorithm x Random					-0.303 (0.185)	-0.277* (0.165)		-1.363 (0.855)	-1.570** (0.661)		-1.041 (0.947)	-0.754 (0.628)
N	54	54	108	73	111	184	67	78	145	61	88	149
R2	0.193	0.407	0.285	0.125	0.407	0.261	0.145	0.527	0.336	0.102	0.343	0.238
Mean outcome	12.85	15.64	14.25	0.09	0.06	0.07	6.74	6.50	6.61	3.39	3.40	3.40

Notes: OLS results for four taxpayer-survey outcomes: Corruption in general (Columns 1 to 3), Corruption experience (Columns 4 to 6), Grade on honesty (Columns 7 to 9), and Friend at tax authority (Columns 10 to 12). The specification is $y_{ig} = \beta_0 + \beta_1 W_i^{\neq 0} + \beta_2 \text{Algorithm}_i + \beta_3 \text{Overlap}_i + \beta_4 \text{Random}_i + \beta_5 \text{Safeties}_i + \beta_6 \text{Horsprogramme}_i + \gamma_g + \varepsilon_{ig}$. Here $W_i^{\neq 0} = \mathbf{1}\{W_{\text{main},i} \neq 0\}$. The control set follows Table D7 (algorithm-selection, overlap, random-selection, replacement, and ad hoc indicators), with the addition of the discrepancy term $W_i^{\neq 0}$. Panel A includes surveyed firms that report having been audited recently; Panel B includes all surveyed firms. Within each panel, columns report full audits, desk audits, and all audits. Corruption in general is the declared belief about the percentage of audits in Senegal affected by corruption; Corruption experience indicates that the respondent declared an instance of bribery to obtain tax favors; Grade on honesty is a 0-to-10 grade for the honesty of the inspector with whom the respondent last interacted during a full audit; and Friend at tax authority captures agreement with the statement that if the firm's owner has a friend at the tax authority, the firm will rarely be audited. Fixed effects are at the inspector-cluster-year level. Robust standard errors (Huber-White) are in parentheses. Stars denote significance at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 43: Robustness: $W_{>\text{med}}$ and survey indexes (Recent audit / All surveyed firms)

A: Surveyed Firms (Self-Reporting a Recent Audit)	Efficiency Index			Corruption Index		
	(1) Full audits	(2) Desk audits	(3) All audits	(4) Full audits	(5) Desk audits	(6) All audits
Downward revision (% of notification) > median	0.44 (0.36)	0.06 (0.24)	0.29 (0.19)	0.07 (0.32)	-0.02 (0.23)	-0.07 (0.17)
Algorithm	-0.51 (0.36)	0.43 (0.31)	-0.11 (0.24)	0.30 (0.28)	-0.41 (0.28)	-0.04 (0.21)
Inspectors x Overlap	0.22 (0.41)	-1.13* (0.60)	-0.72 (0.52)	0.66* (0.36)	-0.39 (0.42)	0.27 (0.26)
Algorithm x Random		-0.04 (0.33)	-0.00 (0.31)		-0.10 (0.33)	-0.09 (0.25)
N	55	81	138	54	81	137
R2	0.16	0.25	0.10	0.27	0.33	0.18
Mean outcome	-0.01	-0.01	0.00	-0.13	-0.10	-0.11
B: All Surveyed Firms	(1)	(2)	(3)	(4)	(5)	(6)
Downward revision (% of notification) > median	0.51* (0.30)	0.16 (0.21)	0.34** (0.17)	-0.12 (0.28)	0.05 (0.21)	-0.09 (0.16)
Algorithm	-0.50 (0.35)	0.18 (0.28)	-0.18 (0.22)	0.36 (0.28)	-0.38 (0.28)	-0.04 (0.21)
Inspectors x Overlap	0.25 (0.36)	-1.44*** (0.55)	-0.79 (0.51)	0.59* (0.34)	-0.22 (0.41)	0.39 (0.24)
Algorithm x Random		-0.01 (0.31)	0.02 (0.28)		-0.18 (0.32)	-0.14 (0.25)
N	61	96	159	60	97	159
R2	0.20	0.20	0.11	0.30	0.21	0.12
Mean outcome	0.01	0.03	0.03	-0.22	-0.05	-0.11

Notes: OLS results for the Efficiency Index and Corruption Index. The specification is $y_{ig} = \beta_0 + \beta_1 W_i^{>\text{med}} + \beta_2 \text{Algorithm}_i + \beta_3 \text{Overlap}_i + \beta_4 \text{Random}_i + \beta_5 \text{Safeties}_i + \beta_6 \text{Horsprogramme}_i + \gamma_g + \varepsilon_{ig}$. Here $W_i^{>\text{med}} = \mathbf{1}\{W_{\text{main},i} > \widetilde{W}\}$, where $\widetilde{W}$ is the pooled median of non-missing W_{main} in the regression-eligible sample. The control set follows Table D7 (algorithm-selection, overlap, random-selection, replacement, and ad hoc indicators), with the addition of the discrepancy term $W_i^{>\text{med}}$. Panel A includes surveyed firms that report having been audited recently; Panel B includes all surveyed firms. Within each panel, columns report full audits, desk audits, and all audits. Fixed effects are at the selection-year $\times$ tax-office level. Robust standard errors (Huber-White) are in parentheses. Stars denote significance at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 44: Robustness: $W_{>med}$ and survey questions (Recent audit / All surveyed firms)

A: Surveyed Firms (Self-Reporting a Recent Audit)	Corruption in General			Corruption Experience			Grade on Honesty			Friend at tax Auth.		
	(1) Full audits	(2) Desk audits	(3) All audits	(4) Full audits	(5) Desk audits	(6) All audits	(7) Full audits	(8) Desk audits	(9) All audits	(10) Full audits	(11) Desk audits	(12) All audits
Downward revision (% of notification) > median	12.917* (7.228)	5.271 (11.889)	8.785 (6.359)	-0.017 (0.077)	0.004 (0.072)	-0.017 (0.047)	-0.026 (0.613)	-0.276 (0.609)	-0.084 (0.465)	0.452 (0.435)	0.320 (0.487)	0.324 (0.315)
Algorithm	-3.681 (6.664)	-28.964 (21.789)	-7.639 (7.663)	0.024 (0.089)	-0.042 (0.072)	0.008 (0.075)	-0.886 (0.859)	-1.062 (1.250)	-0.908 (0.767)	0.230 (0.490)	0.569 (0.898)	0.266 (0.461)
Inspectors x Overlap	38.773*** (2.486)		37.454*** (3.538)	0.008 (0.020)		0.005 (0.017)	-0.972 (1.567)		-0.977 (1.669)	0.077 (0.493)		0.089 (0.523)
Algorithm x Random					-0.274 (0.227)	-0.262 (0.199)		0.071 (1.845)	-1.288 (1.022)		-1.587 (1.016)	-0.735 (0.671)
N	49	45	94	65	89	154	61	67	128	57	76	133
R2	0.301	0.431	0.313	0.136	0.367	0.237	0.135	0.540	0.323	0.098	0.393	0.239
Mean outcome	12.63	17.33	14.88	0.07	0.04	0.05	6.68	6.44	6.56	3.45	3.35	3.39
B: All Surveyed Firms	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Downward revision (% of notification) > median	8.636 (7.434)	-2.923 (11.470)	4.235 (6.259)	-0.063 (0.083)	-0.083 (0.064)	-0.076 (0.049)	-0.025 (0.561)	-0.145 (0.556)	-0.041 (0.414)	0.354 (0.387)	0.136 (0.389)	0.228 (0.274)
Algorithm	-5.573 (6.444)	-20.132 (14.723)	-8.615 (6.352)	0.019 (0.088)	0.031 (0.071)	0.023 (0.063)	-0.864 (0.837)	-0.344 (0.854)	-0.731 (0.674)	0.272 (0.480)	0.462 (0.713)	0.323 (0.414)
Inspectors x Overlap	21.448 (16.514)		21.788 (16.340)	-0.142 (0.153)		-0.138 (0.169)	-0.568 (1.308)		-0.538 (1.373)	-0.021 (0.394)		0.024 (0.393)
Algorithm x Random					-0.295* (0.175)	-0.278* (0.165)		-1.420 (0.908)	-1.643** (0.691)		-0.940 (0.945)	-0.654 (0.605)
N	54	54	108	73	111	184	67	78	145	61	88	149
R2	0.199	0.408	0.285	0.108	0.423	0.270	0.143	0.528	0.335	0.093	0.325	0.223
Mean outcome	12.85	15.64	14.25	0.09	0.06	0.07	6.74	6.50	6.61	3.39	3.40	3.40

Notes: OLS results for four taxpayer-survey outcomes: Corruption in general (Columns 1 to 3), Corruption experience (Columns 4 to 6), Grade on honesty (Columns 7 to 9), and Friend at tax authority (Columns 10 to 12). The specification is $y_{ig} = \beta_0 + \beta_1 W_i^{>med} + \beta_2 \text{Algorithm}_i + \beta_3 \text{Overlap}_i + \beta_4 \text{Random}_i + \beta_5 \text{Safeties}_i + \beta_6 \text{Horsprogramme}_i + \gamma_g + \varepsilon_{ig}$. Here $W_i^{>med} = \mathbf{1}\{W_{\text{main},i} > \bar{W}\}$, where $\bar{W}$ is the pooled median of non-missing W_{main} in the regression-eligible sample. The control set follows Table D7 (algorithm-selection, overlap, random-selection, replacement, and ad hoc indicators), with the addition of the discrepancy term $W_i^{>med}$. Panel A includes surveyed firms that report having been audited recently; Panel B includes all surveyed firms. Within each panel, columns report full audits, desk audits, and all audits. Corruption in general is the declared belief about the percentage of audits in Senegal affected by corruption; Corruption experience indicates that the respondent declared an instance of bribery to obtain tax favors; Grade on honesty is a 0-to-10 grade for the honesty of the inspector with whom the respondent last interacted during a full audit; and Friend at tax authority captures agreement with the statement that if the firm's owner has a friend at the tax authority, the firm will rarely be audited. Fixed effects are at the inspector-cluster-year level. Robust standard errors (Huber-White) are in parentheses. Stars denote significance at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 45: Robustness: $W_{\geq q75}$ and survey indexes (Recent audit / All surveyed firms)

A: Surveyed Firms (Self-Reporting a Recent Audit)	Efficiency Index			Corruption Index		
	(1) Full audits	(2) Desk audits	(3) All audits	(4) Full audits	(5) Desk audits	(6) All audits
Downward revision (% of notification) in top quartile	0.24 (0.34)	-0.32 (0.29)	-0.19 (0.21)	0.04 (0.31)	0.29 (0.27)	0.21 (0.19)
Algorithm	-0.57 (0.37)	0.38 (0.32)	-0.13 (0.24)	0.29 (0.30)	-0.37 (0.29)	-0.04 (0.21)
Inspectors x Overlap	0.06 (0.43)	-1.02 (0.67)	-0.65 (0.61)	0.62 (0.41)	-0.49 (0.38)	0.19 (0.25)
Algorithm x Random		0.05 (0.33)	0.08 (0.29)		-0.16 (0.33)	-0.12 (0.25)
N	55	81	138	54	81	137
R2	0.13	0.27	0.09	0.27	0.34	0.19
Mean outcome	-0.01	-0.01	0.00	-0.13	-0.10	-0.11
B: All Surveyed Firms	(1)	(2)	(3)	(4)	(5)	(6)
Downward revision (% of notification) in top quartile	0.30 (0.31)	-0.25 (0.24)	-0.10 (0.18)	0.10 (0.29)	0.47* (0.25)	0.31 (0.19)
Algorithm	-0.60* (0.35)	0.16 (0.28)	-0.21 (0.22)	0.35 (0.29)	-0.37 (0.28)	-0.06 (0.21)
Inspectors x Overlap	0.08 (0.39)	-1.34** (0.63)	-0.74 (0.59)	0.51 (0.39)	-0.39 (0.37)	0.26 (0.24)
Algorithm x Random		0.06 (0.30)	0.07 (0.26)		-0.21 (0.33)	-0.15 (0.25)
N	61	96	159	60	97	159
R2	0.16	0.21	0.08	0.30	0.24	0.13
Mean outcome	0.01	0.03	0.03	-0.22	-0.05	-0.11

Notes: OLS results for the Efficiency Index and Corruption Index. The specification is $y_{ig} = \beta_0 + \beta_1 W_i^{\geq q75} + \beta_2 \text{Algorithm}_i + \beta_3 \text{Overlap}_i + \beta_4 \text{Random}_i + \beta_5 \text{Safeties}_i + \beta_6 \text{Horsprogramme}_i + \gamma_g + \varepsilon_{ig}$. Here $W_i^{\geq q75} = \mathbf{1}\{W_{\text{main},i} \geq \widetilde{W}_{75}\}$, where $\widetilde{W}_{75}$ is the pooled 75th percentile of non-missing W_{main} in the regression-eligible sample. The control set follows Table D7 (algorithm-selection, overlap, random-selection, replacement, and ad hoc indicators), with the addition of the discrepancy term $W_i^{\geq q75}$. Panel A includes surveyed firms that report having been audited recently; Panel B includes all surveyed firms. Within each panel, columns report full audits, desk audits, and all audits. Fixed effects are at the selection-year $\times$ tax-office level. Robust standard errors (Huber-White) are in parentheses. Stars denote significance at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 46: Robustness: $W_{\geq q75}$ and survey questions (Recent audit / All surveyed firms)

A: Surveyed Firms (Self-Reporting a Recent Audit)	Corruption in General			Corruption Experience			Grade on Honesty			Friend at tax Auth.		
	(1) Full audits	(2) Desk audits	(3) All audits	(4) Full audits	(5) Desk audits	(6) All audits	(7) Full audits	(8) Desk audits	(9) All audits	(10) Full audits	(11) Desk audits	(12) All audits
Downward revision (% of notification) in top quartile	11.780 (8.289)	-13.054 (15.636)	3.656 (7.835)	-0.095* (0.052)	-0.034 (0.059)	-0.075* (0.038)	-0.619 (0.631)	-0.620 (0.612)	-0.531 (0.503)	-0.205 (0.458)	0.535 (0.408)	0.080 (0.345)
Algorithm	-6.734 (6.332)	-23.874 (21.692)	-8.631 (7.504)	0.040 (0.089)	-0.036 (0.070)	0.019 (0.076)	-0.779 (0.875)	-0.949 (1.071)	-0.801 (0.765)	0.203 (0.501)	0.544 (0.944)	0.248 (0.479)
Inspectors x Overlap	29.902*** (8.100)		34.686*** (6.991)	0.084 (0.057)		0.064 (0.043)	-0.481 (1.527)		-0.552 (1.627)	0.273 (0.695)		0.002 (0.646)
Algorithm x Random					-0.264 (0.209)	-0.255 (0.181)		0.461 (1.807)	-1.083 (0.920)		-1.605 (1.054)	-0.670 (0.631)
N	49	45	94	65	89	154	61	67	128	57	76	133
R2	0.283	0.444	0.296	0.154	0.369	0.248	0.147	0.546	0.331	0.080	0.399	0.229
Mean outcome	12.63	17.33	14.88	0.07	0.04	0.05	6.68	6.44	6.56	3.45	3.35	3.39
B: All Surveyed Firms	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Downward revision (% of notification) in top quartile	11.482 (8.507)	-16.220 (13.043)	2.365 (7.493)	-0.104** (0.049)	-0.085* (0.050)	-0.096*** (0.035)	-0.671 (0.604)	-0.275 (0.712)	-0.445 (0.487)	-0.148 (0.456)	0.289 (0.459)	0.056 (0.329)
Algorithm	-8.040 (6.046)	-16.979 (14.546)	-9.026 (6.419)	0.041 (0.090)	0.032 (0.068)	0.038 (0.065)	-0.730 (0.860)	-0.264 (0.782)	-0.619 (0.685)	0.252 (0.493)	0.415 (0.720)	0.301 (0.421)
Inspectors x Overlap	14.378 (17.544)		20.969 (16.704)	-0.073 (0.163)		-0.080 (0.180)	-0.009 (1.290)		-0.168 (1.359)	0.211 (0.596)		0.020 (0.506)
Algorithm x Random					-0.303* (0.172)	-0.285* (0.158)		-1.387 (0.892)	-1.650** (0.694)		-0.926 (0.930)	-0.600 (0.570)
N	54	54	108	73	111	184	67	78	145	61	88	149
R2	0.210	0.438	0.281	0.115	0.418	0.270	0.156	0.529	0.341	0.080	0.327	0.217
Mean outcome	12.85	15.64	14.25	0.09	0.06	0.07	6.74	6.50	6.61	3.39	3.40	3.40

Notes: OLS results for four taxpayer-survey outcomes: Corruption in general (Columns 1 to 3), Corruption experience (Columns 4 to 6), Grade on honesty (Columns 7 to 9), and Friend at tax authority (Columns 10 to 12). The specification is $y_{ig} = \beta_0 + \beta_1 W_i^{\geq q75} + \beta_2 \text{Algorithm}_i + \beta_3 \text{Overlap}_i + \beta_4 \text{Random}_i + \beta_5 \text{Safeties}_i + \beta_6 \text{Horsprogramme}_i + \gamma_g + \varepsilon_{ig}$. Here $W_i^{\geq q75} = \mathbf{1}\{W_{\text{main},i} \geq \bar{W}_{75}\}$, where $\bar{W}_{75}$ is the pooled 75th percentile of non-missing W_{main} in the regression-eligible sample. The control set follows Table D7 (algorithm-selection, overlap, random-selection, replacement, and ad hoc indicators), with the addition of the discrepancy term $W_i^{\geq q75}$. Panel A includes surveyed firms that report having been audited recently; Panel B includes all surveyed firms. Within each panel, columns report full audits, desk audits, and all audits. Corruption in general is the declared belief about the percentage of audits in Senegal affected by corruption; Corruption experience indicates that the respondent declared an instance of bribery to obtain tax favors; Grade on honesty is a 0-to-10 grade for the honesty of the inspector with whom the respondent last interacted during a full audit; and Friend at tax authority captures agreement with the statement that if the firm's owner has a friend at the tax authority, the firm will rarely be audited. Fixed effects are at the inspector-cluster-year level. Robust standard errors (Huber-White) are in parentheses. Stars denote significance at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.